EXPONENTIAL SUMS OVER THE RIEMANN ZEROS VERSUS THE RIEMANN–VON MANGOLDT CLOCK, WITH FINITE MÖBIUS INTERPOLATION AND PRIME-POWER SPECTRAL IDENTITIES

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Remark 0.1. This manuscript develops the analytic identities in full theorem–proof form. Standard and literature-based material is cited at the point of use. Original finite identities are proved directly. Heuristics and conjectures are explicitly labelled. Numerical evidence is confined to the final part, immediately before the conclusions, and is never used as a substitute for an analytic proof.

1. Introduction: the exponential zero sum versus the Riemann–von Mangoldt clock

The point of departure is not the finite interpolation introduced later. It is the direct comparison between the exponential walk formed by the actual ordinates of the nontrivial zeros and the walk formed by the smooth Riemann–von Mangoldt clock. Let

$$0 < \gamma_1 \le \gamma_2 \le \cdots \tag{1.1}$$

denote the positive ordinates, repeated according to multiplicity, and fix a frequency $\omega > 0$. The primary object is

$$S_N(\omega) = \sum_{k=1}^{N} e^{i\omega\gamma_k}. \tag{1.2}$$

The smooth comparison sequence is obtained by inverting the principal Riemann–von Mangoldt counting function. Writing

$$N_0(T) = \frac{T}{2\pi}\log\Big(\frac{T}{2\pi e}\Big) + \frac{7}{8}, \tag{1.3}$$

we define $\widetilde{\gamma}_k$ by

$$N_0(\widetilde{\gamma}_k) = k - \frac{1}{2} \tag{1.4}$$

and form

$$S_N^{\mathrm{RVM}}(\omega) = \sum_{k=1}^{N} e^{i\omega\widetilde{\gamma}_k}. \tag{1.5}$$

The discrepancy

$$\mathcal{E}_N(\omega) = S_N(\omega) - S_N^{\mathrm{RVM}}(\omega) \tag{1.6}$$

is the object from which the subsequent prime-power investigation arose. The smooth clock reproduces the main zero density; the remaining coherent frequencies are forced by the explicit formula to be supported on prime powers. The two branches with frequencies

$$\omega - \log n \qquad \text{and} \qquad \omega + \log n \tag{1.7}$$

are therefore not an auxiliary model imposed on the data. They are the Fourier displacements of the two prime terms in the Guinand–Weil explicit formula.

A second construction was found while studying how a finite Möbius section can be completed by the reciprocal zeta function. It begins with an affine interpolation of the terminal prime coefficients. The smallest numerical example already contains the complete mechanism. For $N = 10$,

$$A_{10}(s) = 1 - 2^{-s} - 3^{-s} - 5^{-s} + 6^{-s} + 10^{-s}, \tag{1.8}$$

$$P_{10}(s) = 7^{-s}, \tag{1.9}$$

$$B_{10} = A_{10}(0) = 0. \tag{1.10}$$

If the unknown terminal value is denoted by the letter m, the first interpolation is

$$\mathcal{I}_{10}(s;m) = A_{10}(s) + m7^{-s}. \tag{1.11}$$

Solving

$$\mathcal{I}_{10}(s;m) = \zeta(s)^{-1} \tag{1.12}$$

for the number m produces

$$\Phi_{10}(s) = 7^s \left(\zeta(s)^{-1} - A_{10}(s)\right). \tag{1.13}$$

The general symbolic form is

$$\mathcal{I}_N(s;m) = A_N(s) + \frac{m - B_N}{h_N} P_N(s), \tag{1.14}$$

and solving $\mathcal{I}_N(s;m) = \zeta(s)^{-1}$ yields

$$\Phi_N(s) = B_N + h_N \frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)}. \tag{1.15}$$

This finite interpolation is analytically independent of the phase-walk experiment, but it uses the same reciprocal-zeta and prime-support architecture. It is included because it gives exact finite identities, not because it is the historical origin of the project.

The principal rigorous results established below are the following. First, the Riemann–von Mangoldt comparison is written exactly as a Stieltjes integral and its smooth antiderivative is derived. Second, the finite interpolation is solved completely: its value at the origin, its limit on the positive real axis, its exponential rate, its rounding criterion, its poles, and its finite parts are proved. Third, the two prime-power branches are derived from a regularized explicit formula and their signed invariant

$$\Delta_\omega(Q) = \sum_{n=p^r>Q} \left(|c_n^-|^2 - |c_n^+|^2\right) \tag{1.16}$$

is shown to satisfy

$$\Delta_\omega(Q) \sim \frac{\omega}{\pi^2 \log Q}. \tag{1.17}$$

Fourth, the complete-zero walk and the ordinate-only walk are separated by an atomic positive measure. This gives several exact routes to the Riemann Hypothesis. None is claimed to be completed: each route is stated with the precise missing uniform or local estimate.

Numerical material is placed only in the final part. It supports the analytic structure but is not used in any proof.

Part 1. The exponential walk and the Riemann–von Mangoldt benchmark

2. The zero-counting function and its smooth inverse

Definition 2.1 (Midpoint zero count)**.** Let $N(T)$ be the number of nontrivial zeros $\rho = \beta + i\gamma$ with $0 < \gamma \le T$, counted with multiplicity. At a zero ordinate $T = \tau$, let

$$N^{\#}(\tau) = \frac{1}{2}\left(N(\tau^-) + N(\tau^+)\right). \tag{2.1}$$

For a simple ordered ordinate γ_k, this equals $k - 1/2$.

Theorem 2.2 (Riemann–von Mangoldt formula)**.** *With the standard continuous determination of the argument,*

$$N^{\#}(T) = N_0(T) + S(T) + R(T), \tag{2.2}$$

where

$$S(T) = \frac{1}{\pi} \arg \zeta\left(\frac{1}{2} + iT\right), \tag{2.3}$$

and

$$R(T) = O(T^{-1}) \tag{2.4}$$

away from the conventional endpoint interpretation. The formula is standard; see [12, Chapter 9].

Proof. For T that is not a zero ordinate, apply the argument principle to the completed zeta function

$$\xi(s) = \frac{1}{2} s(s-1)\pi^{-s/2}\Gamma\left(\frac{s}{2}\right)\zeta(s) \tag{2.5}$$

on the standard rectangle with vertical sides to the right of one and to the left of zero, and horizontal sides at heights 0 and T. The functional equation $\xi(s) = \xi(1-s)$ pairs the two vertical sides. After letting the horizontal width tend to infinity, the change of argument gives the exact counting relation

$$N(T) = 1 + \frac{\vartheta(T)}{\pi} + S(T), \tag{2.6}$$

where

$$\vartheta(T) = \Im \log \Gamma\left(\frac{1}{4} + \frac{iT}{2}\right) - \frac{T}{2}\log \pi. \tag{2.7}$$

The choice of the path defining $S(T)$ is the standard one from 2 to $2 + iT$ and then to $1/2 + iT$. At a zero ordinate, taking the average of the two one-sided limits gives the midpoint convention.

Stirling's formula in a fixed sector gives

$$\log \Gamma(z) = \left(z - \frac{1}{2}\right)\log z - z + \frac{1}{2}\log(2\pi) + O(|z|^{-1}). \tag{2.8}$$

With $z = 1/4 + iT/2$, taking imaginary parts yields

$$\vartheta(T) = \frac{T}{2}\log\left(\frac{T}{2\pi}\right) - \frac{T}{2} - \frac{\pi}{8} + O(T^{-1}). \tag{2.9}$$

Substitution into (2.6) gives

$$N^{\#}(T) = \frac{T}{2\pi}\log\left(\frac{T}{2\pi}\right) - \frac{T}{2\pi} + 1 - \frac{1}{8} + S(T) + O(T^{-1}) \tag{2.10}$$

$$= \frac{T}{2\pi}\log\left(\frac{T}{2\pi e}\right) + \frac{7}{8} + S(T) + O(T^{-1}), \tag{2.11}$$

which is the claimed formula. The argument-principle identity and the uniform Stirling expansion are standard; the displayed steps record the normalization and the origin of the constant $7/8$. □

Theorem 2.3 (Exact Lambert–W inverse of the smooth clock)**.** *For every real $x > 7/8$, the positive solution of $N_0(T) = x$ is*

$$\boxed{T = \frac{2\pi(x - 7/8)}{W((x-7/8)/e)}.} \tag{2.12}$$

In particular,

$$\widetilde{\gamma}_k = \frac{2\pi(k - 11/8)}{W((k-11/8)/e)} \tag{2.13}$$

when the midpoint convention $N_0(\widetilde{\gamma}_k) = k - 1/2$ is used.

Proof. Put $u = T/(2\pi e)$ and $y = x - 7/8$. The equation $N_0(T) = x$ becomes

$$y = e\,u\log u. \tag{2.14}$$

Writing $v = \log u$ gives $u = e^v$, hence

$$ve^v = \frac{y}{e}. \tag{2.15}$$

By the defining equation of the principal Lambert function, $v = W(y/e)$. Therefore

$$u = e^{W(y/e)} = \frac{y/e}{W(y/e)}. \tag{2.16}$$

Multiplication by $2\pi e$ proves (2.12). Substituting $x = k - 1/2$ gives (2.13). Positivity and uniqueness follow because $N_0'(T) = (2\pi)^{-1}\log(T/2\pi) > 0$ for $T > 2\pi$, the range relevant to all nontrivial zeros. □

3. Stieltjes representation and the smooth RVM walk

Proposition 3.1 (Exact Stieltjes representation)**.** *For T interpreted with midpoint endpoint weights,*

$$S_\omega^{\#}(T) = \int_{(0,T]} e^{i\omega t}\ \mathrm{d}N^{\#}(t). \tag{3.1}$$

For $0 < T_0 < T$, integration by parts gives

$$\int_{(T_0,T]} e^{i\omega t}\ \mathrm{d}N(t) = e^{i\omega T}N(T) - e^{i\omega T_0}N(T_0) \tag{3.2}$$

$$- i\omega\int_{T_0}^{T} e^{i\omega t}N(t)\ \mathrm{d}t. \tag{3.3}$$

Proof. The measure $\mathrm{d}N$ is the locally finite atomic measure

$$\mathrm{d}N = \sum_{\gamma>0} m(\gamma)\delta_\gamma. \tag{3.4}$$

Integrating $e^{i\omega t}$ against this measure gives the phase sum, with a half atom at an endpoint under the midpoint convention. The integration-by-parts identity is the standard formula for a continuously differentiable function integrated against a bounded-variation function:

$$\int_a^b f\ \mathrm{d}G = f(b)G(b) - f(a)G(a) - \int_a^b G(t)f'(t)\ \mathrm{d}t. \tag{3.5}$$

Take $f(t) = e^{i\omega t}$ and $G(t) = N(t)$. □

Theorem 3.2 (Smooth Riemann–von Mangoldt contribution)**.** *The Stieltjes integral of the smooth density*

$$N_0'(t) = \frac{1}{2\pi}\log\!\left(\frac{t}{2\pi}\right) \tag{3.6}$$

has the antiderivative

$$\boxed{A_{\mathrm{smooth},\omega}(T) = \frac{e^{i\omega T}\log(T/2\pi) - \mathrm{Ei}(i\omega T)}{2\pi i\omega} + C_\omega.} \tag{3.7}$$

Consequently, for any $T_0 > 0$,

$$\int_{T_0}^{T} e^{i\omega t}N_0'(t)\ \mathrm{d}t = A_{\mathrm{smooth},\omega}(T) - A_{\mathrm{smooth},\omega}(T_0). \tag{3.8}$$

Proof. Differentiate the numerator in (3.7). The derivative of the first term is

$$\frac{\mathrm{d}}{\mathrm{d}T}\left[e^{i\omega T}\log(T/2\pi)\right] = i\omega e^{i\omega T}\log(T/2\pi) + \frac{e^{i\omega T}}{T}. \tag{3.9}$$

Since $\frac{\mathrm{d}}{\mathrm{d}z}\,\mathrm{Ei}(z) = e^z/z$, the chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}T}\,\mathrm{Ei}(i\omega T) = \frac{e^{i\omega T}}{T}. \tag{3.10}$$

The two T^{-1} terms cancel. Division by $2\pi i\omega$ leaves exactly $e^{i\omega T}N_0'(T)$. The definite identity follows from the fundamental theorem of calculus on any interval avoiding the chosen branch cut; a change of branch alters only the constant C_ω. □

Corollary 3.3 (Exact RVM remainder decomposition)**.** *For $T_0 < T$,*

$$S_\omega^{\#}(T) - S_\omega^{\#}(T_0) = A_{\mathrm{smooth},\omega}(T) - A_{\mathrm{smooth},\omega}(T_0) \tag{3.11}$$

$$+ \int_{(T_0,T]} e^{i\omega t}\,\mathrm{d}S(t) + \int_{(T_0,T]} e^{i\omega t}\,\mathrm{d}R(t). \tag{3.12}$$

The arithmetic content of the phase walk is therefore contained in the Stieltjes transforms of the argument term and the smaller remainder.

Proof. The midpoint counting formula is the equality of bounded-variation functions

$$N^{\#} = N_0 + S + R. \tag{3.13}$$

Taking Stieltjes differentials preserves this equality as an identity of locally finite distributions. Integration against the continuously differentiable function $e^{i\omega t}$ is linear, so

$$\int e^{i\omega t}\,\mathrm{d}N^{\#} = \int e^{i\omega t}\,\mathrm{d}N_0 + \int e^{i\omega t}\,\mathrm{d}S + \int e^{i\omega t}\,\mathrm{d}R. \tag{3.14}$$

The first term equals the difference of the antiderivative in Theorem 3.2; the left side is the zero phase sum by Proposition 3.1. Restricting the identity to $(T_0, T]$ proves both displayed lines. □

4. Exact comparison of the indexed zero walk with the smooth clock

Definition 4.1. For each k, define the RVM displacement

$$\delta_k = \gamma_k - \widetilde{\gamma}_k. \tag{4.1}$$

Lemma 4.2 (Exact phase perturbation identity)**.** *For real x and h,*

$$e^{i(x+h)} - e^{ix} = ih\int_0^1 e^{i(x+th)}\,\mathrm{d}t. \tag{4.2}$$

In particular,

$$|e^{i(x+h)} - e^{ix}| \le |h|, \tag{4.3}$$

$$\left|e^{i(x+h)} - e^{ix} - ihe^{ix}\right| \le \frac{h^2}{2}. \tag{4.4}$$

Proof. Define $F(t) = e^{i(x+th)}$ on $[0,1]$. Since $F'(t) = ihe^{i(x+th)}$, the fundamental theorem of calculus gives

$$e^{i(x+h)} - e^{ix} = F(1) - F(0) = ih\int_0^1 e^{i(x+th)}\,\mathrm{d}t. \tag{4.5}$$

The integrand has modulus one, so taking absolute values yields the Lipschitz estimate. For the quadratic estimate, subtract ihe^{ix}:

$$e^{i(x+h)} - e^{ix} - ihe^{ix} = ihe^{ix}\int_0^1 (e^{ith} - 1)\,\mathrm{d}t. \tag{4.6}$$

Apply the already proved Lipschitz bound to $e^{ith}-1$, obtaining $|e^{ith}-1|\le t|h|$. Integration of t from zero to one gives $h^2/2$. □

Theorem 4.3 (Exact and first-order decompositions of the discrepancy)**.** *For every N,*

$$\mathcal{E}_N(\omega)=i\omega\sum_{k=1}^{N}\delta_k\int_0^1 e^{i\omega(\widetilde{\gamma}_k+t\delta_k)}\,\mathrm{d}t. \tag{4.7}$$

Moreover,

$$\mathcal{E}_N(\omega)=i\omega\sum_{k=1}^{N}\delta_k e^{i\omega\widetilde{\gamma}_k}+\mathcal{R}_N(\omega), \tag{4.8}$$

where

$$|\mathcal{R}_N(\omega)|\le\frac{\omega^2}{2}\sum_{k=1}^{N}\delta_k^2. \tag{4.9}$$

Proof. For each k, apply Lemma 4.2 to

$$x=\omega\widetilde{\gamma}_k,\qquad h=\omega(\gamma_k-\widetilde{\gamma}_k)=\omega\delta_k. \tag{4.10}$$

The resulting identity is

$$e^{i\omega\gamma_k}-e^{i\omega\widetilde{\gamma}_k}=i\omega\delta_k\int_0^1 e^{i\omega(\widetilde{\gamma}_k+t\delta_k)}\,\mathrm{d}t. \tag{4.11}$$

Summing from 1 to N gives the exact formula. For the first-order form, the quadratic estimate in the lemma gives a remainder r_k with

$$|r_k|\le\frac{\omega^2\delta_k^2}{2}. \tag{4.12}$$

The triangle inequality then gives the stated bound for the sum of all remainders. No restriction on the signs or dependence of the displacements is needed. □

Proposition 4.4 (Displacement in terms of the Riemann–von Mangoldt remainder)**.** *Assume γ_k is interpreted with midpoint counting, and let ξ_k lie between γ_k and $\widetilde{\gamma}_k$. Then*

$$\boxed{\delta_k=-\frac{S^{\#}(\gamma_k)+R(\gamma_k)}{N_0'(\xi_k)}.} \tag{4.13}$$

Equivalently,

$$\delta_k=-\frac{2\pi\,[S^{\#}(\gamma_k)+R(\gamma_k)]}{\log(\xi_k/2\pi)}. \tag{4.14}$$

Proof. By definition of the smooth inverse and midpoint count,

$$N_0(\widetilde{\gamma}_k)=k-\frac{1}{2}=N^{\#}(\gamma_k). \tag{4.15}$$

The Riemann–von Mangoldt formula gives

$$N_0(\widetilde{\gamma}_k)-N_0(\gamma_k)=S^{\#}(\gamma_k)+R(\gamma_k). \tag{4.16}$$

The mean-value theorem supplies a point ξ_k between the two ordinates such that the left-hand side equals

$$N_0'(\xi_k)(\widetilde{\gamma}_k-\gamma_k)=-N_0'(\xi_k)\delta_k. \tag{4.17}$$

Solving for δ_k proves the result. □

Remark 4.5. Equations (4.8) and (4.14) explain why a comparison with the RVM clock is sensitive to the argument of zeta. The explicit formula then rewrites that argument contribution in terms of prime powers. This is the analytic route by which the prime-power branches entered the project.

Part 2. Finite Möbius interpolation

5. The first affine interpolation: numerical and symbolic forms

Definition 5.1 (Terminal affine interpolation)**.** Let $Q_N = \{q_1, \dots, q_{h_N}\}$ be the primes in $(N/2, N]$, and let A_N, P_N, B_N be defined in Section 6. For a prescribed terminal sum $m \in \mathbb{C}$, define

$$\boxed{\mathcal{I}_N(s; m) = A_N(s) + \frac{m - B_N}{h_N} P_N(s).} \tag{5.1}$$

This is the symmetric affine interpolation in which all terminal-prime coefficients are assigned their common mean

$$\lambda_N(m) = \frac{m - B_N}{h_N}. \tag{5.2}$$

Proposition 5.2 (Recovery of the true finite Möbius polynomial)**.** *For the true Mertens value* $m = M(N)$*,*

$$\mathcal{I}_N(s; M(N)) = M_s(N). \tag{5.3}$$

Proof. The set Q_N contains only primes, so every terminal Möbius coefficient is exactly -1. Therefore its total contribution at $s = 0$ is $-h_N$, which proves $M(N) = B_N - h_N$. Hence the affine mean coefficient for the true terminal value is

$$\frac{M(N) - B_N}{h_N} = \frac{-h_N}{h_N} = -1. \tag{5.4}$$

Substitution assigns coefficient -1 to every term in $P_N(s)$, so

$$\mathcal{I}_N(s; M(N)) = A_N(s) - P_N(s). \tag{5.5}$$

The right side is exactly the decomposition of $M_s(N)$. □

Example 5.3 (The numerical model $N = 10$)**.** The Möbius values up to ten are

$$(1, -1, -1, 0, -1, 1, -1, 0, 0, 1). \tag{5.6}$$

The only terminal prime is 7, so

$$Q_{10} = \{7\}, \qquad h_{10} = 1, \qquad P_{10}(s) = 7^{-s}, \tag{5.7}$$

$$A_{10}(s) = 1 - 2^{-s} - 3^{-s} - 5^{-s} + 6^{-s} + 10^{-s}, \qquad B_{10} = 0. \tag{5.8}$$

Thus

$$\mathcal{I}_{10}(s; m) = A_{10}(s) + m7^{-s}. \tag{5.9}$$

The actual terminal value is $M(10) = -1$, which restores the coefficient -1 of 7^{-s}.

Theorem 5.4 (Completion equation)**.** *At every point where* $P_N(s) \neq 0$*, the unique value of* m *satisfying*

$$\mathcal{I}_N(s; m) = \zeta(s)^{-1} \tag{5.10}$$

is

$$\boxed{m = \Phi_N(s) = B_N + h_N \frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)}.} \tag{5.11}$$

Proof. Starting from the affine interpolation, the completion equation is

$$A_N(s) + \frac{m - B_N}{h_N} P_N(s) = \zeta(s)^{-1}. \tag{5.12}$$

Because $h_N \neq 0$ and $P_N(s) \neq 0$, this is a nondegenerate linear equation in m. Subtract $A_N(s)$, multiply by $h_N/P_N(s)$, and add B_N to obtain the displayed solution. If two values m_1, m_2 solved the equation, subtracting their equations would give

$$\frac{m_1 - m_2}{h_N} P_N(s) = 0, \tag{5.13}$$

so $m_1 = m_2$. This proves both existence and uniqueness. □

Remark 5.5. The function $\Phi_N(s)$ is therefore not introduced by an arbitrary quotient. It is the terminal value selected by requiring the affine finite interpolation to coincide with the completed reciprocal-zeta Dirichlet series at the spectral parameter s.

6. Terminal half-interval decomposition

Definition 6.1. For an integer $N \geq 2$, define

$$Q_N = \left\{ q \text{ prime} : \frac{N}{2} < q \leq N \right\}, \tag{6.1}$$

and put

$$h_N = |Q_N|, \qquad q_N = \min Q_N. \tag{6.2}$$

Bertrand's postulate guarantees $h_N \geq 1$.

Definition 6.2. Let

$$M_s(N) = \sum_{n \leq N} \frac{\mu(n)}{n^s}, \tag{6.3}$$

$$P_N(s) = \sum_{q \in Q_N} q^{-s}, \tag{6.4}$$

$$A_N(s) = \sum_{\substack{n \leq N \\ n \notin Q_N}} \frac{\mu(n)}{n^s}, \tag{6.5}$$

$$B_N = A_N(0) = \sum_{\substack{n \leq N \\ n \notin Q_N}} \mu(n). \tag{6.6}$$

Since every $q \in Q_N$ is prime and $\mu(q) = -1$,

$$M_s(N) = A_N(s) - P_N(s) \tag{6.7}$$

and

$$M(N) = B_N - h_N. \tag{6.8}$$

Definition 6.3. For $\Re s > 1$, define the exact tail

$$R_N(s) = \frac{1}{\zeta(s)} - M_s(N) = \sum_{n > N} \frac{\mu(n)}{n^s}. \tag{6.9}$$

The first equality supplies a meromorphic continuation of R_N.

Definition 6.4. Define

$$\Phi_N(s) = B_N + h_N \frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)} \tag{6.10}$$

at points at which the quotient is defined, and meromorphically elsewhere.

7. Canonical identity and endpoint values

Theorem 7.1 (Canonical orientation)**.** *For every $N \geq 2$,*

$$\boxed{\Phi_N(s) = B_N + h_N \frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)} = M(N) + h_N \frac{R_N(s)}{P_N(s)}.} \tag{7.1}$$

Proof. The functions $A_N(s)$ and $P_N(s)$ are finite Dirichlet polynomials and hence entire. In the half-plane $\Re s > 1$, absolute convergence of the reciprocal-zeta Dirichlet series gives

$$\zeta(s)^{-1} = M_s(N) + R_N(s). \tag{7.2}$$

The terminal decomposition $M_s(N) = A_N(s) - P_N(s)$ therefore yields

$$\zeta(s)^{-1} - A_N(s) = M_s(N) + R_N(s) - A_N(s) \tag{7.3}$$

$$= R_N(s) - P_N(s). \tag{7.4}$$

Divide by $P_N(s)$ at a point where it is nonzero, multiply by h_N, and add B_N:

$$\Phi_N(s) = B_N + h_N \left(\frac{R_N(s)}{P_N(s)} - 1 \right) \tag{7.5}$$

$$= B_N - h_N + h_N \frac{R_N(s)}{P_N(s)}. \tag{7.6}$$

Because the terminal primes all have Möbius value -1, $M(N) = B_N - h_N$. This proves the identity on $\Re s > 1$. Both sides are meromorphic continuations of the same function, so the identity extends to every connected component of their common domain and hence meromorphically to the plane. □

Corollary 7.2 (Value at the origin)**.** *The meromorphic interpolation satisfies*

$$\boxed{\Phi_N(0) = -2.} \tag{7.7}$$

Proof. The point $s = 0$ is regular for every finite Dirichlet polynomial in the formula. The classical special value $\zeta(0) = -1/2$ gives

$$\zeta(0)^{-1} = -2. \tag{7.8}$$

Moreover, by definition,

$$A_N(0) = B_N, \qquad P_N(0) = \sum_{q \in Q_N} 1 = h_N. \tag{7.9}$$

Since $h_N \geq 1$, the denominator does not vanish. Direct substitution in the completion solution gives

$$\Phi_N(0) = B_N + h_N \frac{-2 - B_N}{h_N} \tag{7.10}$$

$$= B_N - 2 - B_N = -2. \tag{7.11}$$

No limiting argument is involved. □

Corollary 7.3 (Large positive real limit)**.** *Along the positive real axis,*

$$\boxed{\lim_{\sigma \to +\infty} \Phi_N(\sigma) = M(N).} \tag{7.12}$$

Proof. For $\sigma > 1$, the tail satisfies

$$|R_N(\sigma)| \leq \sum_{n > N} n^{-\sigma} \leq \int_N^\infty x^{-\sigma} \, \mathrm{d}x = \frac{N^{1-\sigma}}{\sigma - 1}. \tag{7.13}$$

The term $q_N^{-\sigma}$ occurs in $P_N(\sigma)$, and all terms are positive, hence

$$P_N(\sigma) \geq q_N^{-\sigma}. \tag{7.14}$$

Consequently,

$$\left|h_N \frac{R_N(\sigma)}{P_N(\sigma)}\right| \le \frac{h_N N}{\sigma - 1}\left(\frac{q_N}{N}\right)^{\sigma}. \tag{7.15}$$

Because $q_N < N < a_N$, the ratio q_N/N is strictly less than one; the exponential decay dominates the factor $h_N N/(\sigma-1)$. The second expression in Theorem 7.1 therefore tends to $M(N)$. □

Experiment 7.4 (The $N = 10$ orientation check)**.** For $N = 10$,

$$Q_{10} = \{7\}, \qquad h_{10} = 1, \qquad M(10) = -1, \qquad B_{10} = 0. \tag{7.16}$$

The inverted quotient P_N/R_N would give $-1/2$ at $s = 0$. The quotient in (7.1) gives

$$0 + \frac{-2}{1} = -2. \tag{7.17}$$

This finite check confirms the orientation of the quotient in Theorem 7.1.

8. Exponential approach, rounding, and recovery of the first tail term

Definition 8.1. Let

$$a_N = \min\{n > N : \mu(n) \neq 0\}. \tag{8.1}$$

Theorem 8.2 (Leading exponential term)**.** *As $\sigma \to +\infty$,*

$$\boxed{\Phi_N(\sigma) - M(N) = h_N \mu(a_N)\left(\frac{q_N}{a_N}\right)^{\sigma}(1 + o(1)).} \tag{8.2}$$

Proof. Factor the least terminal prime from the denominator:

$$P_N(\sigma) = q_N^{-\sigma}\left[1 + \sum_{\substack{q \in Q_N \\ q > q_N}} \left(\frac{q_N}{q}\right)^{\sigma}\right]. \tag{8.3}$$

Every ratio in the bracket is strictly smaller than one, and there are only finitely many of them. Thus the bracket is $1 + o(1)$.

For the numerator, factor the first nonzero Möbius term after N:

$$R_N(\sigma) = \mu(a_N) a_N^{-\sigma}\left[1 + \sum_{n > a_N} \frac{\mu(n)}{\mu(a_N)}\left(\frac{a_N}{n}\right)^{\sigma}\right]. \tag{8.4}$$

For $\sigma \ge 2$, the absolute value of the summand is bounded by $(a_N/n)^2$, whose sum over $n > a_N$ is finite. Each fixed summand tends to zero as $\sigma \to \infty$. Dominated convergence therefore shows that the bracket is $1 + o(1)$. Division gives

$$\frac{R_N(\sigma)}{P_N(\sigma)} = \mu(a_N)\left(\frac{q_N}{a_N}\right)^{\sigma}(1 + o(1)). \tag{8.5}$$

Multiply by h_N and use the canonical identity. □

Remark 8.3. The factor is $(q_N/a_N)^{\sigma}$, not $(a_N/q_N)^{\sigma}$. Since $q_N \le N < a_N$, the correct factor decays. The inverted factor would contradict Corollary 7.3.

Theorem 8.4 (Uniform real-axis error bound)**.** *For every real $\sigma > 1$,*

$$\boxed{|\Phi_N(\sigma) - M(N)| \le \frac{h_N N}{\sigma - 1}\left(\frac{q_N}{N}\right)^{\sigma}.} \tag{8.6}$$

Proof. The reciprocal-zeta tail is absolutely convergent for $\sigma > 1$, so

$$|R_N(\sigma)| \le \sum_{n>N} |\mu(n)| n^{-\sigma} \tag{8.7}$$

$$\le \sum_{n>N} n^{-\sigma} \tag{8.8}$$

$$\le \int_N^\infty x^{-\sigma}\, \mathrm{d}x = \frac{N^{1-\sigma}}{\sigma - 1}. \tag{8.9}$$

The denominator is a sum of positive real terms. Retaining only its largest term gives

$$P_N(\sigma) = \sum_{q \in Q_N} q^{-\sigma} \ge q_N^{-\sigma}. \tag{8.10}$$

The canonical identity now gives

$$|\Phi_N(\sigma) - M(N)| = h_N \frac{|R_N(\sigma)|}{P_N(\sigma)} \tag{8.11}$$

$$\le \frac{h_N N^{1-\sigma} q_N^\sigma}{\sigma - 1} \tag{8.12}$$

$$= \frac{h_N N}{\sigma - 1} \left(\frac{q_N}{N} \right)^\sigma. \tag{8.13}$$

□

Corollary 8.5 (Finite rounding criterion)**.** *If*

$$\boxed{\frac{h_N N}{\sigma - 1} \left(\frac{q_N}{N} \right)^\sigma < \frac{1}{2},} \tag{8.14}$$

then $M(N)$ is the unique nearest integer to the real number $\Phi_N(\sigma)$.

Definition 8.6. Define

$$E_N(s) = \frac{\Phi_N(s) - M(N)}{h_N} = \frac{R_N(s)}{P_N(s)}. \tag{8.15}$$

Corollary 8.7 (Recovery of a_N)**.** *The first nonzero Möbius term after N is recovered by*

$$\boxed{a_N = q_N \lim_{\sigma \to +\infty} |E_N(\sigma)|^{-1/\sigma}.} \tag{8.16}$$

Proof. The assumed inequality and Theorem 8.4 imply

$$|\Phi_N(\sigma) - M(N)| < \frac{1}{2}. \tag{8.17}$$

The open interval of radius $1/2$ centered at an integer contains no other integer. Indeed, for $k \in \mathbb{Z}$, $k \neq M(N)$,

$$|\Phi_N(\sigma) - k| \ge |k - M(N)| - |\Phi_N(\sigma) - M(N)| \tag{8.18}$$

$$> 1 - \frac{1}{2} = \frac{1}{2}. \tag{8.19}$$

Thus $M(N)$ is strictly closer than every other integer and is the unique nearest integer. □

9. Feasible terminal values and non-distinction in $\Re s > 1$

Definition 9.1. For an integer parameter m, define

$$F_N(s;m) = M_s(N) + P_N(s)\left(1 + \frac{m - B_N}{h_N}\right). \tag{9.1}$$

A value m is called terminally feasible when

$$-1 \le \frac{m - B_N}{h_N} \le 1. \tag{9.2}$$

Proposition 9.2 (Exact substitution identity)**.** *The expression* (9.1) *is the exact result of substituting*

$$A_N(s) = M_s(N) + P_N(s) \tag{9.3}$$

into the terminal-value interpolation.

Theorem 9.3 (The half-plane limit does not select $M(N)$)**.** *Fix s with $\Re s > 1$. For every sequence of feasible integers $m(N)$,*

$$\boxed{F_N(s;m(N)) \longrightarrow \frac{1}{\zeta(s)} \qquad (N \to \infty).} \tag{9.4}$$

Thus convergence in the absolutely convergent half-plane does not distinguish $m(N) = M(N)$ from other feasible terminal values.

Proof. Fix s with $\sigma = \Re s > 1$. Absolute convergence of $\sum \mu(n)n^{-s}$ gives

$$M_s(N) \longrightarrow \zeta(s)^{-1}. \tag{9.5}$$

For the terminal prime polynomial,

$$|P_N(s)| \le \sum_{N/2<q\le N} q^{-\sigma} \tag{9.6}$$

$$\le \sum_{N/2<n\le N} n^{-\sigma} \tag{9.7}$$

$$\le \int_{N/2-1}^{N} x^{-\sigma}\,\mathrm{d}x = O_\sigma(N^{1-\sigma}). \tag{9.8}$$

Feasibility means $|(m(N) - B_N)/h_N| \le 1$, so the multiplier $1 + (m(N) - B_N)/h_N$ has absolute value at most two. The entire terminal correction is therefore $O_\sigma(N^{1-\sigma})$, uniformly over all feasible choices. Adding it to $M_s(N)$ proves the common limit. □

Remark 9.4. The selection of the true value $M(N)$ is arithmetical and combinatorial. It is not produced by convergence to $1/\zeta(s)$ in $\Re s > 1$, because the entire feasible family has the same limit.

10. Poles and finite parts at zeta zeros

Definition 10.1. If a meromorphic function has a Laurent expansion

$$f(s) = \frac{a_{-1}}{s - \rho} + a_0 + O(s - \rho), \tag{10.1}$$

its finite part at ρ is

$$\mathrm{FP}_{s=\rho} f(s) = a_0. \tag{10.2}$$

Proposition 10.2 (Finite part of the reciprocal zeta function)**.** *Let ρ be a simple zero of ζ. Then*

$$\frac{1}{\zeta(s)} = \frac{1}{\zeta'(\rho)(s-\rho)} - \frac{\zeta''(\rho)}{2\zeta'(\rho)^2} + O(s-\rho), \tag{10.3}$$

and hence

$$\mathrm{FP}_{s=\rho}\,\frac{1}{\zeta(s)} = -\frac{\zeta''(\rho)}{2\zeta'(\rho)^2}. \tag{10.4}$$

Proof. Write $z = s - \rho$. Since the zero is simple,

$$\zeta(\rho+z) = \zeta'(\rho)z + \frac{1}{2}\zeta''(\rho)z^2 + O(z^3). \tag{10.5}$$

Factor the linear term:

$$\zeta(\rho+z) = \zeta'(\rho)z\left[1 + \frac{\zeta''(\rho)}{2\zeta'(\rho)}z + O(z^2)\right]. \tag{10.6}$$

Using $(1+w)^{-1} = 1 - w + O(w^2)$,

$$\frac{1}{\zeta(\rho+z)} = \frac{1}{\zeta'(\rho)z}\left[1 - \frac{\zeta''(\rho)}{2\zeta'(\rho)}z + O(z^2)\right] \tag{10.7}$$

$$= \frac{1}{\zeta'(\rho)z} - \frac{\zeta''(\rho)}{2\zeta'(\rho)^2} + O(z). \tag{10.8}$$

The constant Laurent coefficient is the finite part. □

Proposition 10.3 (Finite part of Φ_N)**.** *Assume that ρ is a simple zero of ζ and $P_N(\rho) \neq 0$. Then*

$$\mathrm{FP}_{s=\rho}\,\Phi_N(s) = B_N + h_N\frac{-\dfrac{\zeta''(\rho)}{2\zeta'(\rho)^2} - A_N(\rho)}{P_N(\rho)} \tag{10.9}$$

$$- h_N\frac{P_N'(\rho)}{\zeta'(\rho)P_N(\rho)^2}. \tag{10.10}$$

Proof. Set $z = s - \rho$. Proposition 10.2 and the Taylor expansion of the entire function A_N give

$$\zeta(s)^{-1} - A_N(s) = \frac{a_{-1}}{z} + a_0 + O(z), \tag{10.11}$$

where

$$a_{-1} = \frac{1}{\zeta'(\rho)}, \qquad a_0 = -\frac{\zeta''(\rho)}{2\zeta'(\rho)^2} - A_N(\rho). \tag{10.12}$$

Likewise,

$$P_N(s) = p_0 + p_1 z + O(z^2), \qquad p_0 = P_N(\rho), \quad p_1 = P_N'(\rho). \tag{10.13}$$

Because $p_0 \neq 0$,

$$\frac{1}{P_N(s)} = \frac{1}{p_0} - \frac{p_1}{p_0^2}z + O(z^2). \tag{10.14}$$

Multiplication yields

$$\frac{\zeta(s)^{-1} - A_N(s)}{P_N(s)} = \frac{a_{-1}}{p_0 z} + \frac{a_0}{p_0} - \frac{a_{-1}p_1}{p_0^2} + O(z). \tag{10.15}$$

Multiplying by h_N and adding B_N gives the asserted finite part. □

Definition 10.4. For a simple zero ρ, define the normalized finite-part observable

$$\mathcal{F}_N(\rho) = N^{-\rho}\,\mathrm{FP}_{s=\rho}\,\Phi_N(s). \tag{10.16}$$

Experiment 10.5 (Scale scan)**.** A finite scan over

$$N = 2000, 4000, 8000, 16000, 32000, 64000 \tag{10.17}$$

produced successive magnitudes

$$|\mathcal{F}_{2N}(\rho) - \mathcal{F}_N(\rho)| \approx 1.12,\ 0.75,\ 0.80,\ 0.75,\ 0.44. \tag{10.18}$$

The differences do not decrease monotonically and remain of the same order as the values themselves. With six scales this is a negative finite result: no scale cancellation law is established.

Experiment 10.6 (Algebraic finite-part audit)**.** In the numerical implementation, the direct tail expression and the finite-part reconstruction agreed to

$$|R_{\text{from FP}} - R_{\text{direct}}| = 4.043 \times 10^{-10}. \tag{10.19}$$

This checks the algebra and numerical implementation; it is not an asymptotic result.

Part 3. Prime-power phase walks over zeta ordinates

11. Phase walks, endpoint conventions, and a universal half

Definition 11.1. Fix $q > 1$ and write

$$\omega = \log q. \tag{11.1}$$

For the positive ordinates, counted with multiplicity, define

$$S_q(T) = \sum_{0<\gamma\le T} m(\gamma) e^{i\omega\gamma}. \tag{11.2}$$

At an ordinate γ_k, the right, left, and midpoint values are

$$S_k^{\text{right}} = \sum_{j\le k} b_j, \tag{11.3}$$

$$S_k^{\text{left}} = \sum_{j<k} b_j, \tag{11.4}$$

$$S_k^{\text{mid}} = \sum_{j<k} b_j + \frac{1}{2} b_k, \qquad b_k = e^{i\omega\gamma_k}. \tag{11.5}$$

Proposition 11.2 (Endpoint conversion)**.** *At every index k,*

$$\overline{b_k} S_k^{\text{right}} = \overline{b_k} S_k^{\text{mid}} + \frac{1}{2}, \tag{11.6}$$

$$\overline{b_k} S_k^{\text{left}} = \overline{b_k} S_k^{\text{mid}} - \frac{1}{2}. \tag{11.7}$$

Thus changing the endpoint convention translates the carrier coefficient by $\pm 1/2$ and does not change a centered variance.

Theorem 11.3 (Exact triangular half identity)**.** *Let $|b_k| = 1$ and*

$$S_k = \sum_{j\le k} b_j. \tag{11.8}$$

Then

$$\boxed{\Re\left(\frac{1}{N}\sum_{k=1}^{N} \overline{b_k} S_k\right) = \frac{1}{2} + \frac{|S_N|^2}{2N}.} \tag{11.9}$$

Proof. Let $b = (b_1, \dots, b_N)^{\mathsf{T}}$, and let L be the lower triangular matrix with $L_{kj} = 1$ for $j \le k$. Then the vector of partial sums is Lb, and

$$\sum_{k=1}^{N} \overline{b_k} S_k = b^* L b. \tag{11.10}$$

Let J denote the all-ones matrix. Entrywise, $L + L^*$ equals $J + I$: off the diagonal exactly one of L_{jk}, L_{kj} equals one, while on the diagonal both do. Hence

$$2\Re(b^* L b) = b^*(L + L^*)b \tag{11.11}$$

$$= b^* J b + b^* b \tag{11.12}$$

$$= \left| \sum_{k=1}^{N} b_k \right|^2 + N, \tag{11.13}$$

because $|b_k| = 1$. Division by $2N$ proves the identity. □

Corollary 11.4. *For midpoint evaluation,*

$$\Re\left(\frac{1}{N} \sum_{k=1}^{N} \overline{b_k} S_k^{\mathrm{mid}} \right) = \frac{|S_N|^2}{2N}. \tag{11.14}$$

Warning 11.5. Theorem 11.3 is universal for unit phases. It does not identify prime-power structure and does not imply the Riemann Hypothesis. It applies exactly to the raw full-sample cumulative walk, not automatically to a residual after smooth and prime-power subtraction.

12. Complete-zero weights and the positive horizontal defect

Definition 12.1. For a positive ordinate τ, define

$$W_q(\tau) = \sum_{\rho : \Im\rho = \tau} q^{\beta - \frac{1}{2}}, \tag{12.1}$$

where zeros are counted with multiplicity.

Definition 12.2. Define

$$D_q(\tau) = \sum_{\substack{\rho = \beta + i\tau \\ \beta > 1/2}} \left[\cosh\left((\beta - \tfrac{1}{2}) \log q \right) - 1 \right]. \tag{12.2}$$

Theorem 12.3 (Positive defect at each ordinate)**.** *For every $q > 1$ and every positive ordinate τ,*

$$\boxed{W_q(\tau) = m(\tau) + 2D_q(\tau), \qquad D_q(\tau) \ge 0.} \tag{12.3}$$

Moreover,

$$D_q(\tau) = 0 \tag{12.4}$$

if and only if every zero at ordinate τ lies on the critical line.

Proof. The functional equation and conjugation symmetry imply that, at a fixed positive ordinate τ, every zero $\beta + i\tau$ off the critical line is accompanied with equal multiplicity by $1 - \beta + i\tau$; see [12, Chapter 2]. Put $\delta = \beta - 1/2 > 0$. The pair contributes to $W_q(\tau)$

$$q^{\beta - 1/2} + q^{1/2 - \beta} = q^{\delta} + q^{-\delta} \tag{12.5}$$

$$= 2\cosh(\delta \log q) \tag{12.6}$$

$$= 2 + 2\left[\cosh(\delta \log q) - 1 \right]. \tag{12.7}$$

The first term 2 is exactly the contribution of two unit weights to the multiplicity $m(\tau)$; the second is the excess. A zero on the critical line has $\delta = 0$ and contributes one to both quantities. Summing

all pairs proves $W_q = m + 2D_q$. Since $q > 1$, $\log q \neq 0$, and $\cosh x - 1 \geq 0$ with equality only at $x = 0$, the defect vanishes precisely when no positive δ occurs. □

Definition 12.4. Define the locally finite complex measures

$$\mu_q = \sum_{\tau>0} m(\tau) e^{i\tau \log q} \delta_\tau, \tag{12.8}$$

$$\nu_q = \sum_{\tau>0} W_q(\tau) e^{i\tau \log q} \delta_\tau. \tag{12.9}$$

Corollary 12.5 (Atomic positive criterion)**.** *The exact measure identity*

$$\boxed{\frac{1}{2} e^{-iT \log q}(\nu_q - \mu_q) = \sum_{\tau>0} D_q(\tau)\delta_\tau} \tag{12.10}$$

holds. For any one fixed $q > 1$,

$$\nu_q = \mu_q \quad \Longleftrightarrow \quad \text{RH}. \tag{12.11}$$

Proof. At the atom $T = \tau$, the coefficient of $\nu_q - \mu_q$ is

$$[W_q(\tau) - m(\tau)]\, e^{i\tau \log q} = 2D_q(\tau) e^{i\tau \log q}. \tag{12.12}$$

Multiplication by $\frac{1}{2} e^{-iT \log q}$ cancels the phase at each atom and gives the positive measure in (12.10). If $\nu_q = \mu_q$, this measure is zero, so every nonnegative mass $D_q(\tau)$ is zero. Theorem 12.3 then places every zero on the critical line. Conversely, under RH, every weight $q^{\beta-1/2}$ equals one, so $W_q(\tau) = m(\tau)$ and the two measures coincide. □

Remark 12.6. No value of $q > 1$ is logically privileged. For a small displacement $\delta = \beta - 1/2$,

$$\cosh(\delta \log q) - 1 = \frac{1}{2}\delta^2 (\log q)^2 + O\left(\delta^4 (\log q)^4\right). \tag{12.13}$$

Larger q amplifies small horizontal displacements but also enlarges the analytic weights.

13. The explicit-formula origin of the two prime-power branches

The Guinand explicit formula and its later formulations relate suitable test-function sums over zeta zeros to the von Mangoldt function $\Lambda(n)$, hence to prime powers [10, 13]. Landau's formula and Gonek's uniform refinements isolate the same prime-power resonances in exponential sums over zeros [9, 8, 7].

Definition 13.1. Let $n = p^r$ be a prime power, put

$$\ell_n = \log n, \tag{13.1}$$

and suppose $\ell_n \neq \omega$. Define

$$c_n^- = \frac{i\Lambda(n)}{2\pi\sqrt{n}(\omega - \ell_n)}, \tag{13.2}$$

$$c_n^+ = \frac{i\Lambda(n)}{2\pi\sqrt{n}(\omega + \ell_n)}. \tag{13.3}$$

The associated oscillators are

$$P_n^-(T) = c_n^- e^{i(\omega-\ell_n)T}, \tag{13.4}$$

$$P_n^+(T) = c_n^+ e^{i(\omega+\ell_n)T}. \tag{13.5}$$

Heuristic 13.2 (Sharp-window prime-spectral ansatz)**.** For a sharp counting window, formal integration of the explicit formula suggests a decomposition

$$S_q(T) = C_q + A_{\mathrm{smooth},q}(T) + \frac{1}{2}e^{i\omega T} + \sum_{n=p^r} \left(P_n^-(T) + P_n^+(T)\right), \tag{13.6}$$

where the infinite prime-power sum requires a specified truncation or regularization. This is a heuristic argument when used with the discontinuous indicator $\mathbf{1}_{\gamma\le T}$.

Definition 13.3 (Smooth term used in the computation)**.** The MATLAB program uses

$$A_{\mathrm{smooth}}(T) = \frac{e^{i\omega T}\log(T/2\pi) - \mathrm{Ei}(i\omega T)}{2\pi i\omega}, \tag{13.7}$$

up to a branch-dependent additive constant. The constant is treated by a joint constant–carrier fit.

Warning 13.4. Equation (13.7) is the antiderivative of the smooth Riemann–von Mangoldt density multiplied by $e^{i\omega T}$. It is not, by itself, a complete sharp-window explicit formula. Gamma-factor, pole, endpoint, branch, and regularization conventions must be retained when an exact global identity is claimed.

14. Exact algebra of each prime-power mode

Proposition 14.1 (Sum, difference, and ratio identities)**.** *For every nonresonant prime power n,*

$$c_n^- + c_n^+ = \frac{i\Lambda(n)\omega}{\pi\sqrt{n}(\omega^2 - \ell_n^2)}, \tag{14.1}$$

$$c_n^- - c_n^+ = \frac{i\Lambda(n)\ell_n}{\pi\sqrt{n}(\omega^2 - \ell_n^2)}, \tag{14.2}$$

$$\frac{c_n^-}{c_n^+} = \frac{\omega + \ell_n}{\omega - \ell_n}. \tag{14.3}$$

Proof. Starting from the definitions, combine denominators:

$$c_n^- + c_n^+ = \frac{i\Lambda(n)}{2\pi\sqrt{n}}\left[\frac{1}{\omega - \ell_n} + \frac{1}{\omega + \ell_n}\right] \tag{14.4}$$

$$= \frac{i\Lambda(n)}{2\pi\sqrt{n}}\frac{2\omega}{\omega^2 - \ell_n^2}, \tag{14.5}$$

which is (14.1). Replacing the plus by a minus in the bracket gives $2\ell_n/(\omega^2 - \ell_n^2)$, proving (14.2). Finally the common factor cancels in the quotient:

$$\frac{c_n^-}{c_n^+} = \frac{\omega + \ell_n}{\omega - \ell_n}. \tag{14.6}$$

All identities are exact whenever $\ell_n \neq \omega$. □

Definition 14.2. Write

$$c_n^- = ia_n^-, \qquad c_n^+ = ia_n^+, \qquad a_n^\pm \in \mathbb{R}, \tag{14.7}$$

and define

$$u_n = a_n^- + a_n^+, \tag{14.8}$$

$$v_n = a_n^- - a_n^+, \tag{14.9}$$

$$d_n = |c_n^-|^2 - |c_n^+|^2. \tag{14.10}$$

Proposition 14.3 (Signed mode invariant)**.** *The quantities in Definition 14.2 satisfy*

$$u_n = -\frac{\omega\Lambda(n)}{\pi\sqrt{n}(\ell_n^2-\omega^2)}, \tag{14.11}$$

$$v_n = -\frac{\Lambda(n)\ell_n}{\pi\sqrt{n}(\ell_n^2-\omega^2)}, \tag{14.12}$$

$$\boxed{d_n = u_n v_n = \frac{\omega\Lambda(n)^2\ell_n}{\pi^2 n(\ell_n^2-\omega^2)^2}.} \tag{14.13}$$

Proof. Because the denominators in $c_n^{\pm}$ are real, the coefficients are purely imaginary. Taking their real scalar factors and using the sum and difference identities gives

$$u_n = \frac{\Lambda(n)}{2\pi\sqrt{n}}\left[\frac{1}{\omega-\ell_n}+\frac{1}{\omega+\ell_n}\right] \tag{14.14}$$

$$= -\frac{\omega\Lambda(n)}{\pi\sqrt{n}(\ell_n^2-\omega^2)}, \tag{14.15}$$

$$v_n = \frac{\Lambda(n)}{2\pi\sqrt{n}}\left[\frac{1}{\omega-\ell_n}-\frac{1}{\omega+\ell_n}\right] \tag{14.16}$$

$$= -\frac{\ell_n\Lambda(n)}{\pi\sqrt{n}(\ell_n^2-\omega^2)}. \tag{14.17}$$

The algebraic factorization

$$(a_n^-)^2-(a_n^+)^2 = (a_n^-+a_n^+)(a_n^--a_n^+) = u_n v_n \tag{14.18}$$

then gives the displayed formula for d_n. Its positivity follows from $\omega>0$, $\Lambda(n)^2\geq 0$, $\ell_n>0$, and the squared denominator. □

Proposition 14.4 (Elliptic form and anisotropy)**.** *After demodulation by $e^{-i\omega T}$, the mode is*

$$g_n(T) = c_n^- e^{-i\ell_n T} + c_n^+ e^{i\ell_n T} = \frac{\Lambda(n)}{\pi\sqrt{n}(\omega^2-\ell_n^2)}\left[\ell_n\sin(\ell_n T)+i\omega\cos(\ell_n T)\right]. \tag{14.19}$$

Its unsigned elliptic area is

$$\pi d_n. \tag{14.20}$$

For uniform phase, its anisotropy is

$$\mathcal{A}_n = \frac{\ell_n^2-\omega^2}{\ell_n^2+\omega^2} \longrightarrow 1 \qquad (n\to\infty). \tag{14.21}$$

Proof. Using $e^{ix}=\cos x + i\sin x$, expand $c_n^- e^{-i\ell_n T}+c_n^+ e^{i\ell_n T}$ and collect its real and imaginary parts. This gives

$$g_n(T) = A_n\sin(\ell_n T) + iB_n\cos(\ell_n T), \tag{14.22}$$

with

$$A_n = \frac{\Lambda(n)\ell_n}{\pi\sqrt{n}(\omega^2-\ell_n^2)}, \qquad B_n = \frac{\Lambda(n)\omega}{\pi\sqrt{n}(\omega^2-\ell_n^2)}. \tag{14.23}$$

This is an ellipse with semiaxes $|A_n|$ and $|B_n|$. Their product is $|d_n|=d_n$, so its unsigned area is πd_n. Under uniform phase, the coordinate variances are $A_n^2/2$ and $B_n^2/2$. The normalized difference of principal variances is therefore

$$\frac{A_n^2-B_n^2}{A_n^2+B_n^2} = \frac{\ell_n^2-\omega^2}{\ell_n^2+\omega^2}, \tag{14.24}$$

which tends to one as $\ell_n\to\infty$. □

Lemma 14.5 (Dominance of $n = 3$ in the minus branch for $\omega = 1$)**.** *For $\omega = 1$, the largest coefficient $|c_n^-|$ among prime powers $n \geq 2$ occurs at $n = 3$.*

Proof. For a prime power $n = p^r$, $\Lambda(n) = \log p \leq \log n$. When $n \geq 3$ and $\omega = 1$, $\log n > 1$, so

$$|c_n^-| \leq f(n) := \frac{\log n}{2\pi\sqrt{n}(\log n - 1)}. \tag{14.25}$$

Let $L = \log x > 1$. Logarithmic differentiation gives

$$x\frac{f'(x)}{f(x)} = \frac{1}{L} - \frac{1}{2} - \frac{1}{L-1} \tag{14.26}$$

$$= -\frac{1}{L(L-1)} - \frac{1}{2} < 0. \tag{14.27}$$

Thus f is strictly decreasing on (e, ∞). At $n = 3$, the number is prime and $\Lambda(3) = \log 3$, so equality holds in (14.25); hence $n = 3$ dominates every $n \geq 4$. The remaining comparison with $n = 2$ is direct. Using the certified bounds $0.6931 < \log 2 < 0.6932$ and $1.0986 < \log 3 < 1.0987$ in the two explicit formulas gives $|c_3^-| > |c_2^-|$. Therefore $n = 3$ is the unique maximum. □

15. The convergent signed tail

Definition 15.1. Define

$$D_\infty = \sum_{n=p^r} d_n, \tag{15.1}$$

$$\Delta_\omega(Q) = \sum_{\substack{n=p^r \\ n>Q}} d_n = D_\infty - \sum_{\substack{n=p^r \\ n\leq Q}} d_n. \tag{15.2}$$

Theorem 15.2 (Asymptotic signed tail)**.** *For fixed $\omega > 0$,*

$$\boxed{\Delta_\omega(Q) \sim \frac{\omega}{\pi^2 \log Q} \qquad (Q \to \infty).} \tag{15.3}$$

Proof. Separate primes from proper prime powers. For a prime p, $\Lambda(p) = \ell_p = \log p$, and the exact formula becomes

$$d_p = \frac{\omega}{\pi^2 p \log p}\left(1 - \frac{\omega^2}{(\log p)^2}\right)^{-2}. \tag{15.4}$$

For $p > e^{2\omega}$, expand $(1-z)^{-2} = 1 + O(z)$ with $z = \omega^2/(\log p)^2$. Uniformly in that range,

$$d_p = \frac{\omega}{\pi^2 p \log p} + O_\omega\left(\frac{1}{p(\log p)^3}\right). \tag{15.5}$$

The error tail is $O_\omega((\log Q)^{-3})$ by the prime number theorem and partial summation.

For the main term, write the prime sum as a Stieltjes integral:

$$\sum_{p>Q} \frac{1}{p \log p} = \int_Q^\infty \frac{1}{x \log x}\, \mathrm{d}\pi(x). \tag{15.6}$$

The prime number theorem $\pi(x) = x/\log x + o(x/\log x)$, followed by partial summation, gives

$$\int_Q^\infty \frac{1}{x \log x}\, \mathrm{d}\pi(x) = \int_Q^\infty \frac{\mathrm{d}x}{x(\log x)^2} + o\left(\frac{1}{\log Q}\right) = \frac{1}{\log Q} + o\left(\frac{1}{\log Q}\right). \tag{15.7}$$

It remains to bound $n = p^r$ with $r \geq 2$. Outside a fixed finite set depending on ω, the denominator satisfies $(r^2(\log p)^2 - \omega^2)^2 \gg_\omega r^4(\log p)^4$. Hence

$$d_{p^r} \ll_\omega \frac{1}{r^3 p^r \log p}. \tag{15.8}$$

Summing over $p^r > Q$, $r \geq 2$, gives $O_\omega(Q^{-1/2})$, after separating the finitely many small primes. This is $o(1/\log Q)$. Combining the prime and proper-power parts proves the theorem. □

Remark 15.3. For $\omega = 1$,

$$\pi\Delta_1(Q) \sim \frac{1}{\pi \log Q}, \tag{15.9}$$

$$\frac{3}{4}\Delta_1(Q) \sim \frac{3}{4\pi^2 \log Q}. \tag{15.10}$$

These constants match two finite numerical quantities, but the matching relations themselves remain conjectural.

16. Finite zero-sampled carrier identities

Definition 16.1. Fix a finite multiset of positive ordinates

$$\Gamma_H = \{\gamma_k : 0 < \gamma_k \leq H\}, \qquad N(H) = |\Gamma_H|, \tag{16.1}$$

and let Y be any complex-valued function on Γ_H. For a cutoff Q, define

$$R_Q(\gamma) = Y(\gamma) - \sum_{\substack{n=p^r \\ n\leq Q}} \left(P_n^-(\gamma) + P_n^+(\gamma)\right). \tag{16.2}$$

The raw carrier coefficient and zero kernel are

$$\widehat{c}_{Q,H} = \frac{1}{N(H)} \sum_{\gamma\in\Gamma_H} e^{-i\omega\gamma} R_Q(\gamma), \tag{16.3}$$

$$M_H(x) = \frac{1}{N(H)} \sum_{\gamma\in\Gamma_H} x^{i\gamma}, \qquad x > 0. \tag{16.4}$$

Also put

$$\widehat{c}_{Y,H} = \frac{1}{N(H)} \sum_{\gamma\in\Gamma_H} e^{-i\omega\gamma} Y(\gamma). \tag{16.5}$$

Theorem 16.2 (Exact finite-sample carrier formula)**.** *For the objects in Definition 16.1,*

$$\boxed{\widehat{c}_{Q,H} = \widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\leq Q}} \left(c_n^- \overline{M_H(n)} + c_n^+ M_H(n)\right).} \tag{16.6}$$

Proof. For each sampled ordinate, demodulation gives

$$e^{-i\omega\gamma_k} P_n^-(\gamma_k) = c_n^- e^{-i\ell_n\gamma_k} = c_n^- n^{-i\gamma_k}, \tag{16.7}$$

$$e^{-i\omega\gamma_k} P_n^+(\gamma_k) = c_n^+ e^{i\ell_n\gamma_k} = c_n^+ n^{i\gamma_k}. \tag{16.8}$$

Insert the definition of the residual into the sample average defining $\widehat{c}_{Q,H}$. Finite sums may be interchanged without any convergence issue. The average of $n^{i\gamma_k}$ is $M_H(n)$, and the average of $n^{-i\gamma_k}$ is its complex conjugate because $n > 0$. Collecting the terms yields the formula exactly. □

Theorem 16.3 (Exact imaginary-part formula)**.** *Write*

$$M_H(n) = x_n + iy_n. \tag{16.9}$$

Then

$$\Im\left(c_n^- \overline{M_H(n)} + c_n^+ M_H(n)\right) = u_n x_n, \tag{16.10}$$

and consequently

$$\boxed{\Im\widehat{c}_{Q,H} = \Im\widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} u_n \Re M_H(n).} \tag{16.11}$$

Proof. Write $M_H(n) = x_n + iy_n$ and $c_n^{\pm} = ia_n^{\pm}$ with real $a_n^{\pm}$. Then

$$c_n^- \overline{M_H(n)} + c_n^+ M_H(n) = ia_n^-(x_n - iy_n) + ia_n^+(x_n + iy_n) \tag{16.12}$$

$$= (a_n^- - a_n^+)y_n + i(a_n^- + a_n^+)x_n \tag{16.13}$$

$$= v_n y_n + iu_n x_n. \tag{16.14}$$

Its imaginary part is therefore $u_n x_n = u_n \Re M_H(n)$. Taking imaginary parts in the exact carrier formula and summing proves the result. □

Definition 16.4. Define the Q-independent quantity

$$C_H = \Im\widehat{c}_{Y,H} - \pi D_\infty. \tag{16.15}$$

Theorem 16.5 (Exact global defect identity)**.** *For the finite-sample raw coefficient,*

$$\boxed{\Im\widehat{c}_{Q,H} - \pi\Delta_\omega(Q) = C_H - \sum_{\substack{n=p^r \\ n\le Q}} u_n \left[\Re M_H(n) - \pi v_n\right].} \tag{16.16}$$

Proof. The exact carrier identity gives

$$\Im\widehat{c}_{Q,H} = \Im\widehat{c}_{Y,H} - \sum_{n\le Q} u_n \Re M_H(n). \tag{16.17}$$

On the other hand, $d_n = u_n v_n$ and $\Delta_\omega(Q) = D_\infty - \sum_{n\le Q} d_n$, so

$$\pi\Delta_\omega(Q) = \pi D_\infty - \pi \sum_{n\le Q} u_n v_n. \tag{16.18}$$

Subtract the second equation from the first and group the cutoff-independent terms:

$$\Im\widehat{c}_{Q,H} - \pi\Delta_\omega(Q) = \left(\Im\widehat{c}_{Y,H} - \pi D_\infty\right) \tag{16.19}$$

$$- \sum_{n\le Q} u_n \left[\Re M_H(n) - \pi v_n\right]. \tag{16.20}$$

The parenthesis is exactly C_H. Every manipulation is finite except for D_∞, whose convergence follows from Theorem 15.2. □

Corollary 16.6 (Equivalent transport target)**.** *The conjectural relation*

$$\Im\widehat{c}_{Q,H} = \pi\Delta_\omega(Q) + o\left(\frac{1}{\log Q}\right) \tag{16.21}$$

is exactly equivalent to

$$\boxed{C_H - \sum_{n\le Q} u_n \left[\Re M_H(n) - \pi v_n\right] = o\left(\frac{1}{\log Q}\right).} \tag{16.22}$$

Theorem 16.7 (Exact increment identity)**.** *When Q crosses a prime power n,*

$$\Delta\Im\widehat{c}_{Q,H} = -u_n \Re M_H(n), \tag{16.23}$$

$$\Delta\left(\pi\Delta_\omega(Q)\right) = -\pi u_n v_n. \tag{16.24}$$

If $u_n v_n \neq 0$, then

$$\boxed{\frac{\Delta \Im \widehat{c}_{Q,H}}{\Delta(\pi \Delta_\omega(Q))} = \frac{\Re M_H(n)}{\pi v_n}.} \tag{16.25}$$

Proof. The global defect identity is exact for every finite cutoff:

$$\Im \widehat{c}_{Q,H} - \pi \Delta_\omega(Q) = C_H - \sum_{n \leq Q} u_n (\Re M_H(n) - \pi v_n). \tag{16.26}$$

Therefore the left side is $o(1/\log Q)$ if and only if the right side is $o(1/\log Q)$. Since $\Delta_\omega(Q) \sim \omega/(\pi^2 \log Q)$, the same equivalence holds with $o(\Delta_\omega(Q))$ in place of $o(1/\log Q)$. No estimate or limit interchange is hidden in this reformulation. □

17. Landau scaling and failure of termwise area transport

Landau's formula concerns sums of x^ρ, not sums of $x^{i\gamma}$ without horizontal weights. Passing from the former to the latter requires RH, or a separate control of the real parts. Uniformity when x grows with H requires a uniform Landau–Gonek formula [9, 8].

Heuristic 17.1 (Conditional Landau estimate for the ordinate kernel)**.** Assume RH and a uniform Landau–Gonek estimate in the range under consideration. With

$$L_H = \log(H/2\pi), \tag{17.1}$$

one expects at a prime power n

$$\Re M_H(n) \approx -\frac{\Lambda(n)}{\sqrt{n}\, L_H}. \tag{17.2}$$

Proposition 17.2 (Required termwise target)**.** *Individual transport of the signed area of mode n would require*

$$\Re M_H(n) \sim \pi v_n. \tag{17.3}$$

For a prime p,

$$\pi v_p = -\frac{(\log p)^2}{\sqrt{p}((\log p)^2 - \omega^2)} \sim -\frac{1}{\sqrt{p}}. \tag{17.4}$$

Heuristic 17.3 (Termwise ratio)**.** Combining (17.2) and (17.4) gives

$$\frac{\Re M_H(p)}{\pi v_p} \approx \frac{\log p}{L_H}. \tag{17.5}$$

If $p \asymp H^\theta$, this predicts

$$\frac{\Re M_H(p)}{\pi v_p} \approx \theta. \tag{17.6}$$

Thus termwise transport cannot be asymptotically exact at a fixed $\theta < 1$; it becomes plausible only for $p = H^{1-o(1)}$.

Warning 17.4. The fixed-p Landau formula alone does not prove (17.6). The latter is a falsifiable scaling prediction that requires uniformity in the joint range $p = p(H)$.

Conjecture 17.5 (Local transport near the height scale)**.** *Let*

$$Q_1, Q_2 = H^{1-o(1)}. \tag{17.7}$$

In a suitable uniform regime,

$$\Im \widehat{c}_{Q_2,H} - \Im \widehat{c}_{Q_1,H} \sim \pi \left[\Delta_\omega(Q_2) - \Delta_\omega(Q_1)\right]. \tag{17.8}$$

Proposition 17.6 (Exact local defect)**.** *The defect in Conjecture 17.5 is exactly*

$$\begin{aligned}&[\Im\widehat{c}_{Q_2,H} - \Im\widehat{c}_{Q_1,H}] - \pi\left[\Delta_\omega(Q_2) - \Delta_\omega(Q_1)\right]\\ &\quad = -\sum_{Q_1<n\le Q_2} u_n\left[\Re M_H(n) - \pi v_n\right].\end{aligned} \tag{17.9}$$

Proof. Subtract (16.16) at Q_1 and Q_2. The constant C_H cancels. □

18. Mixed carrier area versus prime self-area

Proposition 18.1 (Mixed form measured by the centroid coefficient)**.** *The imaginary part of the raw carrier coefficient is*

$$\Im\widehat{c}_{Q,H} = \frac{1}{N(H)}\sum_{\gamma_k\le H}\Im\left[e^{-i\omega\gamma_k}R_Q(\gamma_k)\right] \tag{18.1}$$

$$= -\frac{1}{N(H)}\sum_{\gamma_k\le H}\Im\left[\overline{R_Q(\gamma_k)}e^{i\omega\gamma_k}\right]. \tag{18.2}$$

It is a mixed bilinear pairing between the residual position and the actual phase increment.

Definition 18.2. For a finite prime-power model G, define the formal self-area

$$\mathfrak{A}_H(G) = -\frac{\pi}{H}\int \Im\left(\overline{G(t)}\,\mathcal{J}G(t)\right)\,\mathrm{d}t, \tag{18.3}$$

where $\mathcal{J}$ is a specified frequency-sign or Hilbert-type operator.

Warning 18.3. The mixed pairing (18.2) is not identical to the self-pairing (18.3). Such an identification would require a justified replacement

$$e^{i\omega\gamma_k} \longleftrightarrow \mathcal{J}G_Q(\gamma_k). \tag{18.4}$$

The exact termwise defect of this replacement is encoded by $\Re M_H(n) - \pi v_n$.

19. Exact quadratic form for the sampled variance

Definition 19.1 (Sampled inner product and removed Gram matrix)**.** For functions on Γ_H, define

$$\langle f, g\rangle_H = \frac{1}{N(H)}\sum_{\gamma_k\le H}\overline{f(\gamma_k)}g(\gamma_k). \tag{19.1}$$

Put

$$\mu_H = M_H(e^\omega) \tag{19.2}$$

and

$$G_H = \begin{pmatrix} 1 & \mu_H \\ \overline{\mu_H} & 1\end{pmatrix}. \tag{19.3}$$

Definition 19.2. For each prime power n and sign $\sigma\in\{-1,+1\}$, define the sampled frequency vector

$$v_{n,\sigma}(\gamma_k) = n^{i\sigma\gamma_k}. \tag{19.4}$$

Proposition 19.3 (Exact unprojected Gram kernel)**.** *The sampled Gram entries are*

$$\langle v_{n,\sigma}, v_{m,\tau}\rangle_H = M_H\left(m^\tau n^{-\sigma}\right). \tag{19.5}$$

Thus the four branch interactions between p^r and q^s involve

$$M_H(q^s/p^r),\quad M_H(q^sp^r),\quad M_H((q^sp^r)^{-1}),\quad M_H(p^r/q^s). \tag{19.6}$$

Definition 19.4. Let Π_H be the orthogonal projection onto the complement of

$$\operatorname{span}\{1, e^{i\omega\gamma}\}. \tag{19.7}$$

Define the projected kernel

$$\widetilde{K}_H((n,\sigma),(m,\tau)) = \langle \Pi_H v_{n,\sigma}, \Pi_H v_{m,\tau} \rangle_H . \tag{19.8}$$

Proposition 19.5 (Schur-complement formula)**.** *Let*

$$K_H(\lambda,\mu) = \frac{1}{N(H)} \sum_{\gamma_k \le H} e^{i(\mu-\lambda)\gamma_k}, \tag{19.9}$$

and

$$u_H(\lambda) = \begin{pmatrix} K_H(0,\lambda) \\ K_H(\omega,\lambda) \end{pmatrix}. \tag{19.10}$$

Then

$$\widetilde{K}_H(\lambda,\mu) = K_H(\lambda,\mu) - u_H(\lambda)^* G_H^{-1} u_H(\mu), \tag{19.11}$$

where G_H is given by (19.3)*.*

Proof. This is the standard Gram-matrix formula for orthogonal projection away from a finite-dimensional subspace. □

Theorem 19.6 (Exact system variance)**.** *Let a be the vector of all retained branch coefficients c_n^σ. The variance after joint removal of the constant and carrier is exactly*

$$\boxed{V(Q,H) = a^* \widetilde{K}_H a.} \tag{19.12}$$

Grouping by prime bases gives

$$V(Q,H) = \sum_p a_p^* \widetilde{K}_{p,p;H} a_p + \sum_{p \ne q} a_p^* \widetilde{K}_{p,q;H} a_q. \tag{19.13}$$

Warning 19.7 (The diagonal issue). The scalar Parseval diagonal

$$\sum_{p,r} \left(|c_{p^r}^-|^2 + |c_{p^r}^+|^2 \right) \tag{19.14}$$

is not the diagonal of the projected, zero-sampled system (19.12). Inside one prime tower, p^r/p^s and p^{r+s} are themselves prime powers, so the entire two-branch tower block is arithmetically resonant. It is currently unproved that tower blocks dominate, contract, or localize the residual.

Question 19.8. Determine the asymptotic size of

$$\mathfrak{E}_p(H) = a_p^* \widetilde{K}_{p,p;H} a_p \tag{19.15}$$

and the summed cross-tower contribution in (19.13), uniformly in a growing prime-power cutoff.

20. Finite logarithmic shells in Bohr mean

Definition 20.1. For fixed $a > 0$, define the finite multiplicative shell

$$\mathcal{S}_a(Q) = \{n = p^r : Q < n \le e^a Q\} \tag{20.1}$$

and its demodulated sum

$$G_{Q,a}(T) = \sum_{n \in \mathcal{S}_a(Q)} g_n(T). \tag{20.2}$$

Proposition 20.2 (Bohr energy of a fixed logarithmic shell)**.** *For fixed $a > 0$,*

$$\lim_{X\to\infty} \frac{1}{X}\int_0^X |G_{Q,a}(T)|^2\,\mathrm{d}T = \frac{a}{2\pi^2\log Q} + o\left(\frac{1}{\log Q}\right) \tag{20.3}$$

as $Q \to \infty$.

Proof. Distinct frequencies $\pm\log n$ are orthogonal in Bohr mean. For primes,

$$|c_p^-|^2 + |c_p^+|^2 = \frac{1}{2\pi^2 p} + O_\omega\left(\frac{1}{p(\log p)^2}\right). \tag{20.4}$$

The prime number theorem and partial summation give [2, 3]

$$\sum_{Q<p\le e^a Q} \frac{1}{p} = \frac{a}{\log Q} + o\left(\frac{1}{\log Q}\right). \tag{20.5}$$

Proper prime powers contribute a lower-order term. □

Corollary 20.3. *The choice $a = 3/2$ gives*

$$\frac{3}{4\pi^2\log Q} + o\left(\frac{1}{\log Q}\right). \tag{20.6}$$

Warning 20.4. Corollary 20.3 proves only the Bohr energy of a deliberately selected hard shell. It does not prove that the actual residual is supported in $(Q, e^{3/2}Q]$. Many transfer profiles have the same quadratic mass, and the projected zero-sampled variance is not the Bohr norm in Proposition 20.2.

21. Regularization and the remaining explicit-formula bridge

Definition 21.1. A regularized counting window is a family $H_{T,\eta}(u)$ tending to the midpoint indicator of $[0, T]$ as $\eta \to \infty$. A formal Gaussian example is

$$H_{T,\eta}(u) = \int_0^T \frac{\eta}{\sqrt{\pi}} e^{-\eta^2(u-v)^2}\,\mathrm{d}v. \tag{21.1}$$

Heuristic 21.2 (Fourier displacement)**.** For

$$h_{T,\eta,\omega}(u) = e^{i\omega u} H_{T,\eta}(u), \tag{21.2}$$

formal Fourier transformation gives

$$\widehat{h}_{T,\eta,\omega}(v) = e^{-(v-\omega)^2/(4\eta^2)} \frac{1 - e^{-i(v-\omega)T}}{i(v-\omega)}. \tag{21.3}$$

Insertion into the explicit formula generates the two branches in Definition 13.1, multiplied by explicit regularizing weights.

Warning 21.3. To turn Section 21 into a theorem one must specify a test class for which the explicit formula is valid, retain every archimedean and boundary term, and control jointly:

$$H, \qquad Q, \qquad \eta. \tag{21.4}$$

A formal Gaussian manipulation is not a proof of a sharp-window formula.

Conjecture 21.4 (Uniform explicit formula for the ordinate walk)**.** *There exist choices $Q = Q(H) \to \infty$ and $\eta = \eta(H) \to \infty$ for which the regularized prime-power formula, including its complete archimedean and endpoint terms, converges locally to the midpoint ordinate walk, with an error tending to zero in a topology strong enough to identify the atomic measure* (12.10).

Remark 21.5. The ordinary explicit formula reconstructs complete-zero weights, which retain $q^{\beta-1/2}$. Replacing that object by the unweighted ordinate walk is precisely where the positive defect $D_q(\tau)$ enters. Truncation control alone does not remove the horizontal zero information.

22. Energy exhaustion and distributional convergence

Conjecture 22.1 (Energy exhaustion)**.** *For a suitable joint regime $Q = Q(H) \to \infty$,*

$$\mathrm{Var}_{\gamma \le H}(F_{Q,H}) \le \varepsilon(Q,H), \qquad \varepsilon(Q(H),H) \to 0. \tag{22.1}$$

The variance is the projected discrete form (19.12)*, not an unprojected Lebesgue L^2-norm.*

Theorem 22.2 (Abstract local distributional implication)**.** *Suppose a sequence F_j satisfies*

$$F_j - \frac{1}{2} \longrightarrow 0 \quad \text{in } L^2_{\mathrm{loc}}(\mathbb{R}), \tag{22.2}$$

and

$$\frac{\mathrm{d}}{\mathrm{d}T}\left[e^{i\omega T}\left(F_j(T) - \frac{1}{2}\right)\right] \longrightarrow \nu_q - \mu_q \quad \text{in } \mathcal{D}'(\mathbb{R}). \tag{22.3}$$

Then RH *holds.*

Proof. Define

$$G_j(T) = e^{i\omega T}\left(F_j(T) - \frac{1}{2}\right). \tag{22.4}$$

Multiplication by the bounded smooth phase preserves local L^2 convergence, so $G_j \to 0$ in L^2_{loc}. On every compact set, Cauchy–Schwarz then gives $G_j \to 0$ in L^1. For $\varphi \in C_c^\infty(\mathbb{R})$,

$$\langle G_j', \varphi\rangle = -\int_{\mathbb{R}} G_j(T)\varphi'(T)\,\mathrm{d}T \longrightarrow 0. \tag{22.5}$$

The assumed identification of the derivatives therefore implies $\nu_q - \mu_q = 0$ in $\mathcal{D}'$. A locally finite measure that is zero as a distribution is the zero measure. Multiplying by $\frac{1}{2}e^{-iT\log q}$ and applying Corollary 12.5 gives $D_q(\tau) = 0$ at every ordinate, hence RH. □

Warning 22.3. A normalized mean-square limit on intervals $[0,H]$ does not imply Theorem 22.2. A fixed compact defect, or even a sparse family of atomic defects, is diluted by division by H. Local or distributional convergence is indispensable.

23. Further analytic consequences of the finite Möbius interpolation

23.1. Meromorphic structure and residues.

Proposition 23.1 (Sources of poles)**.** *The meromorphic function Φ_N can have poles only at zeros of ζ and at zeros of the exponential polynomial P_N. At a simple zero ρ of ζ satisfying $P_N(\rho) \neq 0$, the residue is*

$$\operatorname*{Res}_{s=\rho} \Phi_N(s) = \frac{h_N}{\zeta'(\rho)P_N(\rho)}. \tag{23.1}$$

At a simple zero s_0 of P_N that is not a zero of ζ and satisfies $\zeta(s_0)^{-1} \neq A_N(s_0)$, the residue is

$$\operatorname*{Res}_{s=s_0} \Phi_N(s) = h_N \frac{\zeta(s_0)^{-1} - A_N(s_0)}{P_N'(s_0)}. \tag{23.2}$$

Proof. Both statements follow directly from the quotient in (6.10). At a simple zeta zero, $\zeta(s)^{-1}$ has principal part $1/(\zeta'(\rho)(s-\rho))$. At a simple zero of P_N, the denominator has principal part $P_N'(s_0)(s-s_0)$. □

Proposition 23.2 (Value at the pole of zeta)**.** *The reciprocal zeta function has a simple zero at $s = 1$, and therefore*

$$\Phi_N(1) = B_N - h_N \frac{A_N(1)}{P_N(1)}. \tag{23.3}$$

Proof. Use $\zeta(s)^{-1} \to 0$ as $s \to 1$ in (6.10). The denominator $P_N(1)$ is positive. □

23.2. **Eventually exact sign and quantitative rounding.**

Corollary 23.3 (Eventually exact sign)**.** *For sufficiently large real σ,*

$$\operatorname{sgn}\big(\Phi_N(\sigma) - M(N)\big) = \mu(a_N). \tag{23.4}$$

Proof. The leading asymptotic has the form

$$\Phi_N(\sigma) - M(N) = A_N(\sigma)(1 + o(1)), \qquad A_N(\sigma) = h_N \mu(a_N) \left(\frac{q_N}{a_N}\right)^{\sigma}. \tag{23.5}$$

The magnitude multiplying $\mu(a_N)$ is strictly positive. Choose σ_0 so that the relative error has modulus less than one half for $\sigma \geq \sigma_0$. Then the full expression cannot change the sign of its leading term, and its sign is $\mu(a_N)$. □

Proposition 23.4 (Existence of a finite rounding threshold)**.** *For every $N \geq 2$ there exists a finite $\sigma_N > 1$ such that*

$$\operatorname{nint} \Phi_N(\sigma) = M(N) \tag{23.6}$$

for all real $\sigma \geq \sigma_N$, where nint *denotes the nearest integer.*

Proof. Let

$$B_N(\sigma) = \frac{h_N N}{\sigma - 1} \left(\frac{q_N}{N}\right)^{\sigma}. \tag{23.7}$$

The base q_N/N lies strictly between zero and one. Hence its exponential power tends to zero, while $h_N N/(\sigma - 1)$ grows at most polynomially and in fact decreases after the constant factor is removed. Thus $B_N(\sigma) \to 0$. Choose σ_N so that $B_N(\sigma) < 1/2$ for every $\sigma \geq \sigma_N$. The rounding criterion then gives the conclusion uniformly for all such σ. □

Proposition 23.5 (A sufficient explicit threshold)**.** *Let*

$$\lambda_N = \log \frac{N}{q_N} > 0. \tag{23.8}$$

Every $\sigma > 1$ satisfying

$$\sigma \lambda_N > \log(2 h_N N) + \log \frac{1}{\sigma - 1} \tag{23.9}$$

is a valid rounding point. In particular, any $\sigma \geq 2$ satisfying

$$\sigma > \frac{\log(2 h_N N)}{\lambda_N} \tag{23.10}$$

is sufficient.

Proof. Rewrite the rounding condition as

$$\frac{h_N N}{\sigma - 1} e^{-\sigma \lambda_N} < \frac{1}{2}. \tag{23.11}$$

All factors are positive. Taking logarithms gives

$$\log(h_N N) - \log(\sigma - 1) - \sigma \lambda_N < -\log 2, \tag{23.12}$$

which rearranges to the implicit condition in the statement. When $\sigma \geq 2$, one has $\log(1/(\sigma - 1)) \leq 0$. The stronger but simpler inequality $\sigma \lambda_N > \log(2 h_N N)$ therefore implies the implicit condition and is sufficient. □

23.3. **Recursive recovery of the nonzero tail.**

Definition 23.6. Let

$$a_{N,1} < a_{N,2} < a_{N,3} < \cdots \tag{23.13}$$

be the increasing sequence of integers exceeding N for which $\mu(a_{N,j}) \neq 0$.

Theorem 23.7 (Recursive exponential extraction)**.** *The real-axis function $E_N(\sigma)$ determines the entire signed sequence*

$$\big(a_{N,j}, \mu(a_{N,j})\big)_{j\geq 1} \tag{23.14}$$

recursively. More precisely, once the first $J-1$ terms are known, define

$$E_N^{(J)}(\sigma) = E_N(\sigma) - \frac{\sum_{j<J} \mu(a_{N,j}) a_{N,j}^{-\sigma}}{P_N(\sigma)}. \tag{23.15}$$

Then

$$a_{N,J} = q_N \lim_{\sigma\to\infty} |E_N^{(J)}(\sigma)|^{-1/\sigma}, \tag{23.16}$$

$$\mu(a_{N,J}) = \lim_{\sigma\to\infty} \operatorname{sgn} E_N^{(J)}(\sigma). \tag{23.17}$$

Proof. By definition,

$$E_N(\sigma) = \frac{1}{P_N(\sigma)} \sum_{j\geq 1} \mu(a_{N,j}) a_{N,j}^{-\sigma}. \tag{23.18}$$

Subtracting the first $J-1$ known terms leaves

$$E_N^{(J)}(\sigma) = \frac{1}{P_N(\sigma)} \sum_{j\geq J} \mu(a_{N,j}) a_{N,j}^{-\sigma}. \tag{23.19}$$

Factor the first remaining term and the least denominator term exactly as in Theorem 8.2. Dominated convergence gives

$$E_N^{(J)}(\sigma) = \mu(a_{N,J}) \left(\frac{q_N}{a_{N,J}}\right)^{\sigma} (1 + o(1)). \tag{23.20}$$

Taking the negative σ-th root recovers the ratio $a_{N,J}/q_N$, and eventual sign recovers $\mu(a_{N,J})$. Induction on J determines the entire signed support sequence uniquely. □

Remark 23.8. Theorem 23.7 is an exact uniqueness statement. It does not by itself yield a computational advantage, because evaluating E_N already requires the reciprocal zeta function with sufficient precision and control of exponentially small differences.

24. Additional structure of the positive horizontal defect

24.1. **Even-moment expansion.**

Proposition 24.1 (Power-series expansion in $\log q$)**.** *At a fixed ordinate τ, put*

$$\delta_\rho = \beta - \frac{1}{2} \tag{24.1}$$

for zeros with $\beta > 1/2$. Then

$$D_q(\tau) = \sum_{m=1}^{\infty} \frac{(\log q)^{2m}}{(2m)!} \sum_{\substack{\rho=\beta+i\tau \\ \beta>1/2}} \delta_\rho^{2m}. \tag{24.2}$$

Proof. At a fixed ordinate there are finitely many zeros, counted with finite multiplicity. Put $y = \log q$ and $\delta_\rho = \beta - 1/2 > 0$. The entire expansion

$$\cosh(\delta_\rho y) - 1 = \sum_{m=1}^{\infty} \frac{\delta_\rho^{2m} y^{2m}}{(2m)!} \tag{24.3}$$

converges absolutely for every complex y. Because the outer zero sum is finite at the fixed ordinate, it may be interchanged with the power series. Summing the coefficients gives the asserted even-moment expansion, with no uniform-convergence subtlety. □

Corollary 24.2 (Quadratic lower bound)**.** *For every $q > 1$,*

$$D_q(\tau) \geq \frac{(\log q)^2}{2} \sum_{\substack{\rho = \beta + i\tau \\ \beta > 1/2}} \left(\beta - \frac{1}{2}\right)^2. \tag{24.4}$$

Proof. The Taylor series

$$\cosh x - 1 = \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots \tag{24.5}$$

has nonnegative terms for real x. Dropping every term except the first gives $\cosh x - 1 \geq x^2/2$. Apply this with $x = (\beta - 1/2)\log q$ to every off-line pair at the fixed ordinate. Since all summands remain nonnegative, summing preserves the inequality and produces the stated quadratic moment. □

Proposition 24.3 (Monotonicity and convexity)**.** *For fixed τ, the function*

$$t \longmapsto D_{e^t}(\tau), \qquad t \geq 0, \tag{24.6}$$

is nonnegative, increasing, and convex. It is identically zero if and only if every zero at ordinate τ lies on the critical line.

Proof. Set $t = \log q \geq 0$. For a horizontal displacement $\delta > 0$, define $f_\delta(t) = \cosh(\delta t) - 1$. Then

$$f_\delta'(t) = \delta \sinh(\delta t) \geq 0, \qquad f_\delta''(t) = \delta^2 \cosh(\delta t) > 0. \tag{24.7}$$

Thus every summand is increasing and strictly convex. Their finite sum has the same properties whenever at least one summand is present. If no off-line pair is present, the sum is identically zero. Conversely, an identically zero sum of nonnegative summands cannot contain a positive summand, so no off-line pair exists. □

Corollary 24.4 (One parameter and many parameters)**.** *One fixed $q > 1$ suffices for the logical criterion* (12.11)*. A continuum of values of q additionally encodes all even power sums of the horizontal displacements at each ordinate through* (24.2)*.*

Proof. The fixed-parameter assertion is Corollary 12.5. For the family assertion, Proposition 24.1 writes the analytic function of $y = \log q$ as

$$D_{e^y}(\tau) = \sum_{m \geq 1} \frac{y^{2m}}{(2m)!} P_{2m}(\tau), \tag{24.8}$$

where $P_{2m}(\tau)$ is the $2m$-th power sum of the positive horizontal displacements. Differentiating $2m$ times at zero gives $P_{2m}(\tau)$. Thus a neighborhood of one in the q-parameter encodes all even moments, whereas positivity makes a single nontrivial parameter enough merely to test whether they all vanish. □

24.2. **Jump formulation.**

Definition 24.5. Define the right-continuous cumulative functions

$$U_q(T) = \sum_{0<\tau\le T} m(\tau)e^{i\tau\log q}, \tag{24.9}$$

$$V_q(T) = \sum_{0<\tau\le T} W_q(\tau)e^{i\tau\log q}. \tag{24.10}$$

Proposition 24.6 (Exact jump defect)**.** *At every ordinate τ,*

$$\boxed{\frac{1}{2}e^{-i\tau\log q}\Delta(V_q-U_q)(\tau) = D_q(\tau).} \tag{24.11}$$

Proof. The jump of the complete-zero walk at τ is

$$\Delta V_q(\tau) = W_q(\tau)e^{i\tau\log q}, \tag{24.12}$$

whereas the jump of the ordinate-only walk is

$$\Delta U_q(\tau) = m(\tau)e^{i\tau\log q}. \tag{24.13}$$

Subtracting and using $W_q - m = 2D_q$ gives

$$\Delta(V_q-U_q)(\tau) = 2D_q(\tau)e^{i\tau\log q}. \tag{24.14}$$

Demodulation by $\frac{1}{2}e^{-i\tau\log q}$ leaves the nonnegative scalar $D_q(\tau)$. □

25. Regularized explicit formula with explicit prime-power weights

25.1. **The explicit formula in the Fourier normalization used here.**

Theorem 25.1 (Guinand–Weil explicit formula)**.** *Let h be holomorphic in a strip $|\Im z| \le 1/2+\epsilon$, real on the real axis after the appropriate conjugation, and decreasing there sufficiently rapidly together with its Fourier transform*

$$\widehat{h}(v) = \int_{\mathbb{R}} h(t)e^{-ivt}\,\mathrm{d}t. \tag{25.1}$$

Then, with zeros counted with multiplicity,

$$\sum_{\rho} h\Big(\frac{\rho-1/2}{i}\Big) = h\Big(\frac{1}{2i}\Big) + h\Big(-\frac{1}{2i}\Big) - \frac{\widehat{h}(0)}{2\pi}\log\pi \tag{25.2}$$

$$+ \frac{1}{2\pi}\int_{-\infty}^{\infty} h(u)\Re\frac{\Gamma'}{\Gamma}\Big(\frac{1}{4}+\frac{iu}{2}\Big)\,\mathrm{d}u \tag{25.3}$$

$$- \frac{1}{2\pi}\sum_{n=2}^{\infty}\frac{\Lambda(n)}{\sqrt{n}}\Big[\widehat{h}(\log n)+\widehat{h}(-\log n)\Big]. \tag{25.4}$$

The formula is understood by the usual limiting procedure whenever the chosen test lies at the boundary of the stated class [10, 13].

Proof. Consider the completed function $\xi(s)$ from (2.5) and the contour integral

$$\frac{1}{2\pi i}\int_{\partial\mathcal{R}}\frac{\xi'}{\xi}(s)h\Big(\frac{s-1/2}{i}\Big)\,\mathrm{d}s \tag{25.5}$$

over a rectangle whose vertical sides are $\Re s = 1/2\pm a$, with $a>1/2$, and whose horizontal sides tend to infinite height through values avoiding zeros. The decay assumptions on h make the horizontal integrals vanish. By the residue theorem, the enclosed zeros of ξ contribute the spectral sum on the left, while the factors $s(s-1)$ produce the two evaluations at $\pm 1/(2i)$.

On the right vertical side, decompose the logarithmic derivative as

$$\frac{\xi'}{\xi}(s) = \frac{1}{s} + \frac{1}{s-1} - \frac{1}{2}\log\pi + \frac{1}{2}\frac{\Gamma'}{\Gamma}\left(\frac{s}{2}\right) + \frac{\zeta'}{\zeta}(s). \tag{25.6}$$

The functional equation $\xi(s) = \xi(1-s)$ transforms the left vertical side into the conjugate contribution of the right side. After the two sides are combined and parametrized by $s = 1/2 + iu$, the constant $-\frac{1}{2}\log\pi$ gives $-\widehat{h}(0)\log\pi/(2\pi)$, and the gamma logarithmic derivative gives the integral in (25.3).

For $\Re s > 1$, the absolutely convergent Dirichlet series

$$-\frac{\zeta'}{\zeta}(s) = \sum_{n=2}^{\infty} \frac{\Lambda(n)}{n^s} \tag{25.7}$$

permits termwise integration. The contribution of n^{-s} on the two paired vertical sides is

$$\frac{\Lambda(n)}{\sqrt{n}}\left[\widehat{h}(\log n) + \widehat{h}(-\log n)\right], \tag{25.8}$$

with the prefactor $-1/(2\pi)$ arising from the contour normalization and the parametrization $\mathrm{d}s = i\,\mathrm{d}u$. Summing the contributions gives (25.4). Absolute convergence in the initial test class justifies all interchanges; the standard limiting argument extends the identity to the stated boundary cases. □

Corollary 25.2 (Why only prime powers occur)**.** *The arithmetic side of the formula is supported exactly on prime powers, because*

$$\Lambda(n) \neq 0 \quad \Longleftrightarrow \quad n = p^r \tag{25.9}$$

for a prime p and integer $r \geq 1$.

Proof. This is the defining support property of the von Mangoldt function: $\Lambda(p^r) = \log p$ and $\Lambda(n) = 0$ otherwise. Applying it to the prime sum (25.4) removes every non-prime-power integer. □

25.2. **Gaussian convolution of a finite interval.**

Definition 25.3. For $\eta > 0$, define

$$g_\eta(t) = \frac{\eta}{\sqrt{\pi}} e^{-\eta^2 t^2}, \tag{25.10}$$

and

$$H_{T,\eta}(z) = \int_0^T g_\eta(z-u)\,\mathrm{d}u. \tag{25.11}$$

The oscillatory test is

$$h_{T,\eta,\omega}(z) = e^{i\omega z} H_{T,\eta}(z). \tag{25.12}$$

Proposition 25.4 (Fourier transform)**.** *With the convention*

$$\widehat{f}(v) = \int_{\mathbb{R}} f(t) e^{-ivt}\,\mathrm{d}t, \tag{25.13}$$

one has

$$\boxed{\widehat{h}_{T,\eta,\omega}(v) = e^{-(v-\omega)^2/(4\eta^2)} \frac{1 - e^{-i(v-\omega)T}}{i(v-\omega)},} \tag{25.14}$$

where the quotient is interpreted continuously as T at $v = \omega$.

Proof. With the Fourier convention $\widehat{f}(v) = \int_{\mathbb{R}} f(t)e^{-ivt}\,\mathrm{d}t$, the normalized Gaussian has transform

$$\widehat{g}_\eta(v) = e^{-v^2/(4\eta^2)}. \tag{25.15}$$

The regularized interval is the convolution $H_{T,\eta} = \mathbf{1}_{[0,T]} * g_\eta$, so the convolution theorem gives

$$\widehat{H}_{T,\eta}(v) = e^{-v^2/(4\eta^2)}\frac{1-e^{-ivT}}{iv}, \tag{25.16}$$

with the quotient interpreted continuously at $v = 0$. Multiplication by $e^{i\omega t}$ translates the Fourier transform: $\widehat{e^{i\omega t}H}(v) = \widehat{H}(v-\omega)$. Substitution proves the formula. □

Proposition 25.5 (Regularized two-branch prime contribution)**.** *In the prime-power side of the Guinand–Weil explicit formula, the test* (25.12) *produces, apart from terms independent of T, the oscillatory sum*

$$\mathcal{P}^{\mathrm{osc}}_{\eta,\omega}(T) = \sum_{n=p^r} c_n^- e^{-(\ell_n-\omega)^2/(4\eta^2)}e^{i(\omega-\ell_n)T} \tag{25.17}$$

$$+ \sum_{n=p^r} c_n^+ e^{-(\ell_n+\omega)^2/(4\eta^2)}e^{i(\omega+\ell_n)T}. \tag{25.18}$$

Proof. Insert $v = \ell_n$ and $v = -\ell_n$ into the exact Fourier transform. For the first argument,

$$\widehat{h}(\ell_n) = w_{n,-}\frac{1-e^{-i(\ell_n-\omega)T}}{i(\ell_n-\omega)}, \tag{25.19}$$

where $w_{n,-} = e^{-(\ell_n-\omega)^2/(4\eta^2)}$. Multiplication by the prime-side factor $-\Lambda(n)/(2\pi\sqrt{n})$ separates a constant and the oscillatory term

$$w_{n,-}c_n^- e^{i(\omega-\ell_n)T}. \tag{25.20}$$

The argument $-\ell_n$ similarly gives

$$w_{n,+}c_n^+ e^{i(\omega+\ell_n)T}, \qquad w_{n,+} = e^{-(\ell_n+\omega)^2/(4\eta^2)}. \tag{25.21}$$

The remaining terms are independent of T and belong to the additive constant that must be retained or fitted before demodulation. □

25.3. **A uniform bound for the regularized prime tail.**

Theorem 25.6 (Uniform Gaussian tail)**.** *Assume $Q > e^{2\omega}$ and put $L = \log Q$. If $L > \omega + \eta^2$, then*

$$\sup_{T\in\mathbb{R}}\left|\sum_{\substack{n=p^r\\ n>Q}}\left[c_n^- e^{-(\ell_n-\omega)^2/(4\eta^2)}e^{i(\omega-\ell_n)T} + c_n^+ e^{-(\ell_n+\omega)^2/(4\eta^2)}e^{i(\omega+\ell_n)T}\right]\right|$$

$$\ll_\omega \frac{\eta^2}{L-\omega-\eta^2}\exp\left(\frac{\eta^2}{4} - \frac{(L-\omega-\eta^2)^2}{4\eta^2}\right). \tag{25.22}$$

Proof. For $n > Q > e^{2\omega}$,

$$|\ell_n - \omega| \geq \frac{1}{2}\ell_n, \qquad \ell_n + \omega \geq \ell_n. \tag{25.23}$$

Using $|1-e^{ix}| \leq 2$ and $\Lambda(n) \leq \log n$, both branches of the omitted prime side are bounded, up to a constant depending on ω, by

$$\sum_{n>Q} n^{-1/2}\exp\left[-\frac{(\log n-\omega)^2}{4\eta^2}\right]. \tag{25.24}$$

When $L = \log Q > \omega + \eta^2$, the corresponding continuous integrand is decreasing beyond Q, so the sum is bounded by a constant multiple of

$$I = \int_Q^\infty x^{-1/2} \exp\left[-\frac{(\log x - \omega)^2}{4\eta^2}\right] \, \mathrm{d}x. \tag{25.25}$$

Put $u = \log x$. Completing the square gives

$$I = \int_L^\infty \exp\left[\frac{u}{2} - \frac{(u-\omega)^2}{4\eta^2}\right] \, \mathrm{d}u \tag{25.26}$$

$$= \exp\left(\frac{\omega}{2} + \frac{\eta^2}{4}\right) \int_L^\infty \exp\left[-\frac{(u-\omega-\eta^2)^2}{4\eta^2}\right] \, \mathrm{d}u. \tag{25.27}$$

For $r = L - \omega - \eta^2 > 0$, the elementary Gaussian-tail estimate

$$\int_r^\infty e^{-v^2/(4\eta^2)} \, \mathrm{d}v \le \frac{2\eta^2}{r} e^{-r^2/(4\eta^2)} \tag{25.28}$$

follows by comparing $1 \le v/r$ and integrating the derivative of the Gaussian. Substitution yields the asserted uniform bound. The estimate is independent of T because only $|1 - e^{ix}| \le 2$ was used. □

Corollary 25.7 (One explicit joint choice)**.** *If*

$$\log Q = \omega + 3\eta^2, \tag{25.29}$$

then the right-hand side of (25.22) *is*

$$O_\omega(e^{-3\eta^2/4}). \tag{25.30}$$

Warning 25.8. Theorem 25.6 controls the prime side of a regularized explicit formula. It does not identify the complete-zero spectral side with the unweighted ordinate walk. That identification is exactly where the positive measure (12.10) appears.

26. Mean-square statements and their exact logical strength

26.1. **Finite Dirichlet-polynomial tails in Lebesgue mean.**

Proposition 26.1 (Finite-tail mean-value estimate)**.** *Let*

$$E_{Q,R}(T) = \sum_{\substack{Q < n \le R \\ n = p^r}} \left(c_n^- n^{-iT} + c_n^+ n^{iT}\right). \tag{26.1}$$

Then

$$\frac{1}{X}\int_0^X |E_{Q,R}(T)|^2 \, \mathrm{d}T \ll_\omega \left(1 + \frac{R}{X}\right)\left[\log\frac{\log R}{\log Q} + \frac{1}{\log Q} + Q^{-1/2}\right] \tag{26.2}$$

for $e^{2\omega} < Q < R$.

Proof. Apply the Montgomery–Vaughan mean-value theorem for Dirichlet polynomials [4, Chapter 7] separately to

$$\sum_{Q<n\le R} c_n^- n^{-iT} \quad \text{and} \quad \sum_{Q<n\le R} c_n^+ n^{iT}. \tag{26.3}$$

It gives an upper bound of the form

$$\int_0^X \left|\sum_{n\le R} a_n n^{-iT}\right|^2 \, \mathrm{d}T \ll (X+R)\sum_{n\le R}|a_n|^2. \tag{26.4}$$

The inequality $|u+v|^2 \le 2|u|^2 + 2|v|^2$ combines the branches. For $n = p^r > e^{2\omega}$,

$$|c_n^-|^2 + |c_n^+|^2 \ll_\omega \frac{\Lambda(n)^2}{n(\log n)^2} = \frac{1}{r^2 p^r}. \tag{26.5}$$

The prime terms sum to $\log(\log R/\log Q) + O(1/\log Q)$ by Mertens' theorem for primes, while proper powers contribute $O(Q^{-1/2})$. Division by X yields the stated estimate. □

Corollary 26.2 (A finite logarithmic tail can be exhausted)**.** *Let*

$$R(X) = X^{1/2}, \qquad Q(X) = R(X)^{1-1/\sqrt{\log\log X}}. \tag{26.6}$$

Then

$$\frac{1}{X}\int_0^X |E_{Q(X),R(X)}(T)|^2 \,\mathrm{d}T \longrightarrow 0. \tag{26.7}$$

Warning 26.3. Corollary 26.2 concerns a finite tail in Lebesgue measure. The numerical variance is sampled on zeta ordinates and contains the projected kernel (19.8); the two norms are not interchangeable.

26.2. **Normalized global convergence does not detect fixed atoms.**

Proposition 26.4 (Counterexample to global normalized sufficiency)**.** *There exist functions F_X such that*

$$\frac{1}{X}\int_0^X \left|F_X(T) - \frac{1}{2}\right|^2 \mathrm{d}T \longrightarrow 0, \tag{26.8}$$

but whose distributional derivatives retain a nonzero fixed atomic part.

Proof. Take $F_X(T) = 1/2 + \mathbf{1}_{[0,1]}(T)$. Then for every $X \ge 1$,

$$\frac{1}{X}\int_0^X \left|F_X(T) - \frac{1}{2}\right|^2 \mathrm{d}T = \frac{1}{X} \longrightarrow 0. \tag{26.9}$$

However, on the fixed compact interval $[0,1]$, the defect remains identically one. In the sense of distributions,

$$\frac{\mathrm{d}}{\mathrm{d}T}\mathbf{1}_{[0,1]} = \delta_0 - \delta_1, \tag{26.10}$$

which does not tend to zero. Thus normalization by the length of an expanding interval can erase a fixed local or atomic obstruction. □

Corollary 26.5. *Any route from energy exhaustion to the vanishing of the positive atomic measure* (12.10) *must use a local topology, atomwise convergence, or distributional convergence against compactly supported test functions.*

27. Weighted prime-power profiles

Definition 27.1. Let $W : [0,\infty) \to \mathbb{C}$ be bounded and compactly supported. Define

$$G_{Q,W}(T) = \sum_{n=p^r} W\left(\log\frac{n}{Q}\right) g_n(T). \tag{27.1}$$

Theorem 27.2 (Weighted Bohr energy)**.** *As $Q \to \infty$,*

$$\lim_{X\to\infty}\frac{1}{X}\int_0^X |G_{Q,W}(T)|^2 \,\mathrm{d}T = \frac{1}{2\pi^2\log Q}\int_0^\infty |W(u)|^2 \,\mathrm{d}u + o\left(\frac{1}{\log Q}\right). \tag{27.2}$$

Proof. Expand the squared modulus of the finite weighted sum. Every diagonal term contributes

$$|W(\log(n/Q))|^2 \left(|c_n^-|^2 + |c_n^+|^2\right). \tag{27.3}$$

Every unequal pair has a nonzero frequency difference and therefore zero Bohr mean. For primes, use

$$|c_p^-|^2 + |c_p^+|^2 = \frac{1}{2\pi^2 p} + O_\omega\left(\frac{1}{p(\log p)^2}\right). \tag{27.4}$$

Write $p = Qe^u$ and apply the prime number theorem by partial summation. For compactly supported continuous W,

$$\sum_p \frac{1}{p}\left|W\left(\log\frac{p}{Q}\right)\right|^2 = \frac{1}{\log Q}\int_0^\infty |W(u)|^2\,\mathrm{d}u + o\left(\frac{1}{\log Q}\right). \tag{27.5}$$

The coefficient error and all proper prime powers are lower order. This proves the formula. □

Corollary 27.3 (Interpretation of the constant 3/4)**.** *A weighted Bohr model has leading energy* $3/(4\pi^2\log Q)$ *precisely when*

$$\int_0^\infty |W(u)|^2\,\mathrm{d}u = \frac{3}{2}. \tag{27.6}$$

The constant determines a quadratic profile mass, not a unique hard support.

28. Detailed projected Gram algebra

28.1. **Four branch expansion.**

Proposition 28.1 (Full unprojected quadratic expansion)**.** *For a finite set* $\mathcal{N}$ *of prime powers, put*

$$G_{\mathcal{N}}(\gamma) = \sum_{n\in\mathcal{N}}\left(c_n^- n^{-i\gamma} + c_n^+ n^{i\gamma}\right). \tag{28.1}$$

Then

$$\|G_{\mathcal{N}}\|_H^2 = \sum_{m,n\in\mathcal{N}} \overline{c_m^-}\,c_n^- M_H(m/n) \tag{28.2}$$

$$+ \sum_{m,n\in\mathcal{N}} \overline{c_m^-}\,c_n^+ M_H(mn) \tag{28.3}$$

$$+ \sum_{m,n\in\mathcal{N}} \overline{c_m^+}\,c_n^-\,\overline{M_H(mn)} \tag{28.4}$$

$$+ \sum_{m,n\in\mathcal{N}} \overline{c_m^+}\,c_n^+ M_H(n/m). \tag{28.5}$$

Proof. Write the finite model as

$$G(\gamma) = \sum_n \left(c_n^- n^{-i\gamma} + c_n^+ n^{i\gamma}\right). \tag{28.6}$$

Multiplying by its conjugate produces four sign choices. For example,

$$\overline{c_m^- m^{-i\gamma}}\,c_n^- n^{-i\gamma} = \overline{c_m^-}\,c_n^-\left(\frac{m}{n}\right)^{i\gamma}, \tag{28.7}$$

whose sample average is $\overline{c_m^-}\,c_n^- M_H(m/n)$. The other three products give the arguments mn, $(mn)^{-1}$, and n/m. Summing all pairs yields the displayed expansion exactly. □

28.2. **Prime-tower blocks.**

Proposition 28.2 (Same-tower resonance closure)**.** *For $m = p^r$ and $n = p^s$, every argument of M_H in* (28.2)–(28.5) *is one of*

$$p^{s-r}, \qquad p^{r-s}, \qquad p^{r+s}, \qquad p^{-(r+s)}. \tag{28.8}$$

Thus an entire two-branch prime tower is an arithmetically resonant Gram block; it is not reduced to the scalar diagonal $r = s$.

Proof. Substitute $m = p^r$ and $n = p^s$ into the four multiplicative arguments. They become

$$p^{r-s}, \qquad p^{r+s}, \qquad p^{-(r+s)}, \qquad p^{s-r}. \tag{28.9}$$

Each is one, a prime power, or the reciprocal of a prime power. Thus all entries inside one tower lie on arithmetic resonance sets for the Landau–Gonek kernel; only the scalar equality $r = s$ is the ordinary Parseval diagonal. □

Proposition 28.3 (Projection correction)**.** *Let V be the matrix whose columns are the sampled prime-power modes, and let B be the two-column matrix with columns* 1 *and $e^{i\omega\gamma}$. Then the residual Gram matrix is*

$$\widetilde{K}_H = \frac{1}{N(H)} V^*V - \frac{1}{N(H)^2} V^*B \left(\frac{1}{N(H)} B^*B \right)^{-1} B^*V. \tag{28.10}$$

In particular, even the norm of a single frequency is altered by the constant–carrier projection.

Proof. Let B be the matrix whose columns are the sampled constant and carrier vectors. Since these columns are linearly independent when $|M_H(e^\omega)| < 1$, the orthogonal projection onto their span is

$$P_B = B(B^*B)^{-1}B^*. \tag{28.11}$$

The complementary projection is $I - P_B$. If V contains the sampled prime frequency vectors, the projected Gram matrix is

$$\frac{1}{N(H)} V^*(I - P_B)V = \frac{1}{N(H)} V^*V \tag{28.12}$$

$$- \frac{1}{N(H)} V^*B(B^*B)^{-1}B^*V. \tag{28.13}$$

Writing $B^*B = N(H)G_H$ gives the normalized Schur-complement form in the proposition. In particular, projection changes both diagonal and off-diagonal entries. □

29. Analytic targets isolated by the exact identities

Conjecture 29.1 (Collective carrier law)**.** *In a specified joint range $Q = Q(H) \to \infty$,*

$$\Im \widehat{c}_{Q,H} = \pi \Delta_\omega(Q) + o\left(\Delta_\omega(Q)\right). \tag{29.1}$$

By Corollary 16.6, this is equivalent to a weighted collective cancellation in the exact kernel $M_H(n)$.

Conjecture 29.2 (Projected variance law)**.** *For a suitable joint range,*

$$V(Q, H) = \frac{3}{4} \Delta_\omega(Q) + o\left(\Delta_\omega(Q)\right). \tag{29.2}$$

The left-hand side is the projected discrete quadratic form (19.12)*; no ordinary Parseval diagonal is assumed.*

Question 29.3 (Tower and cross-tower separation)**.** Can one prove, uniformly in a growing cutoff, asymptotic formulas for

$$\sum_{p} a_p^* \widetilde{K}_{p,p;H} a_p \tag{29.3}$$

and

$$\sum_{p \neq q} a_p^* \widetilde{K}_{p,q;H} a_q \tag{29.4}$$

separately? This is the exact decomposition required before any assertion about a "diagonal" can be made.

30. Exact routes from the exponential zero walk to the Riemann Hypothesis

This section records the routes that arise directly from the phase-walk construction. Each route is an exact implication or equivalence. What is missing in every incomplete route is stated separately; no numerical observation is used as a premise.

30.1. **Route I: equality of the complete and ordinate-only atomic measures.**

Theorem 30.1 (Fixed-q atomic criterion)**.** *For any fixed $q > 1$,*

$$\nu_q = \mu_q \quad \Longleftrightarrow \quad \text{RH}. \tag{30.1}$$

Equivalently, it is enough to prove equality of the jumps of the two walks at every positive ordinate.

Proof. The measure difference is given exactly by

$$\frac{1}{2} e^{-iT \log q} (\nu_q - \mu_q) = \sum_{\tau > 0} D_q(\tau) \delta_\tau. \tag{30.2}$$

Every coefficient on the right is nonnegative. Equality of the measures forces every coefficient to vanish, and Theorem 12.3 then places every zero on the critical line. Conversely, under RH, all complete-zero weights equal one and the measures are identical. Since a locally finite atomic measure is determined by its jumps, equality of jumps is equivalent to equality of measures. □

Corollary 30.2 (Midpoint residual criterion)**.** *Suppose an explicit-formula reconstruction is proved to converge locally to the complete-zero midpoint walk, and suppose its residual at every ordinate τ is shown to equal*

$$\frac{m(\tau)}{2} e^{i\tau \log q}. \tag{30.3}$$

Then RH *follows.*

Proof. The complete midpoint jump is $\frac{1}{2} W_q(\tau) e^{i\tau \log q}$. Equality with (30.3) gives $W_q(\tau) = m(\tau)$ for every τ. Apply Theorem 30.1. □

30.2. **Route II: local distributional convergence.**

Theorem 30.3 (Test-function criterion)**.** *Let η_j be a sequence of locally finite complex measures such that*

$$\eta_j \longrightarrow \nu_q - \mu_q \quad \text{in } \mathcal{D}'(\mathbb{R}). \tag{30.4}$$

If, for every $\varphi \in C_c^\infty(\mathbb{R})$,

$$\langle \eta_j, \varphi \rangle \longrightarrow 0, \tag{30.5}$$

then RH *holds.*

Proof. For every test function φ, convergence (30.4) gives

$$\langle \eta_j, \varphi \rangle \longrightarrow \langle \nu_q - \mu_q, \varphi \rangle. \tag{30.6}$$

The second hypothesis says that the same left-hand side tends to zero. Uniqueness of limits in $\mathbb{C}$ yields

$$\langle \nu_q - \mu_q, \varphi \rangle = 0 \tag{30.7}$$

for every $\varphi \in C_c^\infty$. Therefore the measure difference is the zero distribution, hence the zero locally finite measure. The fixed-q atomic criterion then implies RH. □

Remark 30.4. For the prime-power construction, the missing theorem is a joint (H, Q, η) limit that identifies the derivative of the regularized prime side with $\nu_q - \mu_q$, while the corresponding residual tends locally to zero. A normalized average over $[0, H]$ is not sufficient, because it can dilute fixed atoms.

30.3. **Route III: small-q curvature at every ordinate.**

Theorem 30.5 (Quadratic curvature criterion)**.** *Put $q = e^y$. At each ordinate τ,*

$$D_{e^y}(\tau) = \frac{y^2}{2} \sum_{\substack{\rho=\beta+i\tau \\ \beta>1/2}} \left(\beta - \frac{1}{2}\right)^2 + O_\tau(y^4) \qquad (y \to 0). \tag{30.8}$$

Consequently,

$$\mathrm{RH} \iff D_{e^y}(\tau) = o(y^2) \text{ for every } \tau. \tag{30.9}$$

Proof. At a fixed ordinate, enumerate the positive horizontal displacements as $\delta_1, \dots, \delta_m$, repeated with multiplicity. Taylor's theorem at zero gives, uniformly over this finite set,

$$\cosh(\delta_j y) - 1 = \frac{\delta_j^2 y^2}{2} + \frac{\delta_j^4 y^4}{24} + O_\tau(y^6). \tag{30.10}$$

Summing proves the expansion. If the total is $o(y^2)$, division by y^2 and passage to the limit gives

$$\frac{1}{2} \sum_{j=1}^{m} \delta_j^2 = 0. \tag{30.11}$$

Every term is nonnegative, so every displacement is zero. This holds at every ordinate and yields RH. Under RH, the defect is identically zero, proving the converse. □

Corollary 30.6 (Second-derivative formulation)**.** *The horizontal second moment at ordinate τ is recovered by*

$$\frac{\mathrm{d}^2}{\mathrm{d}y^2} D_{e^y}(\tau)\Big|_{y=0} = \sum_{\substack{\rho=\beta+i\tau \\ \beta>1/2}} \left(\beta - \frac{1}{2}\right)^2. \tag{30.12}$$

Proof. Differentiate the finite sum

$$D_{e^y}(\tau) = \sum_j [\cosh(\delta_j y) - 1] \tag{30.13}$$

twice. The second derivative of the j-th term is $\delta_j^2 \cosh(\delta_j y)$, which at $y = 0$ equals δ_j^2. Finite summation commutes with differentiation, proving the formula. □

30.4. **Route IV: large-q amplification.**

Definition 30.7. At a positive ordinate τ, define

$$\delta_{\max}(\tau) = \max_{\rho=\beta+i\tau} \left|\beta - \frac{1}{2}\right|, \tag{30.14}$$

with value zero when all zeros at the ordinate are on the critical line.

Theorem 30.8 (Recovery of the maximal horizontal displacement)**.** *For every positive ordinate τ,*

$$\boxed{\lim_{q\to\infty} \frac{\log\left(1 + D_q(\tau)\right)}{\log q} = \delta_{\max}(\tau).} \tag{30.15}$$

Proof. If $\delta_{\max} = 0$, then $D_q(\tau) = 0$ identically and the limit is zero. Otherwise, let $m_{\max} \geq 1$ denote the number of paired zeros with displacement $\delta_{\max}$. Since

$$\cosh(\delta \log q) - 1 = \frac{1}{2}q^{\delta} + \frac{1}{2}q^{-\delta} - 1, \tag{30.16}$$

we have the lower bound

$$D_q(\tau) \geq m_{\max}\left(\frac{1}{2}q^{\delta_{\max}} - 1\right). \tag{30.17}$$

If $M(\tau)$ is the finite number of off-line pairs at the ordinate, every summand is at most $q^{\delta_{\max}}$ for sufficiently large q, so

$$D_q(\tau) \leq M(\tau) q^{\delta_{\max}}. \tag{30.18}$$

Taking logarithms, dividing by $\log q$, and applying the squeeze theorem gives the result. □

Corollary 30.9 (Subpolynomial-growth route)**.** *If one can prove from the prime side that, for every ordinate τ,*

$$D_q(\tau) = q^{o(1)} \qquad (q \to \infty), \tag{30.19}$$

then RH *follows.*

Proof. The growth exponent in Theorem 30.8 must then be zero at every ordinate. □

30.5. **Route V: reconstruction from a family of frequencies.**

Theorem 30.10 (Finite exponential-polynomial reconstruction)**.** *At a fixed ordinate τ, the function*

$$\mathcal{W}_\tau(y) = W_{e^y}(\tau) = \sum_{\rho:\Im\rho=\tau} e^{(\beta-1/2)y} \tag{30.20}$$

is an entire exponential polynomial. Knowledge of $\mathcal{W}_\tau(y)$ on any real set having an accumulation point uniquely determines the multiset of horizontal displacements $\{\beta - 1/2\}$. In particular,

$$\mathcal{W}_\tau(y) \equiv m(\tau) \quad \Longleftrightarrow \quad \textit{all zeros at } \tau \textit{ lie on } \Re s = \frac{1}{2}. \tag{30.21}$$

Proof. The sum is finite, so $\mathcal{W}_\tau$ is entire. If two finite exponential polynomials agree on a set with an accumulation point, the identity theorem makes them equal everywhere. Their derivatives at zero are the power sums

$$\mathcal{W}_\tau^{(k)}(0) = \sum_{\rho:\Im\rho=\tau} \left(\beta - \frac{1}{2}\right)^k. \tag{30.22}$$

The finite multiset of exponents is determined by these power sums through Newton identities, after combining repeated values with multiplicity. If the function is the constant $m(\tau)$, its second derivative is zero; by symmetry and nonnegativity this forces every squared displacement to be zero. The converse is immediate. □

30.6. **Route VI: projected energy exhaustion plus local atomic control.**

Theorem 30.11 (A sufficient two-part energy route)**.** *Let F_j be regularized residuals and suppose:*

(i) *for every compact interval I,*

$$\int_I \left| F_j(T) - \frac{1}{2} \right|^2 \, \mathrm{d}T \longrightarrow 0; \tag{30.23}$$

(ii) *the distributional derivatives satisfy*

$$\frac{\mathrm{d}}{\mathrm{d}T} \left[e^{i\omega T} \left(F_j(T) - \frac{1}{2} \right) \right] \longrightarrow \nu_q - \mu_q. \tag{30.24}$$

Then RH *holds.*

Proof. Condition (i) is local L^2-convergence of $F_j - 1/2$ to zero. Multiplication by the smooth unit-modulus factor $e^{i\omega T}$ preserves that convergence. On compact sets, local L^2 implies local L^1 by Cauchy–Schwarz. Consequently, the distributional derivative of the product tends to zero when tested against a compactly supported smooth function. Condition (ii) identifies that same derivative limit with $\nu_q - \mu_q$. Hence the measure difference is zero, and the positive atomic criterion gives RH. □

Remark 30.12. The measured quantity $N(H)^{-1} \sum_{\gamma \le H} |F(\gamma)|^2$ is a global normalized energy. Even if it tends to zero, it does not imply (30.23). The route requires either local windows, compactly supported test functions, or an independent argument excluding a finite or sparse atomic defect.

30.7. **Route VII: a one-sided positive test family.**

Theorem 30.13 (Exhausting positive tests)**.** *Let*

$$\mathcal{D}_q = \sum_{\tau > 0} D_q(\tau) \delta_\tau. \tag{30.25}$$

Suppose there is a sequence of nonnegative test functions $\varphi_j \in C_c^\infty(0, \infty)$ such that for every $T > 0$,

$$\inf_{0 < t \le T} \varphi_j(t) > 0 \tag{30.26}$$

for all sufficiently large j, and

$$\langle \mathcal{D}_q, \varphi_j \rangle \longrightarrow 0. \tag{30.27}$$

Then RH *holds.*

Proof. Fix an ordinate τ. For all sufficiently large j, $\varphi_j(\tau) > 0$. Positivity of all masses gives

$$0 \le D_q(\tau) \varphi_j(\tau) \le \langle \mathcal{D}_q, \varphi_j \rangle. \tag{30.28}$$

The right-hand side tends to zero, while the test value is bounded below by a positive number once a compact interval containing τ is fixed. Hence $D_q(\tau) = 0$. Since τ was arbitrary, the positive defect measure is zero and RH follows. □

Remark 30.14. The exact algebra does not choose among these routes. The analytic work still required is one of the following: local reconstruction of the ordinate-only measure, a sufficiently strong uniform bound in q, or a local projected-energy theorem accompanied by derivative identification. The finite numerical decay of a global variance is evidence for none of these implications by itself.

31. Open problems and falsifiable next computations

Question 31.1 (Increment audit)**.** For every prime power n in a growing range, compute the exact ratio

$$\mathcal{I}_H(n) = \frac{\Re M_H(n)}{\pi \upsilon_n} \tag{31.1}$$

and compare it with

$$\frac{\log n}{\log(H/2\pi)}. \tag{31.2}$$

This tests the local Landau mechanism without fitting a global law.

Question 31.2 (Height–cutoff matrix)**.** Repeat the complete computation for a two-dimensional grid of (H, Q), including regimes

$$Q = H^{\theta} \tag{31.3}$$

with several fixed θ and a near-height regime $Q = H^{1-o(1)}$. Record raw and joint coefficients separately.

Question 31.3 (Projected tower blocks)**.** Compute

$$a_p^* \widetilde{K}_{p,p;H} a_p \tag{31.4}$$

for every prime tower and compare its cumulative sum with the cross-tower term in (19.13).

Question 31.4 (Local test-function convergence)**.** For compactly supported smooth φ, determine whether the regularized residual satisfies

$$\int \left(F_{Q,H}(T) - \frac{1}{2} \right) \varphi(T)\, \mathrm{d}T \longrightarrow 0 \tag{31.5}$$

under a joint limiting regime, while the prime-side derivative is identified with the complete spectral measure.

Part 4. Numerical support

32. Computational object and exact program identities

32.1. Input data and sampled height.

Experiment 32.1 (Zero data)**.** The full-data run used 2,001,052 positive ordinates from `zeross.bin`, with

$$\gamma_1 = 14.134725142, \qquad \gamma_{2{,}001{,}052} = 1{,}132{,}490.6587144111. \tag{32.1}$$

The phase parameter was

$$\omega = 1, \qquad q = e. \tag{32.2}$$

All ordinates were used; the stride was one.

32.2. **Exact definition of the program.**

Definition 32.2 (Program variables)**.** At each sampled ordinate, the code forms

$$Y(\gamma_k) = S_q(\gamma_k) - A_{\mathrm{smooth}}(\gamma_k), \tag{32.3}$$

and

$$R_Q(\gamma_k) = Y(\gamma_k) - P_Q^-(\gamma_k) - P_Q^+(\gamma_k), \tag{32.4}$$

where

$$P_Q^-(\gamma) = \sum_{\substack{n=p^r \\ n\le Q}} c_n^- e^{i(\omega-\ell_n)\gamma}, \tag{32.5}$$

$$P_Q^+(\gamma) = \sum_{\substack{n=p^r \\ n\le Q}} c_n^+ e^{i(\omega+\ell_n)\gamma}. \tag{32.6}$$

The raw carrier statistic is

$$\widehat{c}_{Q,H} = \frac{1}{N(H)} \sum_{\gamma_k \le H} e^{-i\omega\gamma_k} R_Q(\gamma_k). \tag{32.7}$$

Proposition 32.3 (Identity evaluated by the raw program)**.** *The program variables satisfy exactly*

$$\widehat{c}_{Q,H} = \widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} \left(c_n^- \overline{M_H(n)} + c_n^+ M_H(n) \right). \tag{32.8}$$

Its imaginary part is

$$\Im\widehat{c}_{Q,H} = \Im\widehat{c}_{Y,H} - \sum_{\substack{n=p^r \\ n\le Q}} u_n \Re M_H(n). \tag{32.9}$$

Proof. The arrays used by the program are finite, so the abstract identity in Theorem 16.2 applies without any limiting operation. The code generates exactly the prime powers $n \le Q$, uses the coefficients in Definition 13.1, and forms the sample kernel by averaging $n^{i\gamma_k}$. Substitution of these arrays in the theorem gives the program identity entry by entry. □

32.3. **Joint constant–carrier fit used for the reported values.**

Definition 32.4. For every cutoff, the code fits

$$R_Q(\gamma_k) = \alpha_{Q,H} + c_{Q,H} e^{i\omega\gamma_k} + E_{Q,H}(\gamma_k) \tag{32.10}$$

by least squares in the sampled inner product (19.1). The reported variance is

$$V(Q,H) = \frac{1}{N(H)} \sum_{\gamma_k \le H} |E_{Q,H}(\gamma_k)|^2. \tag{32.11}$$

Proposition 32.5 (Closed form of the reported carrier)**.** *Let*

$$\mu_H = M_H(e^{\omega}). \tag{32.12}$$

Then

$$\boxed{c_{Q,H} = \frac{\langle e^{i\omega\gamma}, R_Q\rangle_H - \overline{\mu_H}\langle 1, R_Q\rangle_H}{1-|\mu_H|^2}.} \tag{32.13}$$

The fitted constant is

$$\alpha_{Q,H} = \frac{\langle 1, R_Q\rangle_H - \mu_H\langle e^{i\omega\gamma}, R_Q\rangle_H}{1-|\mu_H|^2}. \tag{32.14}$$

Proof. The fit minimizes

$$\sum_{k=1}^{N(H)} |R_k - a - cb_k|^2 \tag{32.15}$$

over $a, c \in \mathbb{C}$. Differentiating with respect to the conjugate parameters, or equivalently imposing orthogonality of the residual to the basis vectors 1 and b, gives the two normal equations. Their Gram matrix is $\left(\begin{smallmatrix} 1 & \mu_H \\ \bar{\mu}_H & 1 \end{smallmatrix}\right)$. Its inverse is explicit, and multiplication by the right-hand side gives the displayed formulas for the constant and carrier coefficients. □

32.4. **Prime-power generation and exact coefficients.**

Experiment 32.6 (Prime powers)**.** The cutoffs and counts were

Q	number of prime powers $n = p^r \le Q$
0	0
3	2
10	7
100	35
1000	193
10000	1280

Every coefficient was generated from (13.2)–(13.3); no frequency or amplitude was fitted individually.

33. Radial geometry and low prime-power resonances

33.1. **Direct radial audit.**

Experiment 33.1 (Original walk and theoretical subtractions)**.** After subtraction of the empirical complex center, the radial statistics were

case	mean radius	radial standard deviation	smoothed radial modes
original	1.9203104476	0.7265477726	1.35731059, 2.49789956
after $n = 3$	1.8079247921	0.3083120307	1.68017926
after $n = 2, 3$	1.7884459085	0.2214048337	1.81300495

The theoretical $n = 3$ coefficients were

$$c_3^- = -1.02370050729i, \qquad c_3^+ = 0.0481029537851i. \tag{33.1}$$

Subtracting $n = 3$ reduced the radial variance by approximately 81.99%; subtracting $n = 2, 3$ reduced it by approximately 90.71%. The finite conclusion is that the observed global double radial mode is dominated by the theoretical $n = 3$ oscillator.

33.2. **Geometric interpretation.**

Experiment 33.2 (Anisotropy)**.** At $Q = 10000$, the centered residual anisotropy was

$$0.981212. \tag{33.2}$$

This is consistent with the exact single-mode limit (14.21): after low frequencies are removed, large-n modes are strongly elongated in one direction.

34. Cutoff sweep and carrier stabilization

34.1. **Global cutoff table.**

Experiment 34.1 (Full-data cutoff sweep)**.** The jointly fitted carrier and residual energy were

Q	terms	$\Re c_{Q,H}$	$\Im c_{Q,H}$	$V(Q,H)$	$\sqrt{V(Q,H)}$
0	0	0.4999989123	0.1381673175	1.2420277147	1.1144630
3	2	0.4999998679	0.0956865241	0.1339324657	0.3659678
10	7	0.4999999862	0.0755224108	0.0596099186	0.2441514
100	35	0.5000000416	0.0545919879	0.0220396104	0.1484574
1000	193	0.5000000755	0.0426449212	0.0112822119	0.1062178
10000	1280	0.4999998710	0.0343125459	0.0082559398	0.0908622

At the largest cutoff the finite variance reduction relative to $Q = 0$ was

$$1 - \frac{0.0082559398}{1.2420277147} = 0.99335\ldots. \tag{34.1}$$

34.2. **Normalized** $1/\log Q$ **comparisons.**

Experiment 34.2 (Two normalized quantities)**.** For $Q \geq 3$, the finite normalized values were

Q	$\pi \log Q\, \Im c_{Q,H}$	$\frac{4\pi^2}{3} \log Q\, V(Q,H)$
3	0.33025	1.93628
10	0.54631	1.80623
100	0.78981	1.33564
1000	0.92545	1.02558
10000	0.99284	1.00065

At $Q = 10000$,

$$\frac{1}{\pi \log Q} = 0.0345601\ldots, \tag{34.2}$$

$$\frac{3}{4\pi^2 \log Q} = 0.00825061\ldots. \tag{34.3}$$

The corresponding measured values were 0.0343125459 and 0.0082559398. These are finite coincidences to be interpreted through the exact defect identities, not proofs of asymptotic laws.

35. Height dependence

35.1. **Twenty-four block audit.**

Experiment 35.1 (Block ranges)**.** For $Q = 10000$ and 24 consecutive height blocks,

$$\Re \operatorname{mean}_{\text{block}} F_Q \in [0.499994991361, 0.500003656384], \tag{35.1}$$

$$\Im \operatorname{mean}_{\text{block}} F_Q \in [0.033825249952, 0.035105891286], \tag{35.2}$$

$$V_{\text{block}} \in [0.0054744557, 0.0089851833]. \tag{35.3}$$

The standard deviation of the block means was

$$0.0003812589. \tag{35.4}$$

The carrier is stable across height at this scale, while the block variance has a visible drift. Consequently the global factor 3/4 cannot yet be regarded as height-uniform.

35.2. **Termwise increment scale.**

Experiment 35.2 (Present range of the Landau ratio)**.** For

$$H = 1.13249 \times 10^6, \qquad Q = 10^4, \tag{35.5}$$

one has approximately

$$\frac{\log Q}{\log(H/2\pi)} \approx 0.76. \tag{35.6}$$

The global ratio

$$\frac{\Im c_{Q,H}}{\pi \Delta_1(Q)} \approx 0.986 \tag{35.7}$$

is therefore not explained by independent termwise transport at the leading Landau scale. The exact increment statistic (16.25) is the direct falsifiable test.

36. Branch decomposition and omitted frequencies

36.1. **Branch energies.**

Experiment 36.1 (Minus and plus branches)**.** At $Q = 10000$,

$$\|P_Q^-\|^2 = 1.2433344613, \tag{36.1}$$

$$\|P_Q^+\|^2 = 0.0309962015, \tag{36.2}$$

$$2\Re\langle P_Q^-, P_Q^+\rangle = -0.0005963448. \tag{36.3}$$

The minus branch dominates, and direct plus–minus energy cancellation is small.

36.2. **Omitted prime-power shell.**

Experiment 36.2 (Exact-frequency omitted fit)**.** For the 56 omitted prime powers immediately above $Q = 10000$, the median amplitude-to-null ratios were

$$3.35349 \quad \text{and} \quad 2.67559 \tag{36.4}$$

for the minus and plus branches. The absolute correlation between the jointly fitted coefficient vector and the theoretical coefficient vector was

$$0.916474. \tag{36.5}$$

The narrow shell explained

$$0.0144446 \tag{36.6}$$

of the remaining residual energy. Thus the next prime-power frequencies are present, but this narrow shell does not account for the full residual.

37. Signed invariant and finite-area comparisons

37.1. **Direct evaluation of $\Delta_1(Q)$.**

Experiment 37.1 (Common invariant)**.** At $Q = 10000$,

$$\Delta_1(10000) \approx 0.01108. \tag{37.1}$$

Hence

$$\pi\Delta_1(10000) \approx 0.03481, \qquad \frac{3}{4}\Delta_1(10000) \approx 0.00831. \tag{37.2}$$

The same exact signed invariant therefore reproduces both observed scales to finite accuracy. The analytic claims that the centroid and projected variance equal these multiples remain Conjectures 29.1 and 29.2.

37.2. **Coefficient prefix sums.**

Experiment 37.2 (Prefix sums)**.** The coefficient combinations

$$A_Q = \sum_{n\le Q} (c_n^- + c_n^+), \qquad B_Q = \sum_{n\le Q} (c_n^- - c_n^+) \tag{37.3}$$

were, at $Q = 10000$,

$$A_Q = -2.5552137617i, \qquad B_Q = -10.6932626383i. \tag{37.4}$$

They do not account for the real carrier value $1/2$.

38. Finite-part computations for Φ_N

38.1. **Direct algebraic verification.**

Experiment 38.1 (Finite-part identity)**.** A numerical comparison between the direct tail expression and the value reconstructed from Proposition 10.3 gave

$$|R_{\text{from FP}} - R_{\text{direct}}| = 4.043 \times 10^{-10}. \tag{38.1}$$

This verifies the implemented identity at the working precision.

38.2. **Scale scan.**

Experiment 38.2 (Normalized finite-part differences)**.** For

$$N = 2000, 4000, 8000, 16000, 32000, 64000, \tag{38.2}$$

the successive magnitudes

$$|\mathcal{F}_{2N}(\rho) - \mathcal{F}_N(\rho)| \tag{38.3}$$

were approximately

$$1.12, \quad 0.75, \quad 0.80, \quad 0.75, \quad 0.44. \tag{38.4}$$

No monotone cancellation law is established by these six scales.

39. Numerical tests dictated by the analytic identities

39.1. **Prime-power increment audit.**

Experiment 39.1 (Required output)**.** For every prime power n in the selected range, compute and store

$$\mathcal{I}_H(n) = \frac{\Re M_H(n)}{\pi v_n}, \tag{39.1}$$

together with

$$\theta_H(n) = \frac{\log n}{\log(H/2\pi)}. \tag{39.2}$$

The comparison is pointwise and requires no global fitting.

39.2. **Two-parameter height–cutoff scan.**

Experiment 39.2 (Required grid)**.** For several heights H_j and exponents θ, set

$$Q_{j,\theta} = H_j^\theta \tag{39.3}$$

and record separately:

$$\widehat{c}_{Q,H}, \quad c_{Q,H}, \quad V(Q,H), \quad \Delta_\omega(Q), \quad \sum_p a_p^* \widetilde{K}_{p,p;H} a_p, \quad \sum_{p\neq q} a_p^* \widetilde{K}_{p,q;H} a_q. \tag{39.4}$$

This directly separates raw carrier, projected carrier, tower blocks, and cross-tower interactions.

40. Complete MATLAB implementation

40.1. **Self-contained implementation.**

Remark 40.1. The following program is the complete implementation used for the prime-power audit. It reads only the original binary ordinate file, generates all prime powers internally, retains all figures, performs the joint constant–carrier fit, and exports every table used in the preceding numerical sections.

Listing 1. Complete MATLAB implementation of the prime-power phase audit.

```
%% ZETA_PRIME_POWER_ANALYTIC_AUDIT
%
% Script autocontenido para los seis tests de la descomposicion
%
%   S_q(T) = sum_{gamma_j <= T} exp(i*gamma_j*log(q))
%
% No usa ningun .mat, .csv ni resultado anterior.
% El unico dato externo necesario es el fichero original de ordenadas:
%
%   zeross.bin
%
% Si no esta en la carpeta actual, MATLAB abre un selector para localizarlo.
%
% No resta 0.5*exp(i*T). Ajusta conjuntamente una constante aditiva y
% el coeficiente de exp(i*log(q)*T). La constante se elimina antes de
% demodular.
% Todos los resumenes principales se muestran con fprintf/disp.
% No se cierra ninguna figura.
%
% TESTS
% -----
% 1. Separacion P_Q^- y P_Q^+ y termino cruzado.
% 2. Identidades de c_n^-, c_n^+ y sumas A_Q, B_Q.
% 3. Busqueda de telescopado dentro de p,p^2,...,p^R.
% 4. Funcionales positivos y matrices de Gram por bloques.
% 5. Ecuacion funcional del residuo demodulado F_Q.
% 6. Unicidad: bloques, anisotropia, gaps y frecuencias omitidas.
%
% Requiere una version de MATLAB que permita funciones locales al final
% de un script (R2016b o posterior).

clear;
clc;
format long g;

%% ========================================================================
% CONFIGURACION
% ========================================================================

zeroBinaryFile = 'zeross.bin';

% q=e reproduce exp(i*gamma).
qPhase = exp(1);
omega = log(qPhase);
```

```
% stride=10: usa todos los ceros para construir S_q, pero analiza 1 de cada 10.
% stride=1: analiza los 2,001,052 puntos completos, mucho mas lento para Q=10000.
stride = 1;

Qlist = [0, 3, 10, 100, 1000, 10000];
Qlist = unique(sort(Qlist(:).'));
Qmax = max(Qlist);

numHeightBlocks = 24;
numGramBlocks = 20;
maxGramPoints = 60000;

omittedTestWidth = 500;
maxSpectralPoints = 50000;
numFrequencyGridPoints = 401;

gapMedianWindow = 1001;
resonanceTolerance = 1e-12;
minimumVariance = 1e-30;

% Si se pone true, guarda tablas y MAT al final. No son necesarios para correr.
saveResults = false;
outputDirectory = 'ZETA_PRIME_POWER_STRUCTURE_RESULTS';

%% ========================================================================
% LOCALIZAR Y LEER zeross.bin
% ========================================================================

if ~isfile(zeroBinaryFile)
    [selectedFile,selectedPath] = uigetfile( ...
        {'*.bin','Binary doubles (*.bin)';'*.*','Todos los archivos'}, ...
        'Selecciona zeross.bin');

    if isequal(selectedFile,0)
        error(['No se encontro zeross.bin y no se selecciono ningun archivo. ' ...
            'Este es el unico fichero de datos necesario.']);
    end

    zeroBinaryFile = fullfile(selectedPath,selectedFile);
end

fileID = fopen(zeroBinaryFile,'rb');
if fileID == -1
    error('No se pudo abrir %s.',zeroBinaryFile);
end

cleanupObject = onCleanup(@() fclose(fileID));
gammaAll = fread(fileID,Inf,'double=>double');
clear cleanupObject;

gammaAll = gammaAll(:);
gammaAll = gammaAll(isfinite(gammaAll) & gammaAll>0);
```

```
if numel(gammaAll)<2
    error('El fichero no contiene suficientes ordenadas validas.');
end

if any(diff(gammaAll)<=0)
    error('Las ordenadas no estan estrictamente ordenadas.');
end

Nfull = numel(gammaAll);

fprintf('\n============================================================\n');
fprintf('ZETA PRIME-POWER STRUCTURE: ANALYTIC AUDIT\n');
fprintf('============================================================\n');
fprintf('Archivo de ceros       : %s\n',zeroBinaryFile);
fprintf('Ceros disponibles      : %d\n',Nfull);
fprintf('q                      : %.17g\n',qPhase);
fprintf('omega=log(q)           : %.17g\n',omega);
fprintf('stride                 : %d\n',stride);
fprintf('Qlist                  :');
fprintf(' %g',Qlist);
fprintf('\n');

%% ========================================================================
% CONSTRUIR S_q CON TODOS LOS CEROS; MUESTREAR DESPUES
% ========================================================================

carrierAll = exp(1i*omega*gammaAll);
SphaseAll = cumsum(carrierAll);

% Parte suave analitica, salvo constante aditiva:
%
% int exp(i*omega*T)/(2*pi)*log(T/(2*pi)) dT
%
% = [exp(i omega T)log(T/2pi)-Ei(i omega T)]/(2*pi*i*omega).
%
% MATLAB: Ei(z) = -expint(-z), salvo la constante de rama, que solo
% modifica una constante aditiva y se absorbe en los ajustes.
Ei_iomegaT_all = -expint(-1i*omega*gammaAll);
AsmoothAll = (carrierAll.*log(gammaAll/(2*pi))-Ei_iomegaT_all) ...
    /(2*pi*1i*omega);

indices = (1:stride:Nfull).';
gamma = gammaAll(indices);
carrier = carrierAll(indices);
Sphase = SphaseAll(indices);
Asmooth = AsmoothAll(indices);

gapBeforeAll = [NaN;diff(gammaAll)];
gapLocalMedianAll = movmedian(gapBeforeAll,gapMedianWindow, ...
    'omitmissing','Endpoints','shrink');
gapResidualAll = gapBeforeAll-gapLocalMedianAll;

gapBefore = gapBeforeAll(indices);
gapResidual = gapResidualAll(indices);
```

```
clear carrierAll SphaseAll AsmoothAll Ei_iomegaT_all;
clear gapBeforeAll gapLocalMedianAll gapResidualAll;

Y = Sphase-Asmooth;
N = numel(gamma);

fprintf('Puntos analizados       : %d\n',N);
fprintf('gamma inicial           : %.15g\n',gamma(1));
fprintf('gamma final             : %.15g\n',gamma(end));

%% ========================================================================
% GENERAR TODAS LAS POTENCIAS PRIMAS <= Qmax
% ========================================================================

[qVec,basePrimeVec,powerExponentVec,lambdaVec] = makePrimePowers(Qmax);
numTerms = numel(qVec);

ellVec = log(qVec);
deltaMinusVec = omega-ellVec;
deltaPlusVec = omega+ellVec;

resonantIndex = find(abs(deltaMinusVec)<=resonanceTolerance,1);
if ~isempty(resonantIndex)
    error(['qPhase es resonante con n=%d. Esta version no oculta la ' ...
        'singularidad: debe introducirse el termino resonante regularizado.'], ...
        qVec(resonantIndex));
end

cMinusVec = 1i*lambdaVec ./ ...
    (2*pi*sqrt(qVec).*deltaMinusVec);

cPlusVec = 1i*lambdaVec ./ ...
    (2*pi*sqrt(qVec).*deltaPlusVec);

%% ========================================================================
% TEST 2: IDENTIDADES ALGEBRAICAS DE LOS COEFICIENTES
% ========================================================================

sumIdentityRHS = 1i*lambdaVec.*omega ./ ...
    (pi*sqrt(qVec).*(omega^2-ellVec.^2));

differenceIdentityRHS = 1i*lambdaVec.*ellVec ./ ...
    (pi*sqrt(qVec).*(omega^2-ellVec.^2));

ratioIdentityRHS = (omega+ellVec)./(omega-ellVec);

sumIdentityError = abs((cMinusVec+cPlusVec)-sumIdentityRHS);
differenceIdentityError = abs((cMinusVec-cPlusVec)-differenceIdentityRHS);
ratioIdentityError = abs((cMinusVec./cPlusVec)-ratioIdentityRHS);

AQ = cumsum(cMinusVec+cPlusVec);
BQ = cumsum(cMinusVec-cPlusVec);
```

```
fprintf('\n============================================================\n');
fprintf('TEST 2: IDENTIDADES DE COEFICIENTES\n');
fprintf('============================================================\n');
fprintf('Potencias primas <=Qmax                  : %d\n',numTerms);
fprintf('max |(c^-+c^+)-formula|                 : %.6e\n', ...
    max(sumIdentityError));
fprintf('max |(c^--c^+)-formula|                 : %.6e\n', ...
    max(differenceIdentityError));
fprintf('max |c^-/c^+-formula|                   : %.6e\n', ...
    max(ratioIdentityError));

coefficientCutoffRows = cell(numel(Qlist),9);

for jj = 1:numel(Qlist)
    currentQ = Qlist(jj);
    lastIndex = find(qVec<=currentQ,1,'last');

    if isempty(lastIndex)
        currentA = 0;
        currentB = 0;
        numberTerms = 0;
    else
        currentA = AQ(lastIndex);
        currentB = BQ(lastIndex);
        numberTerms = lastIndex;
    end

    coefficientCutoffRows(jj,:) = { ...
        currentQ,numberTerms, ...
        real(currentA),imag(currentA),abs(currentA-0.5), ...
        real(currentB),imag(currentB),abs(currentB-0.5), ...
        abs(currentB+0.5)};
end

coefficientCutoffDisplay = cell2table(coefficientCutoffRows, ...
    'VariableNames',{ ...
    'Q','num_terms','Re_AQ','Im_AQ','abs_AQ_minus_half', ...
    'Re_BQ','Im_BQ','abs_BQ_minus_half','abs_BQ_plus_half'});

disp(coefficientCutoffDisplay);


%% ========================================================================
% AUDITORIA RADIAL DIRECTA: ORIGINAL, n=3, n=2+3
% ========================================================================

fprintf('\n============================================================\n');
fprintf('AUDITORIA RADIAL DIRECTA DE LA BIMODALIDAD\n');
fprintf('============================================================\n');

numRadialBins = 220;
radialSmoothWidth = 9;

index3 = find(qVec==3,1);
```

```
index23 = find(qVec<=3);

if isempty(index3)
    error('No se encontro n=3 entre las potencias primas.');
end

ell3 = ellVec(index3);
P3minus = cMinusVec(index3).*carrier.*exp(-1i*ell3*gamma);
P3plus = cPlusVec(index3).*carrier.*exp(1i*ell3*gamma);
P3 = P3minus+P3plus;

P23 = complex(zeros(N,1));
for auditIndex = index23(:).'
    ellAudit = ellVec(auditIndex);
    P23 = P23 ...
        +cMinusVec(auditIndex).*carrier.*exp(-1i*ellAudit*gamma) ...
        +cPlusVec(auditIndex).*carrier.*exp(1i*ellAudit*gamma);
end

[radialOriginal,radialCentersOriginal,radialDensityOriginal, ...
    radialPeaksOriginal] = radialDistributionSummary( ...
    Sphase,numRadialBins,radialSmoothWidth);

[radialAfter3,radialCentersAfter3,radialDensityAfter3, ...
    radialPeaksAfter3] = radialDistributionSummary( ...
    Sphase-P3,numRadialBins,radialSmoothWidth);

[radialAfter23,radialCentersAfter23,radialDensityAfter23, ...
    radialPeaksAfter23] = radialDistributionSummary( ...
    Sphase-P23,numRadialBins,radialSmoothWidth);

radialAuditTable = table( ...
    ["original";"after_n3";"after_n2_n3"], ...
    [mean(radialOriginal);mean(radialAfter3);mean(radialAfter23)], ...
    [std(radialOriginal);std(radialAfter3);std(radialAfter23)], ...
    [numel(radialPeaksOriginal);numel(radialPeaksAfter3); ...
        numel(radialPeaksAfter23)], ...
    [joinNumericVector(radialPeaksOriginal); ...
        joinNumericVector(radialPeaksAfter3); ...
        joinNumericVector(radialPeaksAfter23)], ...
    'VariableNames',{'case_name','mean_radius','std_radius', ...
    'num_detected_peaks','detected_peak_radii'});

disp(radialAuditTable);

fprintf('Coeficiente teorico total n=3:\n');
fprintf('  c3^- = %.12g %+.12gi\n',real(cMinusVec(index3)), ...
    imag(cMinusVec(index3)));
fprintf('  c3^+ = %.12g %+.12gi\n',real(cPlusVec(index3)), ...
    imag(cPlusVec(index3)));

figure('Name','Auditoria radial: original y sustracciones pequenas', ...
    'Color','w');
plot(radialCentersOriginal,radialDensityOriginal,'LineWidth',1.3);
```

```
hold on;
plot(radialCentersAfter3,radialDensityAfter3,'LineWidth',1.3);
plot(radialCentersAfter23,radialDensityAfter23,'LineWidth',1.3);
grid on;
xlabel('Radio respecto del centro medio');
ylabel('Densidad suavizada');
legend('Original','Despues de n=3','Despues de n=2,3', ...
    'Location','best');
title('Prueba directa de la bimodalidad radial');

%% ========================================================================
% PREPARAR MATRIZ DE GRAM POR BLOQUES LOGARITMICOS DE n
% ========================================================================

gramIndices = unique(round(linspace(1,N,min(N,maxGramPoints)))).';
numGramPoints = numel(gramIndices);

logEdges = linspace(log(2),log(Qmax)+100*eps(log(Qmax)), ...
    numGramBlocks+1);

gramBlockOfTerm = discretize(ellVec,logEdges);
gramBlockOfTerm(isnan(gramBlockOfTerm)) = numGramBlocks;

gramMinus = complex(zeros(numGramPoints,numGramBlocks,'single'));
gramPlus = complex(zeros(numGramPoints,numGramBlocks,'single'));

%% ========================================================================
% BUCLE PRINCIPAL: TESTS 1, 4, 5 Y PARTE DE 6
% ========================================================================

Pminus = complex(zeros(N,1));
Pplus = complex(zeros(N,1));

nextTerm = 1;
maxIncrementIdentityError = 0;
maxConstantCancellationIdentityError = 0;

summaryRows = cell(numel(Qlist),43);
positivityRows = cell(numel(Qlist),17);

blockRows = cell(numel(Qlist)*numHeightBlocks,16);
blockRow = 0;

[constantY,cY] = fitConstantAndCarrier(Y,carrier);

RbothAtQmax = [];
FAtQmax = [];
EAtQmax = [];
PminusAtQmax = [];
PplusAtQmax = [];
cBothAtQmax = NaN;

for qi = 1:numel(Qlist)
```

```
Q = Qlist(qi);

while nextTerm<=numTerms && qVec(nextTerm)<=Q

    ell = ellVec(nextTerm);
    z = exp(1i*ell*gamma);

    termMinus = cMinusVec(nextTerm).*carrier.*conj(z);
    termPlus = cPlusVec(nextTerm).*carrier.*z;

    oldCombinedResidual = Y-Pminus-Pplus;

    Pminus = Pminus+termMinus;
    Pplus = Pplus+termPlus;

    newCombinedResidual = Y-Pminus-Pplus;

    incrementError = max(abs( ...
        (newCombinedResidual-oldCombinedResidual) ...
        +termMinus+termPlus));

    maxIncrementIdentityError = max( ...
        maxIncrementIdentityError,incrementError);

    gramBlock = gramBlockOfTerm(nextTerm);

    gramMinus(:,gramBlock) = gramMinus(:,gramBlock) ...
        +single(termMinus(gramIndices));

    gramPlus(:,gramBlock) = gramPlus(:,gramBlock) ...
        +single(termPlus(gramIndices));

    nextTerm = nextTerm+1;
end

numTermsUsed = nextTerm-1;

% Cuatro residuos pedidos.
RminusOnlyRaw = Y-Pminus;
RplusOnlyRaw = Y-Pplus;
RbothRaw = Y-Pminus-Pplus;

% CORRECCION: la parte suave esta definida salvo constante aditiva.
% Se ajustan [constante + portadora] conjuntamente en cada residuo.
[constantMinus,cMinusResidual] = ...
    fitConstantAndCarrier(RminusOnlyRaw,carrier);

[constantPlus,cPlusResidual] = ...
    fitConstantAndCarrier(RplusOnlyRaw,carrier);

[jointConstant,cBothResidual] = ...
    fitConstantAndCarrier(RbothRaw,carrier);

RminusOnly = RminusOnlyRaw-constantMinus;
```

```
RplusOnly = RplusOnlyRaw-constantPlus;
Rboth = RbothRaw-jointConstant;

E = Rboth-cBothResidual*carrier;
F = conj(carrier).*Rboth;
Fc = F-cBothResidual;

[constantPminus,projectionMinus] = ...
    fitConstantAndCarrier(Pminus,carrier);

[constantPplus,projectionPlus] = ...
    fitConstantAndCarrier(Pplus,carrier);

carrierCancellationIdentityError = abs( ...
    cBothResidual-(cY-projectionMinus-projectionPlus));

constantCancellationIdentityError = abs( ...
    jointConstant-(constantY-constantPminus-constantPplus));

maxConstantCancellationIdentityError = max( ...
    maxConstantCancellationIdentityError, ...
    constantCancellationIdentityError);

% TEST 1: energias y termino cruzado.
energyMinus = mean(abs(Pminus).^2);
energyPlus = mean(abs(Pplus).^2);
crossMinusPlus = 2*real(mean(conj(Pminus).*Pplus));
energyCombined = mean(abs(Pminus+Pplus).^2);

crossEnergyIdentityError = abs( ...
    energyCombined-(energyMinus+energyPlus+crossMinusPlus));

% Forma centrada.
PminusCentered = Pminus-mean(Pminus);
PplusCentered = Pplus-mean(Pplus);

centeredEnergyMinus = mean(abs(PminusCentered).^2);
centeredEnergyPlus = mean(abs(PplusCentered).^2);

centeredCross = 2*real(mean( ...
    conj(PminusCentered).*PplusCentered));

centeredEnergyCombined = mean(abs( ...
    PminusCentered+PplusCentered).^2);

% TEST 5: ecuacion funcional discreta y diferencial.
dgamma = diff(gamma);

derivativeDefect = diff(Rboth)./dgamma ...
    -1i*omega*0.5*(Rboth(2:end)+Rboth(1:end-1));

discreteDefect = Rboth(2:end) ...
    -exp(1i*omega*dgamma).*Rboth(1:end-1);
```

```
    Fvariance = mean(abs(Fc).^2);
    FmaxDeviation = max(abs(Fc));
    FmeanAbsIncrement = mean(abs(diff(F)));
    FtotalVariation = sum(abs(diff(F)));
    Fanisotropy = complexAnisotropy(Fc,minimumVariance);

    corrGapReal = safeCorrelation(real(Fc),gapBefore);
    corrGapImag = safeCorrelation(imag(Fc),gapBefore);
    corrGapAbs = safeCorrelation(abs(Fc),gapBefore);

    corrGapResidualReal = safeCorrelation(real(Fc),gapResidual);
    corrGapResidualImag = safeCorrelation(imag(Fc),gapResidual);
    corrGapResidualAbs = safeCorrelation(abs(Fc),gapResidual);

    % Bloques de altura.
    heightEdges = round(linspace(1,N+1,numHeightBlocks+1));
    blockMeans = complex(NaN(numHeightBlocks,1));

    for hb = 1:numHeightBlocks

        idx = heightEdges(hb):(heightEdges(hb+1)-1);

        Fb = F(idx);
        Rb = Rboth(idx);
        blockMean = mean(Fb);
        blockMeans(hb) = blockMean;
        Fbc = Fb-blockMean;

        if numel(idx)>=2
            dgb = diff(gamma(idx));

            discreteBlock = Rb(2:end) ...
                -exp(1i*omega*dgb).*Rb(1:end-1);

            derivativeBlock = diff(Rb)./dgb ...
                -1i*omega*0.5*(Rb(2:end)+Rb(1:end-1));

            meanAbsDeltaBlock = mean(abs(diff(Fb)));

            variationPerHeight = sum(abs(diff(Fb))) ...
                /max(gamma(idx(end))-gamma(idx(1)),eps);

            discreteBlockRMS = sqrt(mean(abs(discreteBlock).^2));
            derivativeBlockRMS = sqrt(mean(abs(derivativeBlock).^2));
        else
            meanAbsDeltaBlock = NaN;
            variationPerHeight = NaN;
            discreteBlockRMS = NaN;
            derivativeBlockRMS = NaN;
        end

        blockRow = blockRow+1;

        blockRows(blockRow,:) = { ...
```

```
        Q,hb,indices(idx(1)),indices(idx(end)), ...
        gamma(idx(1)),gamma(idx(end)), ...
        real(blockMean),imag(blockMean),mean(abs(Fbc).^2), ...
        max(abs(Fb-cBothResidual)),max(abs(Fbc)), ...
        meanAbsDeltaBlock,variationPerHeight, ...
        complexAnisotropy(Fbc,minimumVariance), ...
        discreteBlockRMS,derivativeBlockRMS};
end

FblockMeanStd = sqrt(mean(abs(blockMeans-mean(blockMeans)).^2));

% TEST 4: positividad elemental/Gram.
normE2 = mean(abs(E).^2);
normCarrier2 = mean(abs(carrier).^2);

innerCarrierE = mean(conj(carrier).*E);
innerEMinus = mean(conj(E).*Pminus);
innerEPlus = mean(conj(E).*Pplus);
innerMinusPlus = mean(conj(Pminus).*Pplus);

carrierSchur = mean(abs(Rboth).^2) ...
    -abs(mean(conj(carrier).*Rboth))^2/normCarrier2;

cauchyDetEMinus = normE2*energyMinus-abs(innerEMinus)^2;
cauchyDetEPlus = normE2*energyPlus-abs(innerEPlus)^2;
cauchyDetMinusPlus = energyMinus*energyPlus ...
    -abs(innerMinusPlus)^2;

QEplusMinus = mean(abs(E+Pminus).^2);
QEminusMinus = mean(abs(E-Pminus).^2);

polarizationErrorMinus = abs( ...
    0.5*(QEplusMinus+QEminusMinus)-normE2-energyMinus);

QEplusPlus = mean(abs(E+Pplus).^2);
QEminusPlus = mean(abs(E-Pplus).^2);

polarizationErrorPlus = abs( ...
    0.5*(QEplusPlus+QEminusPlus)-normE2-energyPlus);

positivityRows(qi,:) = { ...
    Q,normE2,carrierSchur, ...
    abs(innerCarrierE),real(innerCarrierE),imag(innerCarrierE), ...
    real(innerEMinus),real(innerEPlus), ...
    cauchyDetEMinus,cauchyDetEPlus,cauchyDetMinusPlus, ...
    QEplusMinus,QEminusMinus,polarizationErrorMinus, ...
    QEplusPlus,QEminusPlus,polarizationErrorPlus};

if numTermsUsed>0
    currentA = AQ(numTermsUsed);
    currentB = BQ(numTermsUsed);
else
    currentA = 0;
    currentB = 0;
```

```
    end

    summaryRows(qi,:) = { ...
        Q,numTermsUsed, ...
        real(cMinusResidual),imag(cMinusResidual), ...
        real(cPlusResidual),imag(cPlusResidual), ...
        real(cBothResidual),imag(cBothResidual),abs(cBothResidual), ...
        abs(cBothResidual-0.5), ...
        real(jointConstant),imag(jointConstant), ...
        real(cBothResidual),imag(cBothResidual),abs(cBothResidual-0.5), ...
        real(projectionMinus),imag(projectionMinus), ...
        real(projectionPlus),imag(projectionPlus), ...
        carrierCancellationIdentityError, ...
        sqrt(mean(abs(RminusOnly).^2)), ...
        sqrt(mean(abs(RplusOnly).^2)), ...
        sqrt(mean(abs(Rboth).^2)),sqrt(normE2), ...
        energyMinus,energyPlus,crossMinusPlus,energyCombined, ...
        crossEnergyIdentityError, ...
        centeredEnergyMinus,centeredEnergyPlus,centeredCross, ...
        centeredEnergyCombined, ...
        Fvariance,FmaxDeviation,FmeanAbsIncrement,FtotalVariation, ...
        FblockMeanStd,Fanisotropy, ...
        sqrt(mean(abs(discreteDefect).^2)), ...
        sqrt(mean(abs(derivativeDefect).^2)), ...
        real(currentA),imag(currentA)};

    fprintf(['Q=%-6g terms=%-5d  c^-res=%.7f%+.7fi  ' ...
        'c^+res=%.7f%+.7fi  c^both=%.7f%+.7fi  ' ...
        'Var(F)=%.4e  RMS(E)=%.4e\n'], ...
        Q,numTermsUsed, ...
        real(cMinusResidual),imag(cMinusResidual), ...
        real(cPlusResidual),imag(cPlusResidual), ...
        real(cBothResidual),imag(cBothResidual), ...
        Fvariance,sqrt(normE2));

    if Q==Qmax
        RbothAtQmax = Rboth;
        FAtQmax = F;
        EAtQmax = E;
        PminusAtQmax = Pminus;
        PplusAtQmax = Pplus;
        cBothAtQmax = cBothResidual;
    end
end

summaryVariableNames = { ...
    'Q','num_prime_power_terms', ...
    'c_after_minus_real','c_after_minus_imag', ...
    'c_after_plus_real','c_after_plus_imag', ...
    'c_after_both_real','c_after_both_imag','c_after_both_abs', ...
    'distance_c_after_both_to_half', ...
    'joint_constant_real','joint_constant_imag', ...
    'joint_carrier_real','joint_carrier_imag', ...
    'distance_joint_carrier_to_half', ...
```

```
    'projection_Pminus_real','projection_Pminus_imag', ...
    'projection_Pplus_real','projection_Pplus_imag', ...
    'carrier_cancellation_identity_error', ...
    'rms_centered_Y_minus_Pminus','rms_centered_Y_minus_Pplus', ...
    'rms_centered_Y_minus_both', ...
    'rms_after_both_fitted_carrier', ...
    'energy_Pminus','energy_Pplus', ...
    'cross_term_2Re_inner_Pminus_Pplus', ...
    'energy_Pminus_plus_Pplus','cross_energy_identity_error', ...
    'centered_energy_Pminus','centered_energy_Pplus', ...
    'centered_cross_term','centered_energy_combined', ...
    'F_variance','F_max_abs_deviation_from_mean', ...
    'F_mean_abs_increment','F_total_variation', ...
    'F_block_mean_std','F_anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms', ...
    'A_Q_real','A_Q_imag'};

summaryTable = cell2table(summaryRows, ...
    'VariableNames',summaryVariableNames);

positivityVariableNames = { ...
    'Q','norm_E2','carrier_schur_complement', ...
    'carrier_projection_abs','carrier_projection_real', ...
    'carrier_projection_imag','Re_inner_E_Pminus', ...
    'Re_inner_E_Pplus','cauchy_det_E_Pminus', ...
    'cauchy_det_E_Pplus','cauchy_det_Pminus_Pplus', ...
    'Q_E_plus_Pminus','Q_E_minus_Pminus', ...
    'polarization_error_minus','Q_E_plus_Pplus', ...
    'Q_E_minus_Pplus','polarization_error_plus'};

positivityTable = cell2table(positivityRows, ...
    'VariableNames',positivityVariableNames);

blockVariableNames = { ...
    'Q','block','k_start','k_end','gamma_start','gamma_end', ...
    'F_mean_real','F_mean_imag','F_variance', ...
    'sup_abs_F_minus_global_c','sup_abs_F_minus_block_c', ...
    'mean_abs_delta_F','variation_per_height','anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms'};

blockTable = cell2table(blockRows(1:blockRow,:), ...
    'VariableNames',blockVariableNames);

%% ========================================================================
% DISPLAY PRINCIPAL: TEST 1 Y TEST 5
% ========================================================================

fprintf('\n============================================================\n');
fprintf('TEST 1: RAMAS; TEST 5: ECUACION FUNCIONAL\n');
fprintf('============================================================\n');

mainDisplayVariables = { ...
    'Q','num_prime_power_terms', ...
    'c_after_minus_real','c_after_minus_imag', ...
```

```
    'c_after_plus_real','c_after_plus_imag', ...
    'c_after_both_real','c_after_both_imag', ...
    'distance_c_after_both_to_half', ...
    'rms_centered_Y_minus_Pminus','rms_centered_Y_minus_Pplus', ...
    'rms_centered_Y_minus_both', ...
    'rms_after_both_fitted_carrier', ...
    'energy_Pminus','energy_Pplus', ...
    'cross_term_2Re_inner_Pminus_Pplus', ...
    'F_variance','F_max_abs_deviation_from_mean', ...
    'F_block_mean_std','F_anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms'};

disp(summaryTable(:,mainDisplayVariables));

fprintf('max error R_j-R_{j-1}=-P_j                  : %.6e\n', ...
    maxIncrementIdentityError);

fprintf('max error ley de energia de ramas           : %.6e\n', ...
    max(summaryTable.cross_energy_identity_error));

fprintf('max error identidad de proyeccion portadora : %.6e\n', ...
    max(summaryTable.carrier_cancellation_identity_error));
fprintf('max error identidad de constantes            : %.6e\n', ...
    maxConstantCancellationIdentityError);

%% ========================================================================
% TEST 3: TELESCOPADO DENTRO DE CADA PRIMO p
% ========================================================================

[groupSummaryTable,groupPartialTable] = primePowerTelescopingTests( ...
    qVec,basePrimeVec,powerExponentVec,cMinusVec,cPlusVec, ...
    gamma,carrier);

fprintf('\n============================================================\n');
fprintf('TEST 3: TELESCOPADO EN p,p^2,...,p^R\n');
fprintf('============================================================\n');

if isempty(groupSummaryTable)
    fprintf('No hay primos con al menos dos potencias <=Qmax.\n');
else
    [~,orderTelescoping] = sort( ...
        groupSummaryTable.boundary_fit_relative_rms_combined,'ascend');

    numDisplay = min(20,height(groupSummaryTable));

    telescopingDisplayVariables = { ...
        'base_prime','num_powers','largest_power', ...
        'combined_energy_ratio_sum_over_individual', ...
        'combined_rms_over_sum_term_rms', ...
        'boundary_fit_relative_rms_combined', ...
        'boundary_fit_relative_rms_minus', ...
        'boundary_fit_relative_rms_plus', ...
        'boundary_alpha_first_real','boundary_alpha_first_imag', ...
        'boundary_alpha_last_real','boundary_alpha_last_imag', ...
```

```
        'tail_primitive_start_abs_minus', ...
        'tail_primitive_start_abs_plus'};

    disp(groupSummaryTable( ...
        orderTelescoping(1:numDisplay), ...
        telescopingDisplayVariables));

    fprintf(['boundary_fit_relative_rms_combined cercano a cero seria ' ...
        'evidencia de colapso a los dos bordes del bloque p^r.\n']);
end

%% ========================================================================
% TEST 4: FUNCIONALES POSITIVOS Y MATRICES DE GRAM
% ========================================================================

Gminus = double(gramMinus);
Gplus = double(gramPlus);

GcombinedCentered = Gminus+Gplus;
GcombinedCentered = GcombinedCentered-mean(GcombinedCentered,1);

GbranchCentered = [ ...
    Gminus-mean(Gminus,1), ...
    Gplus-mean(Gplus,1)];

gramCombined = (GcombinedCentered'*GcombinedCentered)/numGramPoints;
gramBranch = (GbranchCentered'*GbranchCentered)/numGramPoints;

gramCombined = 0.5*(gramCombined+gramCombined');
gramBranch = 0.5*(gramBranch+gramBranch');

eigCombined = sort(real(eig(gramCombined)));
eigBranch = sort(real(eig(gramBranch)));

fprintf('\n============================================================\n');
fprintf('TEST 4: POSITIVIDAD Y GRAM\n');
fprintf('============================================================\n');

positivityDisplayVariables = { ...
    'Q','norm_E2','carrier_schur_complement', ...
    'carrier_projection_abs','Re_inner_E_Pminus', ...
    'Re_inner_E_Pplus','cauchy_det_E_Pminus', ...
    'cauchy_det_E_Pplus','cauchy_det_Pminus_Pplus', ...
    'polarization_error_minus','polarization_error_plus'};

disp(positivityTable(:,positivityDisplayVariables));

fprintf('Gram combinada: min=%.6e, max=%.6e, negativos<-1e-10=%d\n', ...
    min(eigCombined),max(eigCombined),nnz(eigCombined<-1e-10));

fprintf('Gram por ramas: min=%.6e, max=%.6e, negativos<-1e-10=%d\n', ...
    min(eigBranch),max(eigBranch),nnz(eigBranch<-1e-10));

%% ========================================================================
```

```
% TEST 5 POR BLOQUES PARA Qmax
% ========================================================================

blockQmax = blockTable(blockTable.Q==Qmax,:);

fprintf('\n============================================================\n');
fprintf('TEST 5: F_Q POR BLOQUES DE ALTURA, Q=%g\n',Qmax);
fprintf('============================================================\n');

blockDisplayVariables = { ...
    'block','gamma_start','gamma_end', ...
    'F_mean_real','F_mean_imag','F_variance', ...
    'sup_abs_F_minus_global_c','sup_abs_F_minus_block_c', ...
    'mean_abs_delta_F','variation_per_height','anisotropy', ...
    'carrier_discrete_defect_rms','derivative_defect_rms'};

disp(blockQmax(:,blockDisplayVariables));

fprintf('Rango Re media por bloque : [%.12g, %.12g]\n', ...
    min(blockQmax.F_mean_real),max(blockQmax.F_mean_real));

fprintf('Rango Im media por bloque : [%.12g, %.12g]\n', ...
    min(blockQmax.F_mean_imag),max(blockQmax.F_mean_imag));

fprintf('Max sup global por bloque : %.6e\n', ...
    max(blockQmax.sup_abs_F_minus_global_c));

fprintf('Max sup local por bloque  : %.6e\n', ...
    max(blockQmax.sup_abs_F_minus_block_c));

%% ========================================================================
% TEST 6: FRECUENCIAS OMITIDAS
% ========================================================================

spectralIndices = unique(round(linspace(1,N, ...
    min(N,maxSpectralPoints)))).';

Ts = gamma(spectralIndices);
Fs = FAtQmax(spectralIndices)-cBothAtQmax;
Es = EAtQmax(spectralIndices);

testStart = Qmax+1;
testEnd = Qmax+omittedTestWidth;
nCandidates = (testStart:testEnd).';

[qTestAll,pTestAll,rTestAll,lambdaTestAll] = makePrimePowers(testEnd);

[isPrimePowerCandidate,locPP] = ismember(nCandidates,qTestAll);

candidateLambda = zeros(size(nCandidates));
candidateBasePrime = zeros(size(nCandidates));
candidateExponent = zeros(size(nCandidates));

candidateLambda(isPrimePowerCandidate) = ...
```

```
    lambdaTestAll(locPP(isPrimePowerCandidate));

candidateBasePrime(isPrimePowerCandidate) = ...
    pTestAll(locPP(isPrimePowerCandidate));

candidateExponent(isPrimePowerCandidate) = ...
    rTestAll(locPP(isPrimePowerCandidate));

omittedRows = cell(numel(nCandidates),19);

for jj = 1:numel(nCandidates)

    n = nCandidates(jj);
    ell = log(n);

    basisMinusF = exp(-1i*ell*Ts);
    basisPlusF = exp(1i*ell*Ts);

    fitMinusF = fitCarrier(Fs,basisMinusF);
    fitPlusF = fitCarrier(Fs,basisPlusF);

    basisMinusR = exp(1i*(omega-ell)*Ts);
    basisPlusR = exp(1i*(omega+ell)*Ts);

    fitMinusR = fitCarrier(Es,basisMinusR);
    fitPlusR = fitCarrier(Es,basisPlusR);

    theoryMinus = 0;
    theoryPlus = 0;

    if isPrimePowerCandidate(jj)
        lam = candidateLambda(jj);

        theoryMinus = 1i*lam / ...
            (2*pi*sqrt(n)*(omega-ell));

        theoryPlus = 1i*lam / ...
            (2*pi*sqrt(n)*(omega+ell));
    end

    omittedRows(jj,:) = { ...
        n,isPrimePowerCandidate(jj),candidateBasePrime(jj), ...
        candidateExponent(jj),candidateLambda(jj), ...
        real(fitMinusF),imag(fitMinusF),abs(fitMinusF), ...
        real(fitPlusF),imag(fitPlusF),abs(fitPlusF), ...
        real(fitMinusR),imag(fitMinusR), ...
        real(fitPlusR),imag(fitPlusR), ...
        real(theoryMinus),imag(theoryMinus), ...
        real(theoryPlus),imag(theoryPlus)};
end

omittedVariableNames = { ...
    'n','is_prime_power','base_prime','power_exponent','Lambda_n', ...
    'fit_F_minus_real','fit_F_minus_imag','fit_F_minus_abs', ...
```

```
    'fit_F_plus_real','fit_F_plus_imag','fit_F_plus_abs', ...
    'fit_R_minus_real','fit_R_minus_imag', ...
    'fit_R_plus_real','fit_R_plus_imag', ...
    'theory_minus_real','theory_minus_imag', ...
    'theory_plus_real','theory_plus_imag'};

omittedTable = cell2table(omittedRows, ...
    'VariableNames',omittedVariableNames);

ppMask = omittedTable.is_prime_power;
nullMask = ~ppMask;

medianPPMinus = median(omittedTable.fit_F_minus_abs(ppMask));
medianNullMinus = median(omittedTable.fit_F_minus_abs(nullMask));

medianPPPlus = median(omittedTable.fit_F_plus_abs(ppMask));
medianNullPlus = median(omittedTable.fit_F_plus_abs(nullMask));

omittedPPMask = qTestAll>Qmax & qTestAll<=testEnd;
omittedPP = qTestAll(omittedPPMask);
omittedPPLambda = lambdaTestAll(omittedPPMask);

Xomitted = complex(zeros(numel(Ts),2*numel(omittedPP)));

for jj = 1:numel(omittedPP)
    ell = log(omittedPP(jj));
    Xomitted(:,jj) = exp(-1i*ell*Ts);
    Xomitted(:,numel(omittedPP)+jj) = exp(1i*ell*Ts);
end

if isempty(Xomitted)
    fittedOmittedCoefficients = [];
    FsAfterOmittedFit = Fs;
    omittedExplainedFraction = NaN;
    fittedTheoryCorrelation = NaN;
    relativeTheoryCoefficientError = NaN;
else
    fittedOmittedCoefficients = Xomitted\Fs;
    FsAfterOmittedFit = Fs-Xomitted*fittedOmittedCoefficients;

    omittedExplainedFraction = 1 ...
        -sum(abs(FsAfterOmittedFit).^2)/sum(abs(Fs).^2);

    theoryOmittedCoefficients = complex(zeros(2*numel(omittedPP),1));

    for jj = 1:numel(omittedPP)

        n = omittedPP(jj);
        ell = log(n);
        lam = omittedPPLambda(jj);

        theoryOmittedCoefficients(jj) = 1i*lam / ...
            (2*pi*sqrt(n)*(omega-ell));
```

```
        theoryOmittedCoefficients(numel(omittedPP)+jj) = 1i*lam / ...
            (2*pi*sqrt(n)*(omega+ell));
    end

    fittedTheoryCorrelation = complexVectorCorrelation( ...
        fittedOmittedCoefficients,theoryOmittedCoefficients);

    relativeTheoryCoefficientError = norm( ...
        fittedOmittedCoefficients-theoryOmittedCoefficients) ...
        /max(norm(theoryOmittedCoefficients),eps);
end

nuMax = log(testEnd);
nuGrid = linspace(-nuMax,nuMax,numFrequencyGridPoints).';
frequencyAmplitude = zeros(size(nuGrid));

for jj = 1:numel(nuGrid)
    basis = exp(1i*nuGrid(jj)*Ts);
    frequencyAmplitude(jj) = abs(fitCarrier(Fs,basis));
end

[sortedAmplitude,sortOrder] = sort(frequencyAmplitude,'descend');
topCount = min(30,numel(sortOrder));

topFrequencyTable = table( ...
    (1:topCount).', ...
    nuGrid(sortOrder(1:topCount)), ...
    sortedAmplitude(1:topCount), ...
    'VariableNames',{'rank','demodulated_frequency','fit_amplitude'});

uniquenessTable = table( ...
    Qmax,numel(omittedPP), ...
    medianPPMinus,medianNullMinus,medianPPMinus/medianNullMinus, ...
    medianPPPlus,medianNullPlus,medianPPPlus/medianNullPlus, ...
    omittedExplainedFraction, ...
    real(fittedTheoryCorrelation),imag(fittedTheoryCorrelation), ...
    abs(fittedTheoryCorrelation),relativeTheoryCoefficientError, ...
    mean(abs(Fs).^2),mean(abs(FsAfterOmittedFit).^2), ...
    complexAnisotropy(Fs,minimumVariance), ...
    safeCorrelation(real(Fs),gapBefore(spectralIndices)), ...
    safeCorrelation(imag(Fs),gapBefore(spectralIndices)), ...
    safeCorrelation(abs(Fs),gapBefore(spectralIndices)), ...
    'VariableNames',{ ...
    'Qmax','num_omitted_prime_powers_joint_fit', ...
    'median_minus_amplitude_prime_power', ...
    'median_minus_amplitude_null', ...
    'minus_prime_power_to_null_ratio', ...
    'median_plus_amplitude_prime_power', ...
    'median_plus_amplitude_null', ...
    'plus_prime_power_to_null_ratio', ...
    'joint_omitted_explained_fraction', ...
    'fit_theory_correlation_real','fit_theory_correlation_imag', ...
    'fit_theory_correlation_abs','relative_theory_coefficient_error', ...
    'F_energy_before_omitted_fit','F_energy_after_omitted_fit', ...
```

```
    'F_anisotropy','corr_gap_realF','corr_gap_imagF','corr_gap_absF'});

fprintf('\n============================================================\n');
fprintf('TEST 6: UNICIDAD Y FRECUENCIAS OMITIDAS\n');
fprintf('============================================================\n');

disp(uniquenessTable);

omittedPrimePowerOnly = omittedTable(omittedTable.is_prime_power,:);

if ~isempty(omittedPrimePowerOnly)

    omittedPrimePowerOnly.max_branch_fit_abs = max( ...
        omittedPrimePowerOnly.fit_F_minus_abs, ...
        omittedPrimePowerOnly.fit_F_plus_abs);

    omittedPrimePowerOnly = sortrows( ...
        omittedPrimePowerOnly,'max_branch_fit_abs','descend');

    numDisplay = min(20,height(omittedPrimePowerOnly));

    fprintf('\nPotencias primas omitidas con mayor amplitud:\n');

    disp(omittedPrimePowerOnly(1:numDisplay,{ ...
        'n','base_prime','power_exponent','Lambda_n', ...
        'fit_F_minus_real','fit_F_minus_imag','fit_F_minus_abs', ...
        'fit_F_plus_real','fit_F_plus_imag','fit_F_plus_abs', ...
        'theory_minus_real','theory_minus_imag', ...
        'theory_plus_real','theory_plus_imag', ...
        'max_branch_fit_abs'}));
end

fprintf('\nFrecuencias no nulas dominantes de F_Q-mean(F_Q):\n');
disp(topFrequencyTable);

fprintf('\nDIAGNOSTICO FINAL PARA Q=%g\n',Qmax);
fprintf('mean(F_Q)                          = %.12g %+.12gi\n', ...
    real(cBothAtQmax),imag(cBothAtQmax));
fprintf('|mean(F_Q)-1/2|                  = %.6e\n', ...
    abs(cBothAtQmax-0.5));
fprintf('Var(F_Q)                           = %.6e\n', ...
    mean(abs(FAtQmax-cBothAtQmax).^2));
fprintf('sup |F_Q-mean(F_Q)|               = %.6e\n', ...
    max(abs(FAtQmax-cBothAtQmax)));
fprintf('RMS E_Q                            = %.6e\n', ...
    sqrt(mean(abs(EAtQmax).^2)));
fprintf('razon omitidas potencia/null (-) = %.6f\n', ...
    medianPPMinus/medianNullMinus);
fprintf('razon omitidas potencia/null (+) = %.6f\n', ...
    medianPPPlus/medianNullPlus);
fprintf('fraccion explicada ajuste omitido= %.6f\n', ...
    omittedExplainedFraction);

%% ========================================================================
```

```
% FIGURAS: NO SE CIERRA NINGUNA
% ========================================================================

figure('Name','Coeficientes de portadora por rama','Color','w');
tiledlayout(2,1,'TileSpacing','compact','Padding','compact');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_minus_real,'-o','LineWidth',1.2);
hold on;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_plus_real,'-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_both_real,'-o','LineWidth',1.2);
yline(0.5,':');
grid on;
xlabel('Q');
ylabel('Parte real');
legend('Y-P^-','Y-P^+','Y-P^--P^+','1/2','Location','best');
title('Coeficiente ajustado de exp(i omega T)');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_minus_imag,'-o','LineWidth',1.2);
hold on;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_plus_imag,'-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.c_after_both_imag,'-o','LineWidth',1.2);
yline(0,':');
grid on;
xlabel('Q');
ylabel('Parte imaginaria');
legend('Y-P^-','Y-P^+','Y-P^--P^+','0','Location','best');

figure('Name','Energia y termino cruzado de ramas','Color','w');
semilogx(max(summaryTable.Q,1), ...
    summaryTable.energy_Pminus,'-o','LineWidth',1.2);
hold on;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.energy_Pplus,'-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.cross_term_2Re_inner_Pminus_Pplus, ...
    '-o','LineWidth',1.2);
semilogx(max(summaryTable.Q,1), ...
    summaryTable.energy_Pminus_plus_Pplus,'-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('Energia');
legend('||P^-||^2','||P^+||^2','2Re<P^-,P^+>', ...
    '||P^-+P^+||^2','Location','best');
title('Cancelacion entre ramas');

figure('Name','Ecuacion funcional del residuo','Color','w');
```

```
tiledlayout(2,2,'TileSpacing','compact','Padding','compact');

nexttile;
semilogx(max(summaryTable.Q,1),summaryTable.F_variance, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('Var(F_Q)');
title('Dependencia residual de T');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.distance_c_after_both_to_half, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('|mean(F_Q)-1/2|');
title('Distancia del coeficiente a 1/2');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.carrier_discrete_defect_rms, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('RMS defecto discreto');
title('R_{k+1}-e^{i omega Delta T}R_k');

nexttile;
semilogx(max(summaryTable.Q,1), ...
    summaryTable.F_block_mean_std, ...
    '-o','LineWidth',1.2);
grid on;
xlabel('Q');
ylabel('STD de medias por bloque');
title('Deriva por altura');

figure('Name','Sumas parciales A_Q y B_Q','Color','w');
tiledlayout(2,1,'TileSpacing','compact','Padding','compact');

nexttile;
plot(qVec,real(AQ),'LineWidth',1);
hold on;
plot(qVec,imag(AQ),'LineWidth',1);
yline(0.5,':');
yline(-0.5,':');
set(gca,'XScale','log');
grid on;
xlabel('Q');
ylabel('A_Q');
legend('Re A_Q','Im A_Q','1/2','-1/2','Location','best');

nexttile;
plot(qVec,real(BQ),'LineWidth',1);
```

```
hold on;
plot(qVec,imag(BQ),'LineWidth',1);
yline(0.5,':');
yline(-0.5,':');
set(gca,'XScale','log');
grid on;
xlabel('Q');
ylabel('B_Q');
legend('Re B_Q','Im B_Q','1/2','-1/2','Location','best');

blockSupMatrix = reshape( ...
    blockTable.sup_abs_F_minus_global_c, ...
    numHeightBlocks,numel(Qlist)).';

figure('Name','Desviacion suprema por bloques','Color','w');
imagesc(1:numHeightBlocks,1:numel(Qlist),log10(blockSupMatrix));
colorbar;
yticks(1:numel(Qlist));
yticklabels(string(Qlist));
xlabel('Bloque de altura');
ylabel('Q');
title('log10 sup_{T en I_j}|F_Q(T)-c_Q|');

figure('Name','Eigenvalores de Gram','Color','w');
tiledlayout(1,2,'TileSpacing','compact','Padding','compact');

nexttile;
semilogy(max(eigCombined,eps),'-o');
grid on;
xlabel('Indice');
ylabel('Eigenvalor');
title('Gram combinada');

nexttile;
semilogy(max(eigBranch,eps),'-o');
grid on;
xlabel('Indice');
ylabel('Eigenvalor');
title('Gram por ramas');

figure('Name','Frecuencias omitidas','Color','w');
semilogy(omittedTable.n,omittedTable.fit_F_minus_abs,'.');
hold on;
semilogy(omittedTable.n(ppMask), ...
    omittedTable.fit_F_minus_abs(ppMask),'o');
semilogy(omittedTable.n,omittedTable.fit_F_plus_abs,'.');
semilogy(omittedTable.n(ppMask), ...
    omittedTable.fit_F_plus_abs(ppMask),'s');
grid on;
xlabel('n');
ylabel('Amplitud ajustada');
legend('Rama - todos','Rama - potencias primas', ...
    'Rama + todos','Rama + potencias primas','Location','best');
```

```
figure('Name','Espectro no nulo de F_Q','Color','w');
plot(nuGrid,frequencyAmplitude,'LineWidth',1);
grid on;
xlabel('nu en exp(i nu T)');
ylabel('Amplitud');
title(sprintf('F_Q-mean(F_Q), Q=%g',Qmax));

%% ========================================================================
% GUARDADO OPCIONAL; NUNCA ES NECESARIO PARA EJECUTAR
% ========================================================================

if saveResults

    if ~isfolder(outputDirectory)
        mkdir(outputDirectory);
    end

    writetable(summaryTable, ...
        fullfile(outputDirectory,'SIX_TESTS_SUMMARY.csv'));

    writetable(coefficientCutoffDisplay, ...
        fullfile(outputDirectory,'COEFFICIENT_CUTOFF_SUMMARY.csv'));

    writetable(groupSummaryTable, ...
        fullfile(outputDirectory,'TELESCOPING_SUMMARY.csv'));

    writetable(groupPartialTable, ...
        fullfile(outputDirectory,'TELESCOPING_PARTIALS.csv'));

    writetable(positivityTable, ...
        fullfile(outputDirectory,'POSITIVITY_FUNCTIONALS.csv'));

    writetable(blockTable, ...
        fullfile(outputDirectory,'BLOCKWISE_FUNCTIONAL_EQUATION.csv'));

    writetable(omittedTable, ...
        fullfile(outputDirectory,'OMITTED_FREQUENCIES.csv'));

    writetable(uniquenessTable, ...
        fullfile(outputDirectory,'UNIQUENESS_SUMMARY.csv'));

    save(fullfile(outputDirectory,'ALL_RESULTS.mat'), ...
        'summaryTable','coefficientCutoffDisplay','radialAuditTable', ...
        'groupSummaryTable','groupPartialTable', ...
        'positivityTable','blockTable','omittedTable', ...
        'uniquenessTable','topFrequencyTable', ...
        'gramCombined','gramBranch','eigCombined','eigBranch', ...
        'RbothAtQmax','FAtQmax','EAtQmax', ...
        'PminusAtQmax','PplusAtQmax','cBothAtQmax','-v7.3');

    fprintf('\nResultados guardados en %s\n',outputDirectory);
end

fprintf('\n============================================================\n');
```

```
fprintf('TERMINADO\n');
fprintf('Tablas disponibles directamente en el workspace:\n');
fprintf('  radialAuditTable\n');
fprintf('  summaryTable\n');
fprintf('  coefficientCutoffDisplay\n');
fprintf('  groupSummaryTable\n');
fprintf('  groupPartialTable\n');
fprintf('  positivityTable\n');
fprintf('  blockQmax\n');
fprintf('  omittedTable\n');
fprintf('  uniquenessTable\n');
fprintf('  topFrequencyTable\n');
fprintf('============================================================\n');

%% ========================================================================
% FUNCIONES LOCALES
% ========================================================================

function c = fitCarrier(z,b)

    z = z(:);
    b = b(:);

    denominator = b'*b;

    if abs(denominator)<=eps
        c = NaN;
    else
        c = (b'*z)/denominator;
    end
end

function [a,c] = fitConstantAndCarrier(z,b)

    z = z(:);
    b = b(:);

    X = [ones(size(b)),b];
    coefficients = X\z;

    a = coefficients(1);
    c = coefficients(2);
end

function value = complexAnisotropy(z,minimumVariance)

    z = z(:);
    z = z-mean(z,'omitnan');

    denominator = mean(abs(z).^2,'omitnan');

    if ~isfinite(denominator) || denominator<=minimumVariance
        value = NaN;
    else
```

```
        value = abs(mean(z.^2,'omitnan'))/denominator;
    end
end

function r = safeCorrelation(x,y)

    x = x(:);
    y = y(:);

    valid = isfinite(x) & isfinite(y);

    if nnz(valid)<3
        r = NaN;
        return;
    end

    x = x(valid);
    y = y(valid);

    x = x-mean(x);
    y = y-mean(y);

    denominator = sqrt(sum(abs(x).^2)*sum(abs(y).^2));

    if denominator<=eps
        r = NaN;
    else
        r = real((x'*y)/denominator);
    end
end

function r = complexVectorCorrelation(x,y)

    x = x(:);
    y = y(:);

    valid = isfinite(real(x)) & isfinite(imag(x)) ...
        & isfinite(real(y)) & isfinite(imag(y));

    x = x(valid);
    y = y(valid);

    denominator = norm(x)*norm(y);

    if denominator<=eps
        r = NaN;
    else
        r = (x'*y)/denominator;
    end
end

function [qVec,basePrimeVec,powerExponentVec,lambdaVec] = ...
    makePrimePowers(Qmax)
```

```
    if Qmax<2
        qVec = zeros(0,1);
        basePrimeVec = zeros(0,1);
        powerExponentVec = zeros(0,1);
        lambdaVec = zeros(0,1);
        return;
    end

    pList = primes(Qmax);

    qList = zeros(0,1);
    baseList = zeros(0,1);
    exponentList = zeros(0,1);
    lambdaList = zeros(0,1);

    for jj = 1:numel(pList)

        p = pList(jj);
        currentPower = p;
        exponent = 1;

        while currentPower<=Qmax

            qList(end+1,1) = currentPower; %#ok<AGROW>
            baseList(end+1,1) = p; %#ok<AGROW>
            exponentList(end+1,1) = exponent; %#ok<AGROW>
            lambdaList(end+1,1) = log(p); %#ok<AGROW>

            if currentPower>floor(Qmax/p)
                break;
            end

            currentPower = currentPower*p;
            exponent = exponent+1;
        end
    end

    [qVec,order] = sort(qList);
    basePrimeVec = baseList(order);
    powerExponentVec = exponentList(order);
    lambdaVec = lambdaList(order);

    [qVec,uniqueIndex] = unique(qVec,'stable');
    basePrimeVec = basePrimeVec(uniqueIndex);
    powerExponentVec = powerExponentVec(uniqueIndex);
    lambdaVec = lambdaVec(uniqueIndex);
end

function [summaryTable,partialTable] = primePowerTelescopingTests( ...
    qVec,basePrimeVec,powerExponentVec,cMinusVec,cPlusVec, ...
    gamma,carrier)

    uniquePrimes = unique(basePrimeVec);
```

```
keep = false(size(uniquePrimes));

for jj = 1:numel(uniquePrimes)
    keep(jj) = nnz(basePrimeVec==uniquePrimes(jj))>=2;
end

uniquePrimes = uniquePrimes(keep);

summaryNames = { ...
    'base_prime','num_powers','largest_power', ...
    'combined_energy_ratio_sum_over_individual', ...
    'minus_energy_ratio_sum_over_individual', ...
    'plus_energy_ratio_sum_over_individual', ...
    'combined_rms_over_sum_term_rms', ...
    'boundary_fit_relative_rms_combined', ...
    'boundary_fit_relative_rms_minus', ...
    'boundary_fit_relative_rms_plus', ...
    'boundary_alpha_first_real','boundary_alpha_first_imag', ...
    'boundary_alpha_last_real','boundary_alpha_last_imag', ...
    'tail_primitive_error_minus','tail_primitive_error_plus', ...
    'tail_primitive_start_abs_minus', ...
    'tail_primitive_start_abs_plus'};

partialNames = { ...
    'base_prime','power_exponent','n_prime_power', ...
    'term_rms_minus','term_rms_plus','term_rms_combined', ...
    'partial_rms_minus','partial_rms_plus','partial_rms_combined', ...
    'partial_coefficient_minus_real', ...
    'partial_coefficient_minus_imag', ...
    'partial_coefficient_plus_real', ...
    'partial_coefficient_plus_imag'};

if isempty(uniquePrimes)
    summaryTable = cell2table(cell(0,numel(summaryNames)), ...
        'VariableNames',summaryNames);

    partialTable = cell2table(cell(0,numel(partialNames)), ...
        'VariableNames',partialNames);

    return;
end

totalPartialRows = 0;

for jj = 1:numel(uniquePrimes)
    totalPartialRows = totalPartialRows ...
        +nnz(basePrimeVec==uniquePrimes(jj));
end

summaryRows = cell(numel(uniquePrimes),numel(summaryNames));
partialRows = cell(totalPartialRows,numel(partialNames));

partialRow = 0;
```

```
for jp = 1:numel(uniquePrimes)

    p = uniquePrimes(jp);
    idxTerms = find(basePrimeVec==p);

    [~,order] = sort(powerExponentVec(idxTerms));
    idxTerms = idxTerms(order);

    R = numel(idxTerms);

    sumMinus = complex(zeros(size(gamma)));
    sumPlus = complex(zeros(size(gamma)));

    sumEnergyMinus = 0;
    sumEnergyPlus = 0;
    sumEnergyCombined = 0;
    sumTermRMSCombined = 0;

    firstMinus = [];
    firstPlus = [];
    lastMinus = [];
    lastPlus = [];

    cumulativeCoefficientMinus = 0;
    cumulativeCoefficientPlus = 0;

    for rr = 1:R

        j = idxTerms(rr);
        ell = log(qVec(j));
        z = exp(1i*ell*gamma);

        termMinus = cMinusVec(j).*carrier.*conj(z);
        termPlus = cPlusVec(j).*carrier.*z;
        termCombined = termMinus+termPlus;

        if rr==1
            firstMinus = termMinus;
            firstPlus = termPlus;
        end

        if rr==R
            lastMinus = termMinus;
            lastPlus = termPlus;
        end

        sumMinus = sumMinus+termMinus;
        sumPlus = sumPlus+termPlus;

        termEnergyMinus = mean(abs(termMinus).^2);
        termEnergyPlus = mean(abs(termPlus).^2);
        termEnergyCombined = mean(abs(termCombined).^2);

        sumEnergyMinus = sumEnergyMinus+termEnergyMinus;
```

```
        sumEnergyPlus = sumEnergyPlus+termEnergyPlus;
        sumEnergyCombined = sumEnergyCombined+termEnergyCombined;

        sumTermRMSCombined = sumTermRMSCombined ...
            +sqrt(termEnergyCombined);

        cumulativeCoefficientMinus = cumulativeCoefficientMinus ...
            +cMinusVec(j);

        cumulativeCoefficientPlus = cumulativeCoefficientPlus ...
            +cPlusVec(j);

        partialRow = partialRow+1;

        partialRows(partialRow,:) = { ...
            p,powerExponentVec(j),qVec(j), ...
            sqrt(termEnergyMinus),sqrt(termEnergyPlus), ...
            sqrt(termEnergyCombined), ...
            sqrt(mean(abs(sumMinus).^2)), ...
            sqrt(mean(abs(sumPlus).^2)), ...
            sqrt(mean(abs(sumMinus+sumPlus).^2)), ...
            real(cumulativeCoefficientMinus), ...
            imag(cumulativeCoefficientMinus), ...
            real(cumulativeCoefficientPlus), ...
            imag(cumulativeCoefficientPlus)};
    end

    sumCombined = sumMinus+sumPlus;

    energyRatioCombined = mean(abs(sumCombined).^2) ...
        /max(sumEnergyCombined,eps);

    energyRatioMinus = mean(abs(sumMinus).^2) ...
        /max(sumEnergyMinus,eps);

    energyRatioPlus = mean(abs(sumPlus).^2) ...
        /max(sumEnergyPlus,eps);

    rmsOverSumRMS = sqrt(mean(abs(sumCombined).^2)) ...
        /max(sumTermRMSCombined,eps);

    firstCombined = firstMinus+firstPlus;
    lastCombined = lastMinus+lastPlus;

    [relativeCombined,alphaCombined] = boundaryFit( ...
        sumCombined,firstCombined,lastCombined);

    [relativeMinus,~] = boundaryFit( ...
        sumMinus,firstMinus,lastMinus);

    [relativePlus,~] = boundaryFit( ...
        sumPlus,firstPlus,lastPlus);

    cM = cMinusVec(idxTerms);
```

```
        cP = cPlusVec(idxTerms);

        % Toda sucesion admite esta primitiva de cola. La identidad es
        % tautologica; lo informativo es su tamano y si existe forma cerrada.
        aM = [-flipud(cumsum(flipud(cM)));0];
        aP = [-flipud(cumsum(flipud(cP)));0];

        primitiveErrorM = max(abs(cM-(aM(2:end)-aM(1:end-1))));
        primitiveErrorP = max(abs(cP-(aP(2:end)-aP(1:end-1))));

        summaryRows(jp,:) = { ...
            p,R,max(qVec(idxTerms)), ...
            energyRatioCombined,energyRatioMinus,energyRatioPlus, ...
            rmsOverSumRMS, ...
            relativeCombined,relativeMinus,relativePlus, ...
            real(alphaCombined(1)),imag(alphaCombined(1)), ...
            real(alphaCombined(2)),imag(alphaCombined(2)), ...
            primitiveErrorM,primitiveErrorP, ...
            abs(aM(1)),abs(aP(1))};
    end

    summaryTable = cell2table(summaryRows, ...
        'VariableNames',summaryNames);

    partialTable = cell2table(partialRows(1:partialRow,:), ...
        'VariableNames',partialNames);
end

function [relativeRMS,alpha] = boundaryFit(target,firstTerm,lastTerm)

    target = target(:);
    firstTerm = firstTerm(:);
    lastTerm = lastTerm(:);

    X = [firstTerm,lastTerm];
    alpha = X\target;

    residual = target-X*alpha;

    relativeRMS = sqrt(mean(abs(residual).^2)) ...
        /max(sqrt(mean(abs(target).^2)),eps);
end


function [radius,centers,densitySmooth,peakRadii] = ...
    radialDistributionSummary(z,numBins,smoothWidth)

    z = z(:);
    z = z(isfinite(real(z)) & isfinite(imag(z)));

    center = mean(z);
    radius = abs(z-center);

    rMin = min(radius);
```

```
    rMax = max(radius);

    if rMax<=rMin
        rMax = rMin+eps(rMin+1);
    end

    edges = linspace(rMin,rMax,numBins+1);
    density = histcounts(radius,edges,'Normalization','pdf');
    centers = ((edges(1:end-1)+edges(2:end))/2).';

    densitySmooth = smoothdata(density(:),'gaussian',smoothWidth);

    candidates = find( ...
        densitySmooth(2:end-1)>densitySmooth(1:end-2) & ...
        densitySmooth(2:end-1)>=densitySmooth(3:end))+1;

    if isempty(candidates)
        peakRadii = zeros(0,1);
        return;
    end

    candidates = candidates( ...
        densitySmooth(candidates)>=0.12*max(densitySmooth));

    if isempty(candidates)
        peakRadii = zeros(0,1);
        return;
    end

    % Conserva maximos separados al menos seis bins, ordenados por altura.
    [~,heightOrder] = sort(densitySmooth(candidates),'descend');
    selected = zeros(0,1);

    for j = 1:numel(heightOrder)
        candidate = candidates(heightOrder(j));

        if isempty(selected) || all(abs(candidate-selected)>=6)
            selected(end+1,1) = candidate; %#ok<AGROW>
        end

        if numel(selected)>=4
            break;
        end
    end

    peakRadii = sort(centers(selected));
end

function s = joinNumericVector(v)

    v = v(:).';

    if isempty(v)
        s = "";
```

```
        return;
    end

    pieces = strings(size(v));

    for j = 1:numel(v)
        pieces(j) = sprintf('%.9g',v(j));
    end

    s = strjoin(pieces,', ');
end
```

41. Complete console output of the full-data run

41.1. **Coefficient identities, radial audit, branches, towers, blocks, and omitted frequencies.**

Remark 41.1. The console transcript below is included as numerical support. The analytic claims of the manuscript are established in the theorem–proof sections and do not depend on numerical rounding in this transcript.

Listing 2. Complete console output for the corrected full-data run.

```
============================================================
ZETA PRIME-POWER STRUCTURE: AUDITORIA ANALITICA
============================================================
Archivo de ceros       : zeross.bin
Ceros disponibles      : 2001052
q                      : 2.7182818284590451
omega=log(q)           : 1
stride                 : 1
Qlist                  : 0 3 10 100 1000 10000
Puntos analizados       : 2001052
gamma inicial           : 14.134725142
gamma final             : 1132490.65871441

============================================================
TEST 2: IDENTIDADES DE COEFICIENTES
============================================================
Potencias primas <=Qmax                 : 1280
max |(c^-+c^+)-formula|                : 2.220446e-16
max |(c^--c^+)-formula|                : 2.220446e-16
max |c^-/c^+-formula|                  : 1.776357e-15
      Q      num_terms    Re_AQ          Im_AQ           abs_AQ_minus_half    Re_BQ          Im_BQ
     abs_BQ_minus_half    abs_BQ_plus_half
    -----    ---------    -----    ------------------    -----------------    -----    ------------------
     -----------------    -----------------

        0         0         0                       0                  0.5      0                       0
                  0.5                 0.5
        3         2         0      -0.675311086221108    0.840264876793701      0      -0.863660742917475
     0.997952844004547    0.997952844004547
       10         7         0       -1.07698676185356      1.1873923046777      0       -1.54062594080643
      1.6197309311999      1.6197309311999
      100        35         0       -1.47655654519435     1.55891604365222      0       -2.87744845271495
     2.92056665700882     2.92056665700882
```

```
     1000       193          0       -1.89626714054451       1.9610785472053       0       -5.31082516388603
     5.33431006985581      5.33431006985581
    10000      1280          0       -2.55521376172467      2.60367382137378       0       -10.6932626383477
     10.7049458593999      10.7049458593999


============================================================
AUDITORIA RADIAL DIRECTA DE LA BIMODALIDAD
============================================================
      case_name         mean_radius           std_radius         num_detected_peaks       detected_peak_radii
    -------------    -----------------    ------------------    ------------------    ------------------------

    "original"       1.92031044763071     0.726547772615351             2             "1.35731059, 2.49789956"
    "after_n3"       1.80792479214104     0.308312030736288             1             "1.68017926"
    "after_n2_n3"    1.78844590851816      0.22140483373138             1             "1.81300495"

Coeficiente teorico total n=3:
  c3^- = 0 -1.02370050729i
  c3^+ = 0 +0.0481029537851i
Q=0      terms=0      c^-res=0.4999989+0.1381673i  c^+res=0.4999989+0.1381673i  c^both=0.4999989+0.1381673i
     Var(F)=1.2420e+00  RMS(E)=1.1145e+00
Q=3      terms=2      c^-res=0.4999999+0.0909043i  c^+res=0.4999989+0.1429495i  c^both=0.4999999+0.0956865i
     Var(F)=1.3393e-01  RMS(E)=3.6597e-01
Q=10     terms=7      c^-res=0.5000000+0.0636593i  c^+res=0.4999989+0.1500304i  c^both=0.5000000+0.0755224i
     Var(F)=5.9610e-02  RMS(E)=2.4415e-01
Q=100    terms=35     c^-res=0.5000000+0.0191267i  c^+res=0.4999989+0.1736326i  c^both=0.5000000+0.0545920i
     Var(F)=2.2040e-02  RMS(E)=1.4846e-01
Q=1000   terms=193    c^-res=0.5000000-0.0206343i  c^+res=0.4999990+0.2014465i  c^both=0.5000001+0.0426449i
     Var(F)=1.1282e-02  RMS(E)=1.0622e-01
Q=10000  terms=1280   c^-res=0.4999999-0.0581400i  c^+res=0.4999989+0.2306199i  c^both=0.4999999+0.0343125i
     Var(F)=8.2559e-03  RMS(E)=9.0862e-02

============================================================
TEST 1: RAMAS; TEST 5: ECUACION FUNCIONAL
============================================================
      Q      num_prime_power_terms    c_after_minus_real    c_after_minus_imag     c_after_plus_real
     c_after_plus_imag    c_after_both_real    c_after_both_imag     distance_c_after_both_to_half
     rms_centered_Y_minus_Pminus    rms_centered_Y_minus_Pplus    rms_centered_Y_minus_both
     rms_after_both_fitted_carrier     energy_Pminus         energy_Pplus
     cross_term_2Re_inner_Pminus_Pplus        F_variance         F_max_abs_deviation_from_mean
     F_block_mean_std        F_anisotropy       carrier_discrete_defect_rms    derivative_defect_rms
    -----    ----------------------    ------------------    -------------------    -----------------
     -----------------    ------------------    -------------------    ------------------------------
     ----------------------------    --------------------------    -------------------------
     -------------------------------    -----------------    --------------------
     ----------------------------------    --------------------    ------------------------------
     --------------------    -----------------    ---------------------------    ---------------------

        0               0             0.499998912312768       0.138167317526428    0.499998912312768
     0.138167317526428    0.499998912312768      0.138167317526428           0.138167317530709
     1.22927492233399              1.22927492233399              1.22927492233399
     1.11446297144985                          0                     0                                0
         1.24202771473284           2.53363749717284            0.0101613448739479    0.164704934429685
       0.828969921883291           2.13681541452671
```

```
     3               2              0.49999988687813       0.0909043288314701     0.499998893284938
  0.142949512755117     0.499999867850236     0.0956865240601598           0.095686524060251
  0.630307565358949               1.22653992517659              0.626967498686433
  0.365967847866614           1.11258876067543     0.00443650350346841             0.00251726408927867
      0.133932465672121             1.3998447731532            0.00612569360136624     0.670875082815371
     0.476587597924875             1.7823665130755
    10               7              0.49999998247139       0.0636593469198101     0.499998916032037
  0.150030381454066     0.499999986190665      0.075522410847451          0.0755224108474523
  0.568023295911874               1.22456854935429              0.561527861557534
  0.244151425501691           1.18492844347144     0.00871311004796696             0.00421177723983395
     0.0596099185745075            1.05195331714233             0.0042120343511029     0.886135136342853
     0.398840669251055            1.69391674505877
   100              35              0.500000017346496       0.0191267412460307      0.49999893653376
  0.173632564206007      0.50000004156745      0.054591987925598          0.0545919879256138
  0.536555599882361               1.22197814824027              0.524423432995709
  0.148457436196022           1.21916077552125      0.0180717973642778             0.00501426487870665
     0.0220396103618959           0.799206246858361             0.0022297937207615     0.958121438757457
      0.2682174006908            1.45913395069404
 1000              193              0.500000036693381      -0.0206342853236017     0.499998951118377
  0.201446524072374     0.500000075498899     0.0426449212223482           0.042644921222415
  0.526364711348993               1.22166555561873               0.51293359872215
  0.10621775695736           1.23384826802886      0.0252917244776332             0.00469116700005195
    0.0112822118930529           0.668993854159613             0.0011112380638497     0.970189423496525
    0.179876374959657            1.21197809635573
10000             1280              0.499999922217885      -0.0581399985655881     0.499998861049832
  0.230619861962331     0.499999870954991     0.0343125458703112           0.0343125458705538
  0.523988825848724               1.22334869951959              0.509345817216466
  0.0908622020249825           1.24333446127337       0.030996201483447           -0.000596344803520444
     0.00825593975682874            0.61202824502835           0.000381258918696766     0.981211983063741
      0.150986530997687             1.0471736199134

max error R_j-R_{j-1}=-P_j                  : 8.962228e-16
max error ley de energia de ramas           : 2.220446e-16
max error identidad de proyeccion portadora : 3.559044e-14
max error identidad de constantes            : 5.446388e-14

============================================================
TEST 3: TELESCOPADO EN p,p^2,...,p^R
============================================================
    base_prime    num_powers    largest_power    combined_energy_ratio_sum_over_individual
     combined_rms_over_sum_term_rms    boundary_fit_relative_rms_combined    boundary_fit_relative_rms_minus
       boundary_fit_relative_rms_plus    boundary_alpha_first_real    boundary_alpha_first_imag
     boundary_alpha_last_real    boundary_alpha_last_imag    tail_primitive_start_abs_minus
     tail_primitive_start_abs_plus
    ----------    ----------    -------------    ----------------------------------------
     ------------------------------    ----------------------------------    ------------------------------
       ------------------------------    -------------------------    -------------------------
     ------------------------    ------------------------    ------------------------------
     ----------------------------

        89            2             7921                       0.99578526600946
     0.951568779802395                 3.25731221427626e-15               1.66529459300868e-14
      5.68490958536155e-15             0.999999999999997           5.38298239788951e-18
     0.999999999999988        -6.13165759929884e-17             0.022712415828171
     0.0146012082752434
```

```
   37            2            1369                      0.992325345063945
0.929272152959258                3.32342530365488e-15              6.90975846079265e-15
 2.57414422667853e-14             0.999999999999997         -8.74885970690982e-18
1.00000000000001        -8.01331294276757e-17             0.0386826763307904
0.0223795089575229
   41            2            1681                      0.992798804843908
0.932179807015267                4.01764974378634e-15              5.71123669041389e-15
 6.00240878492108e-15                            1         -4.41766278752356e-18
0.999999999999997         6.86611361451242e-17             0.0362585450928221
0.0212931769572583
   23            2             529                      0.98981553486963
0.914722655159467                5.29829056019486e-15              3.01909112408219e-14
 5.06477325930078e-14             0.999999999999995          1.5256443527521e-17
1.00000000000002        -4.12721429521546e-17             0.052842635596283
0.0281454412919937
   67            2            4489                      0.994867301216485
0.945073421083208                6.62713306705871e-15              2.92396921729543e-14
 3.16762417963749e-14             0.999999999999993          4.06385013591999e-18
0.999999999999977         -6.6991869985725e-17             0.0268592065905553
0.0167695322528614
   73            2            5329                      0.995124247078512
0.947082293993756                8.02253098923419e-15              4.21876931539434e-14
 2.84327775551962e-14             0.999999999999992          2.29352846836488e-17
0.999999999999967         1.25402372809613e-16             0.0255226923919741
0.0160830067737586
   29            2             841                      0.991065442073751
0.921970578232658                1.69958093184349e-14              3.92022883362443e-15
 4.85298540260265e-14             1.00000000000002          5.31348409790742e-18
0.999999999999986        -1.67495027840586e-16             0.0452613167826162
0.0251764129412383
   59            2            3481                      0.994356231992412
0.941904651888199                1.82233584563793e-14              4.44612460135524e-14
  3.3145884657823e-14             1.00000000000002          3.31967944946604e-17
0.999999999999965         2.13895102136555e-16             0.0289902084009457
0.0178408950865232
   61            2            3721                      0.994484921893845
0.942760057482987                1.89061413493395e-14              4.06345157893362e-14
 1.56552838293448e-14             0.999999999999981         -1.56580841679002e-17
0.999999999999966        -2.22783722808597e-17             0.0284133680209016
0.0175536598492679
   43            2            1849                      0.993018430417279
0.933512335891677                1.91767251608662e-14              9.19221287192386e-15
 2.67694821919958e-14             1.00000000000002         -5.47518533793213e-18
0.999999999999988         1.24241990600724e-16             0.0351952587338588
0.0208067465987886
   97            2            9409                      0.996034159322401
0.953412551798485                2.19836316719303e-14              2.48682193279047e-15
 4.67252549818149e-15             0.999999999999978         -1.96285882336457e-17
0.999999999999975        -2.20623599182162e-16             0.0216013618259659
0.0140005339242615
   71            2            5041                      0.995055469368263
0.94645716703403                 2.45606582770506e-14              1.40537772877289e-14
1.35601376253121e-14             1.00000000000002         -2.17658103478551e-17
1.00000000000007        -3.81439794611675e-16             0.0259471363764858
0.0163022731943209
```

```
   83            2             6889                      0.995556221672419
0.950034257670606                 2.80380400407119e-14               7.38122722146823e-15
 1.69133036510308e-14             1.00000000000003         -1.68047878446504e-17
1.00000000000006        -3.60217187758929e-16            0.0236603972161033
0.0151069817398424
   53            2             2809                      0.993923584391844
0.939166861275507                 3.64856201837887e-14               5.33595501442798e-14
 6.78833363346336e-15             1.00000000000004         -1.38986297087105e-17
0.999999999999991       -4.48286028808063e-16            0.030939494184714
0.0187966802009105
   47            2             2209                      0.99342415440416
0.935964632967202                 4.02588932254384e-14               4.69976317545872e-15
 3.70119612870561e-14             1.00000000000004          3.64557105503385e-17
0.999999999999993       -5.00636376597972e-16            0.0333062074556399
0.0199271852036153
   79            2             6241                      0.995407211972277
0.948922007921855                 4.26585420767673e-14               6.84664346328644e-16
 1.95720970351145e-14            0.999999999999957          4.32727341964049e-17
0.999999999999994        6.93817715660186e-16            0.0243581235750551
0.015475353217867
   31            2              961                      0.99142834108407
0.924010046881153                 4.48560611982839e-14               1.73461849505888e-14
 4.95966705898557e-15             1.00000000000004          1.34960651075476e-18
1.00000000000001        -1.47682556498033e-16            0.0433336826232223
0.0243790159002065
    3            8             6561                      0.994136424160185
0.925379114250609                  0.0526293055865214                 0.0488679460726364
     0.38539545977793            0.997125200778133          1.04928001769985e-08
-0.373319182377488          0.00313372076449291              1.09725325128044
0.080998072339286
   19            3             6859                      0.988180958947934
0.896291448276131                  0.099571908742451                 0.0910436270892582
    0.130813323815928            0.994234206127615          8.21480982437346e-07
0.587653567802991       -6.66788850988072e-05            0.0610579674917809
0.0314116469586121
   17            3             4913                      0.987462564808847
0.891547371679716                  0.104055647033926                 0.0950571818522644
    0.138802345385611            0.993904379532631          7.82474768756128e-08
0.603159942163849       -1.05608473149603e-05            0.0661990286058359
0.0331867205015699
```

boundary_fit_relative_rms_combined cercano a cero seria evidencia de colapso a los dos bordes del bloque p^r.

```
============================================================
TEST 4: POSITIVIDAD Y GRAM
============================================================
      Q           norm_E2          carrier_schur_complement    carrier_projection_abs     Re_inner_E_Pminus
          Re_inner_E_Pplus       cauchy_det_E_Pminus     cauchy_det_E_Pplus     cauchy_det_Pminus_Pplus
    polarization_error_minus    polarization_error_plus
    -----    -------------------    ------------------------    ----------------------    --------------------
        ---------------------    -------------------    --------------------    ------------------------
     ------------------------    ----------------------

       0       1.24202771473284         1.24202771473284         4.1753563397406e-15                          0
                           0                        0                        0                          0
                       0                        0
       3      0.133932465672121        0.133932465672121        3.19449329162395e-14     -0.00516035229366167
         0.000339021642957032     0.148985126760466     0.000594076917452501      0.00493441978003228
    2.22044604925031e-16       8.67361737988404e-19
```

```
       10      0.0596099185745075        0.0596099185745074         7.25002166702646e-15      -0.00596341179707172
          0.000207836720997232     0.0705979257513303      0.000519344584387362        0.0103199771600532
     2.22044604925031e-16        3.46944695195361e-18
      100      0.0220396103618959        0.0220396103618958         1.12998339769851e-14      -0.00697316806144494
        -0.000663780178893137      0.026821203386263      0.000397854767628292         0.022026140776441
                        0                         0
     1000      0.0112822118930529        0.0112822118930528         3.76207946660896e-14       -0.0090127255333809
         -0.00296783860935025     0.0138393083795104      0.000276538526093234        0.0312006486796008
     2.22044604925031e-16        1.38777878078145e-17
    10000     0.00825593975682874       0.00825593975682881         2.86661244710573e-14      -0.00982808606142384
         -0.00476027747414303      0.010168303134064      0.000233242529391743        0.0385385565639377
     2.22044604925031e-16        3.46944695195361e-18

Gram combinada: min=2.291421e-03, max=1.119809e+00, negativos<-1e-10=0
Gram por ramas: min=4.858227e-04, max=1.112574e+00, negativos<-1e-10=0

============================================================
TEST 5: F_Q POR BLOQUES DE ALTURA, Q=10000
============================================================
    block      gamma_start          gamma_end           F_mean_real          F_mean_imag            F_variance
              sup_abs_F_minus_global_c    sup_abs_F_minus_block_c     mean_abs_delta_F
     variation_per_height       anisotropy        carrier_discrete_defect_rms    derivative_defect_rms
    -----    ----------------    ----------------    -----------------    ------------------
     -------------------    ------------------------    ------------------------    ------------------
     --------------------    -----------------    --------------------------    ---------------------

      1          14.134725142      63699.04754739    0.499994991361003    0.0350523539924386
     0.0054744556993458        0.542532351404617           0.542510726459913        0.0963670633500536
     0.126163323741996       0.979858632599489          0.121147135176878            0.538564667956422
      2       63699.700151835     118462.736278749    0.500001451274137     0.035105891285452
     0.00701927614984981        0.530144044699919           0.530116241500027        0.110272510195026
     0.167888441917521       0.980474241382446          0.137953328199234            0.726396110177662
      3       118463.232066064     170651.787459452    0.499998562254379    0.0349210363561646
     0.00745311407766993        0.549893284088027           0.549871555991103        0.113755262822731
     0.181736637024686       0.980873421275848          0.142306171066992            0.7574604344515
      4       170652.38272511     221291.519378741    0.500000256946116    0.0348042049768492
     0.00767564743723714        0.581090952553066           0.581063883641327        0.116011219222976
     0.191009406027093       0.980878273487911          0.144913294713216            0.880192519706131
      5       221292.295838797     270836.328162977    0.500000955646856    0.0347080955882823
     0.007831121437243         0.57962624512367           0.579600207778524        0.117381041928519
     0.197536641502951       0.980905060378457          0.146681991054392            0.923600652953341
      6       270836.931684803     319539.773103185    0.499999624306421    0.0346475825508576
     0.00808884143363537        0.534809090007345           0.53479827792399         0.118850165982451
     0.203463517740728       0.981360633102241          0.14914505740182             1.16913063854399
      7       319540.570678831     367564.435949345     0.49999957921803     0.0345578903969988
     0.00818709113446676         0.52294466209729           0.52292439742989         0.119910420246151
     0.208180893856153       0.981406498730984          0.150029861940657            0.880453279774671
      8       367565.006156693     415022.283240227    0.500000487984706    0.0344519123654481
     0.00814253081563704        0.600282121372596           0.600276208543475        0.119491391046747
     0.209930169452775       0.981010897479751          0.149551484403089            1.00807618824186
      9       415023.15758963     461995.409615676    0.500002125859323    0.0343892982332912
     0.00818298949111592        0.572654690283211           0.572652885861422          0.1200026729604
     0.213007944730247       0.980937348210029          0.150172616772631            1.08111614297039
     10       461995.761482843     508544.337297334    0.499999128775315    0.0343361215632085
     0.00835629092630199        0.542768812320236           0.542768199088721          0.12137168582189
     0.217396246824282       0.981335496066421          0.152037594340513            0.934750513408792
     11       508544.765796632     554719.129189251     0.49999902683957     0.0342841671661181
     0.00838655885140866        0.555269896680713           0.55526995649336         0.121722656912353
     0.219791838956823       0.981261093718569          0.152335939237595            1.12844994151658
```

```
 12       554719.634077261     600557.928334761     0.499999052950246     0.0342521074641879
0.00848384933678091        0.606087165112338           0.606091467206371          0.12213899474892
0.222160553553316       0.981322097240396          0.153080139635091             1.07065793214649
 13       600558.535755246     646093.102175532     0.500000235514038     0.0342171059277988
0.00852631056710716        0.524846602477766           0.524849827921532         0.122203119197641
0.223759839331275       0.981196827744373          0.153515460380432             1.14290197565882
 14        646094.01946502     691351.826509571     0.499999459797555     0.0341425285063966
0.00852459159575678         0.61202824502835           0.612034526321802         0.122717190800213
0.226075215930959        0.98130473863714          0.153711972752659             1.52515702182085
 15       691352.526827545     736357.304575928      0.50000063261387     0.0340996079052541
0.00863390467854271        0.603869638677834           0.603880275368038         0.123435919580137
0.228680535306117         0.9813459888377          0.154802894954523             1.13198994875413
 16       736358.196419715     781127.536440619     0.500000507064813     0.0340763607134163
0.00863949315690915        0.551934114882366           0.551943379429299         0.123479710718261
0.229961941722585       0.981411615889858          0.154800438591316             1.02064882439406
 17       781128.075976077     825679.648395047     0.499999867045727     0.0340594494960409
0.00867051311819898        0.586536403755823           0.586546166766736         0.123341062550961
0.230826520207627       0.981193765149418          0.155214716157765             1.15057126782006
 18       825680.493113046     870029.136314775     0.499999157278646     0.0340083355422251
0.00880860286374675        0.564338258958265           0.564345300438399         0.124310351932475
0.233705005485217       0.981676906683931          0.156532439348918             1.09637171176575
 19       870029.689029286     914187.909332815     0.499999527336173     0.0339700206865135
0.00876154865120847         0.56850486610471           0.568522179523045         0.124185981373753
0.234477981944182       0.981561079878874           0.15588197711697            0.995676199295695
 20       914188.612117913     958168.018537517      0.50000099473625     0.0339345816929237
0.00878369871741152        0.599171890474147           0.599196831871485         0.124030304858425
0.235136204413765        0.98142509043805          0.156206512272264             1.06669516660454
 21        958168.60666595     1001979.98016229     0.499998363067321     0.0338951222311845
0.0088529271020219         0.597221450310013            0.597233477557632         0.125000771021225
0.237887755021112       0.981656183204604          0.157057387478477             1.16995875593502
 22       1001980.68512795     1045632.21311476     0.500003656384172     0.0338638859862856
0.00874571864686109        0.562103795049353           0.562114143907445         0.124273605310392
0.237367088718873        0.98131312028741          0.156023921934697             1.01328485216897
 23       1045632.45556665     1089133.21595891     0.499998860220289     0.0338981896499074
0.00898518332990823        0.571629074187526           0.571636102485886           0.1254657058582
0.240474616933205       0.981670300817732          0.158067171395785             1.12663969590805
 24       1089133.56438868     1132490.65871441     0.500000398441743     0.0338252499519786
0.00892480514817264        0.590816935924078           0.590833242133321         0.125308756813391
0.240969628397672       0.981703647581976          0.157535630788163             1.18442415402658

Rango Re media por bloque : [0.499994991361, 0.500003656384]
Rango Im media por bloque : [0.033825249952, 0.0351058912855]
Max sup global por bloque : 6.120282e-01
Max sup local por bloque  : 6.120345e-01

============================================================
TEST 6: UNICIDAD Y FRECUENCIAS OMITIDAS
============================================================
    Qmax     num_omitted_prime_powers_joint_fit    median_minus_amplitude_prime_power
     median_minus_amplitude_null    minus_prime_power_to_null_ratio    median_plus_amplitude_prime_power
     median_plus_amplitude_null    plus_prime_power_to_null_ratio    joint_omitted_explained_fraction
     fit_theory_correlation_real    fit_theory_correlation_imag    fit_theory_correlation_abs
     relative_theory_coefficient_error    F_energy_before_omitted_fit    F_energy_after_omitted_fit
     F_anisotropy        corr_gap_realF        corr_gap_imagF        corr_gap_absF
```

| | | |

```
10177      10177             1           9.22788555119193     0.000562556014238549     -0.00123881678217888
     0.00136056469488371     0.000525626337530265     0.00105890848795867     0.00118218874659523
   0            -0.00176939258053915           0            0.00142339876359955     0.00136056469488371
10061      10061             1           9.21642184229205     -0.000271184796372421     -0.0012905239456047
     0.00131870893223739     -0.000199026354140983     0.00104720828811089     0.00106595341752386
   0            -0.00177983266568738           0            0.00143140682870411     0.00131870893223739
10247      10247             1           9.23474025879972     0.000296226420166288     -0.00127916398853374
     0.00131301584208496     0.000225798114775531     0.00100579696158174     0.00103083088650042
   0            -0.00176317956235639           0            0.0014186315781825     0.00131301584208496
10457      10457             1           9.25502688959859     -0.00038302950951308     -0.00124755231657541
     0.00130502811760916     -0.000336465701900575     0.0010635359416545     0.00111548996756868
   0            -0.00174492100357133           0            0.00140461550806041     0.00130502811760916
10223      10223             1           9.23239536275701     4.37551338344118e-05     -0.00126550404521048
     0.00126626024188591     6.46513618466203e-05     0.00112056448509538     0.00112242797713069
   0            -0.00176530245864271           0            0.00142026058016567     0.00126626024188591
10099      10099             1           9.22019168802656     3.77615682350026e-05     -0.0012349486588066
     0.00123552584996178     0.000122474628046823     0.00095475772183488     0.000962581083295603
   0            -0.00177639258075131           0            0.00142876845882169     0.00123552584996178
10429      10429             1           9.25234566612121     0.000104275707715879     -0.00122341220360067
     0.00122784805376671     0.000135594574873729     0.000939541687326359     0.000949275761282911
   0            -0.00174732317276288           0            0.00140646007086086     0.00122784805376671
10343      10343             1            9.2440652413778     8.65573758937093e-05     -0.00120755464889939
     0.00121065288559526     0.000151903149928664     0.000998017665518561     0.00100951167781525
   0            -0.00175476293000098           0            0.00141217180262058     0.00121065288559526
10111      10111             1            9.2213792190914     9.1842043748573e-05     -0.00120508425512231
     0.00120857892706419     5.50085507262679e-05     0.000972297490544918     0.00097385232493071
   0            -0.00177531032055336           0            0.00142793835000022     0.00120857892706419
```

```
Frecuencias no nulas dominantes de F_Q-mean(F_Q):
    rank    demodulated_frequency       fit_amplitude
    ----    ---------------------    --------------------

      1         7.22212181819358      0.00102998542892402
      2       -0.509252179488009     0.000972670281121882
      3        -7.22212181819358     0.000962787079088038
      4        0.509252179488009     0.000944393206974707
      5        -7.26841747087431     0.000923900818886456
      6        0.694434790210921     0.000912309613268334
      7       -0.694434790210921     0.000907278710300437
      8         7.26841747087431     0.000828631152433364
      9        -2.82403481352441     0.000809343213775063
     10        -6.89805224942848     0.000799201794454691
     11         6.89805224942848     0.000798947722046012
     12         8.42580878789251     0.000771802707211477
     13        0.879617400933833     0.000761874424348131
     14       0.0462956526807281     0.000760539159031102
     15       -0.879617400933833     0.000742831699098273
     16        -5.32400005828373     0.000727744418239324
     17        -8.42580878789251     0.000703624825291971
     18        -8.56469574593469     0.000703431607127999
     19        -8.61099139861542     0.000697831699989827
     20         8.61099139861542     0.000697444147794987
     21         8.93506096738052     0.000693242426624136
     22        -9.02765227274197     0.000691823571238767
     23      -0.0462956526807281     0.000690811206825879
     24         8.56469574593469     0.000687798036575012
     25         9.02765227274197     0.000680600662464099
     26                        0      0.00067560438002473
```

```
     27        -8.33321748253105     0.000673228670039579
     28         3.84253917250043     0.000672369683136021
     29        -3.84253917250043       0.0006713463801668
     30        -6.24991311189829     0.000667528070597226


DIAGNOSTICO FINAL PARA Q=10000
mean(F_Q)                          = 0.499999870955 +0.03431254587703i
|mean(F_Q)-1/2|                   = 3.431255e-02
Var(F_Q)                           = 8.255940e-03
sup |F_Q-mean(F_Q)|                = 6.120282e-01
RMS E_Q                            = 9.086220e-02
razon omitidas potencia/null (-) = 3.353487
razon omitidas potencia/null (+) = 2.675593
fraccion explicada ajuste omitido= 0.014445

============================================================
TERMINADO
Tablas disponibles directamente en el workspace:
  radialAuditTable
  summaryTable
  coefficientCutoffDisplay
  groupSummaryTable
  groupPartialTable
  positivityTable
  blockQmax
  omittedTable
  uniquenessTable
  topFrequencyTable
============================================================
>>
```

42. Conclusion

The finite Möbius interpolation is completely oriented by

$$\Phi_N(s) = M(N) + h_N \frac{R_N(s)}{P_N(s)}. \tag{42.1}$$

Its value at zero, its integer limit, its exponential rate, its rounding criterion, and its recovery of a_N are mutually consistent. The absolutely convergent half-plane does not select $M(N)$ among feasible terminal values.

For the zeta phase walk, the following components are exact:

(i) the endpoint conversion and triangular half identity;
(ii) the positive atomic horizontal defect $D_q(\tau)$;
(iii) the two-branch coefficient algebra;
(iv) the convergent signed invariant $\Delta_\omega(Q) \sim \omega/(\pi^2 \log Q)$;
(v) the raw zero-kernel identity for the carrier coefficient;
(vi) the exact global and local defects (16.16) and (17.9);
(vii) the joint constant–carrier projection formula;
(viii) the projected zero-sampled quadratic form (19.12).

The finite computation shows strong prime-power structure, including the disappearance of the radial double mode after subtraction of $n = 3$, a 99.335% finite variance reduction through $Q = 10000$, stable real carrier coefficient near $1/2$, and detectable omitted prime-power frequencies. None of these finite observations proves uniform energy exhaustion.

The possible routes to the Riemann Hypothesis are now explicit: equality of atomic jumps for one fixed $q > 1$, local distributional convergence, small-q quadratic curvature, large-q subpolynomial growth, or local projected-energy exhaustion accompanied by derivative identification. The main unresolved statements are not ordinary Parseval estimates. They are uniform estimates for the projected zero-sampled kernel, together with a local explicit-formula limit strong enough to identify the positive atomic measure

$$\sum_{\tau} D_q(\tau)\delta_\tau. \tag{42.2}$$

The termwise area mechanism fails away from the near-height regime. Any stable global relation between the centroid and $\Delta_\omega(Q)$ must therefore be collective.

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